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Yudanov et al.

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(54) **READ BROADCAST OPERATIONS
ASSOCIATED WITH A MEMORY DEVICE**

(58) **Field of Classification Search**
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(Continued)

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(73) Assignee: **Micron Technology, Inc.**, Boise, ID
(US)

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This patent is subject to a terminal dis-
claimer.

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(74) *Attorney, Agent, or Firm* — Holland & Hart LLP

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(Continued)

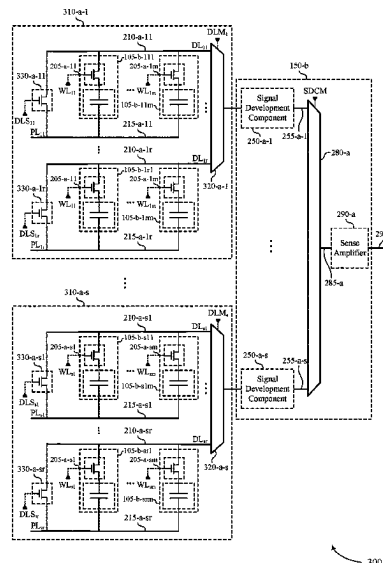
(57) **ABSTRACT**

Methods, systems, and devices related to write broadcast
operations associated with a memory device are described.
In one example, a memory device in accordance with the
described techniques may include a memory array, a sense
amplifier array, and a signal development cache configured
to store signals (e.g., cache signals, signal states) associated
with logic states (e.g., memory states) that may be stored at
the memory array (e.g., according to various read or write
operations). The memory device may enable read broadcast
operations. A read broadcast may occur from the memory
array to multiple locations of the signal development cache,
for example via one or more multiplexers.

(51) **Int. Cl.**
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G06F 9/54 (2006.01)
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(2013.01); **G06F 3/0673** (2013.01);
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20 Claims, 12 Drawing Sheets



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(58) **Field of Classification Search**

USPC 365/191
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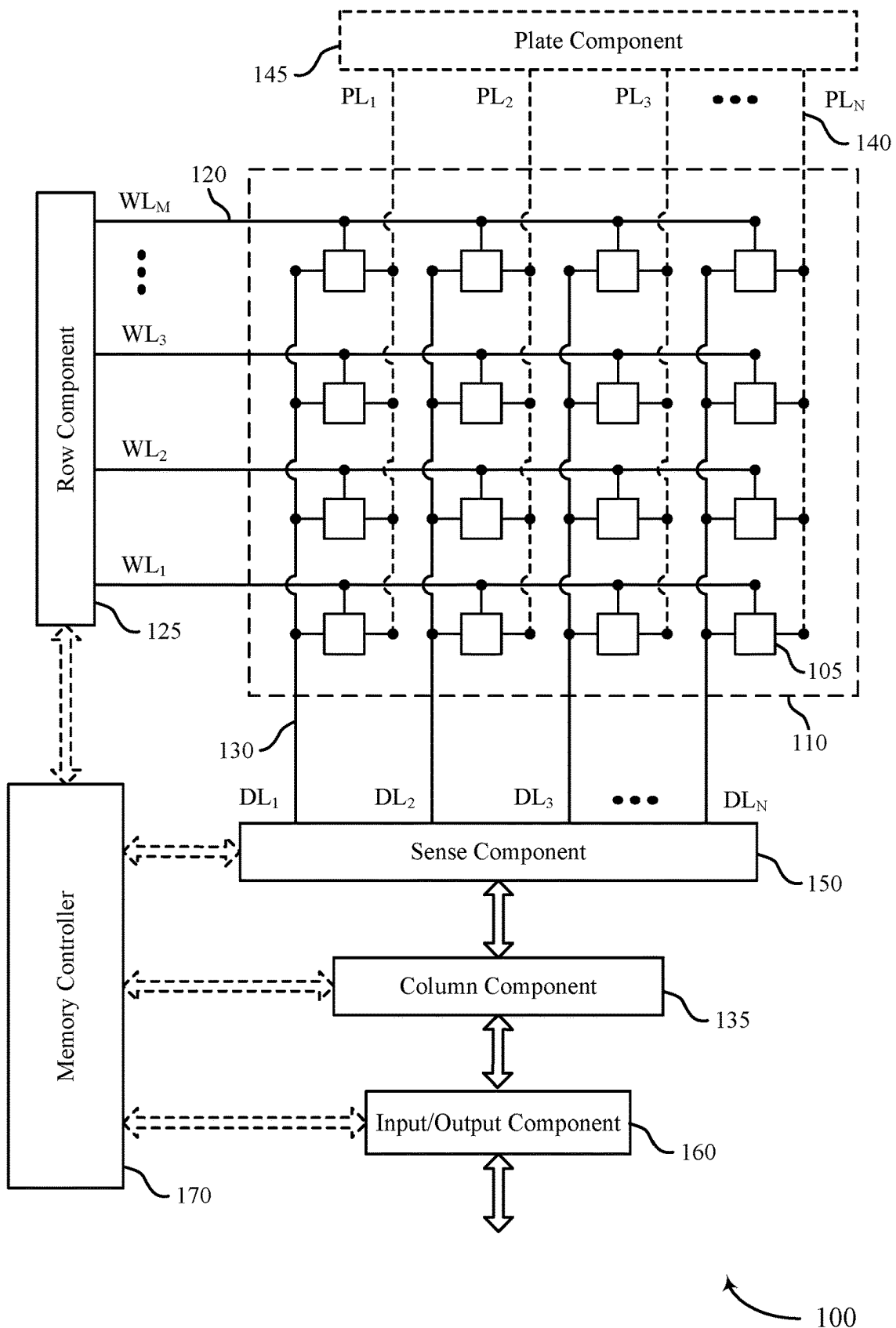


FIG. 1

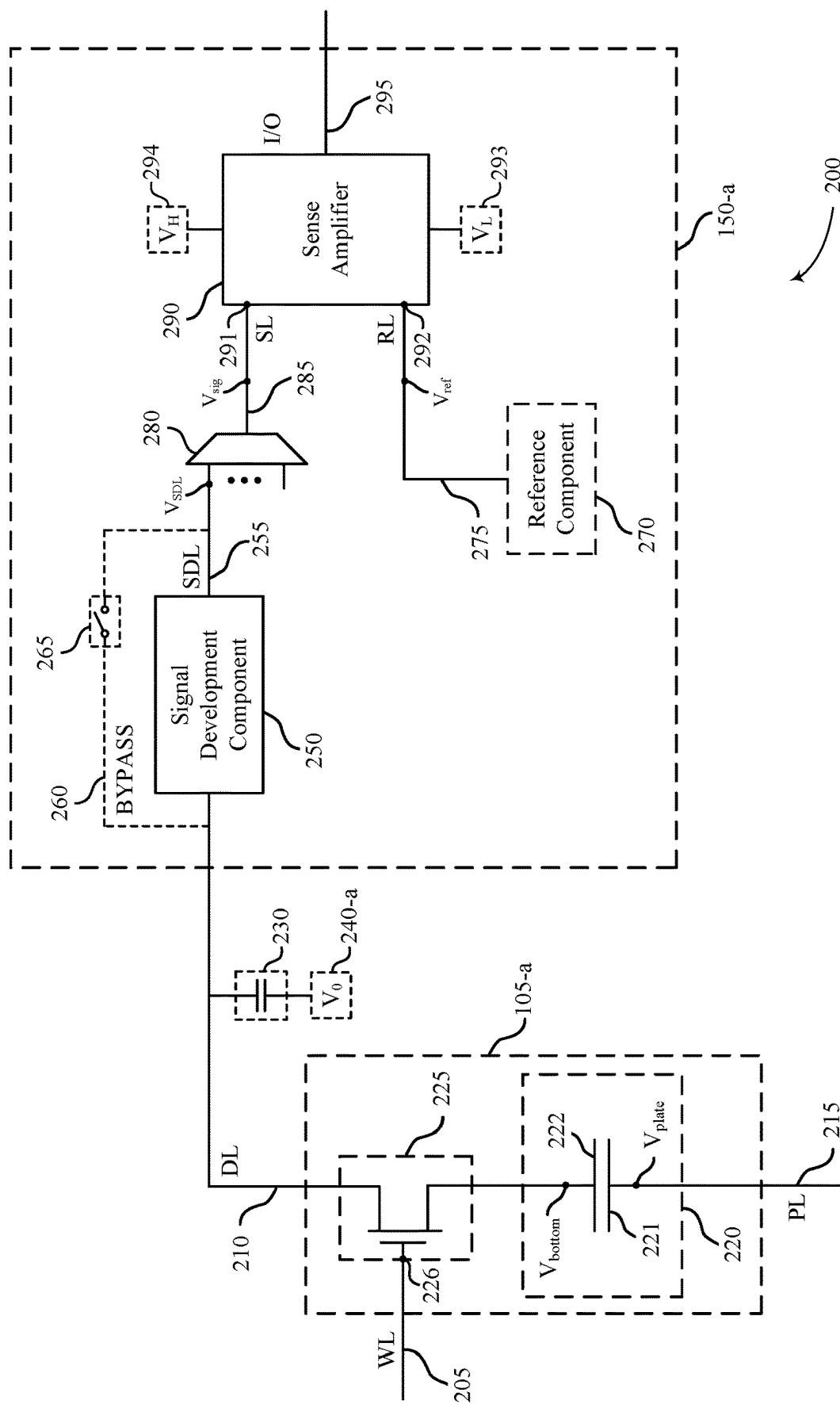


FIG. 2

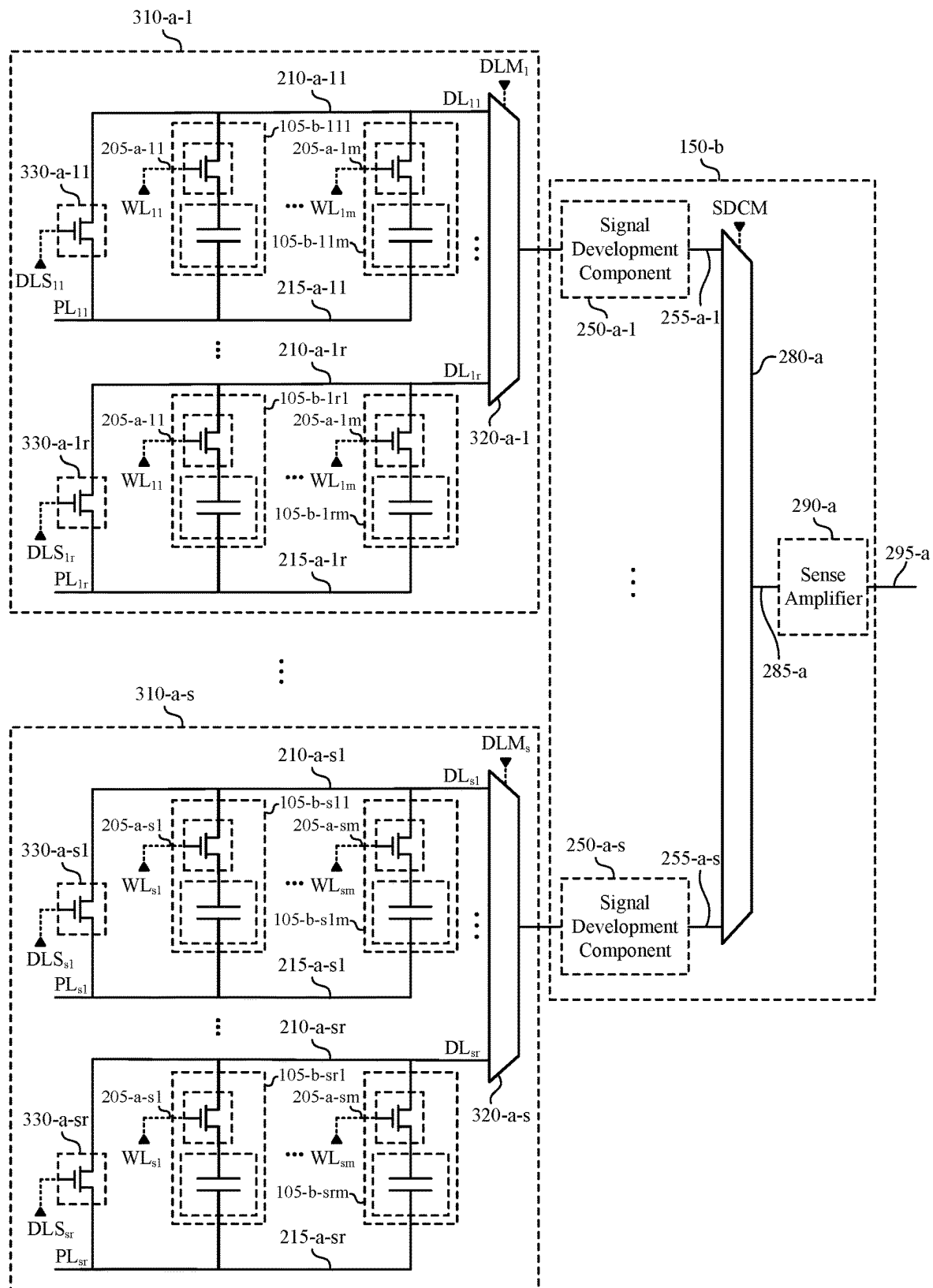


FIG. 3

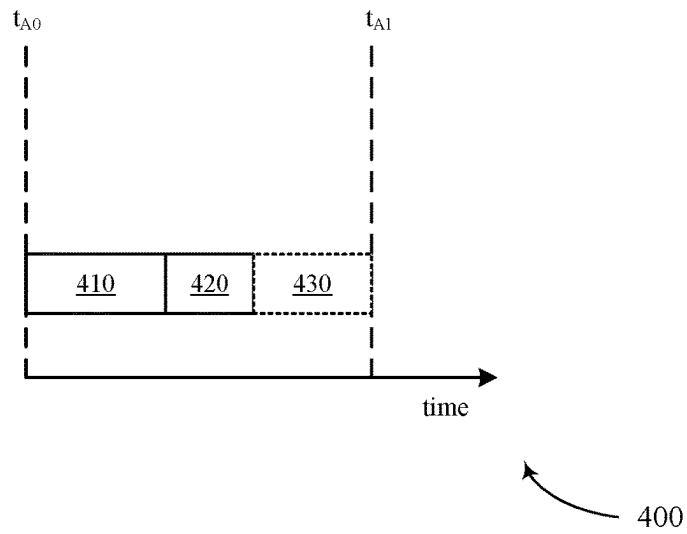


FIG. 4A

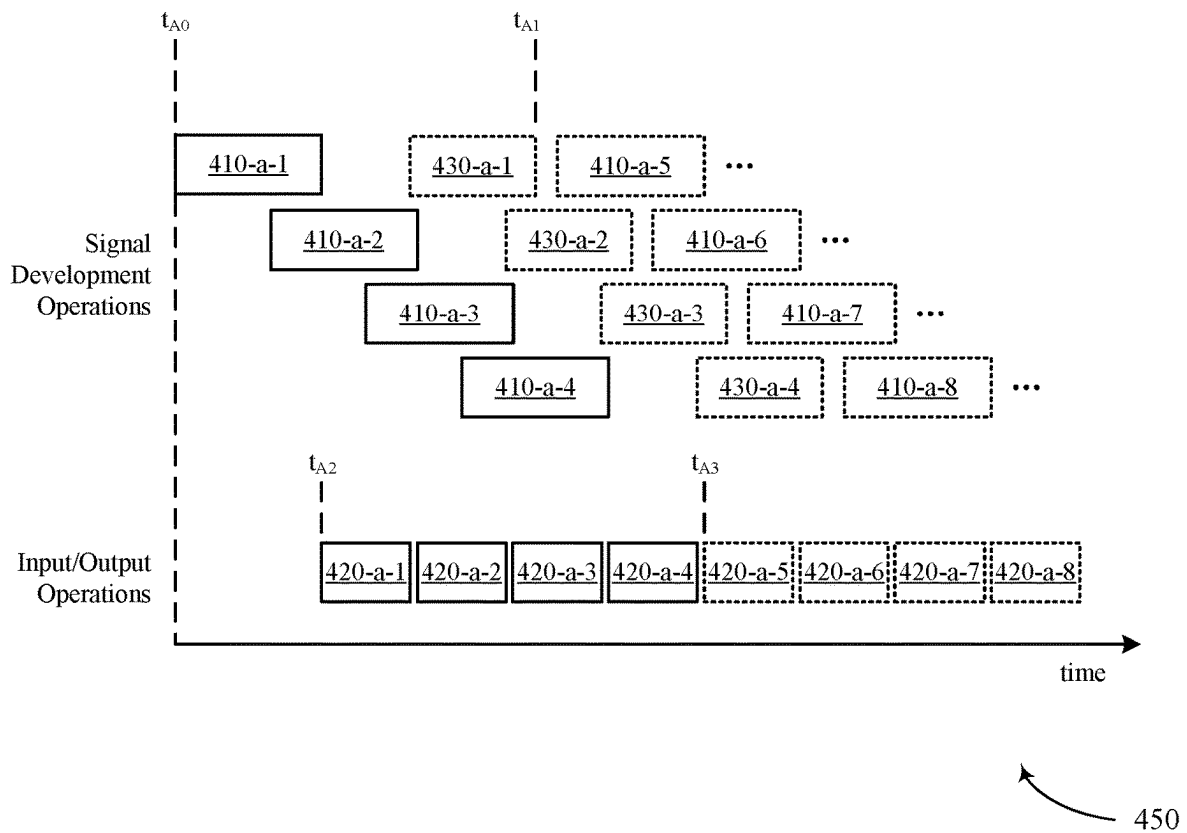


FIG. 4B

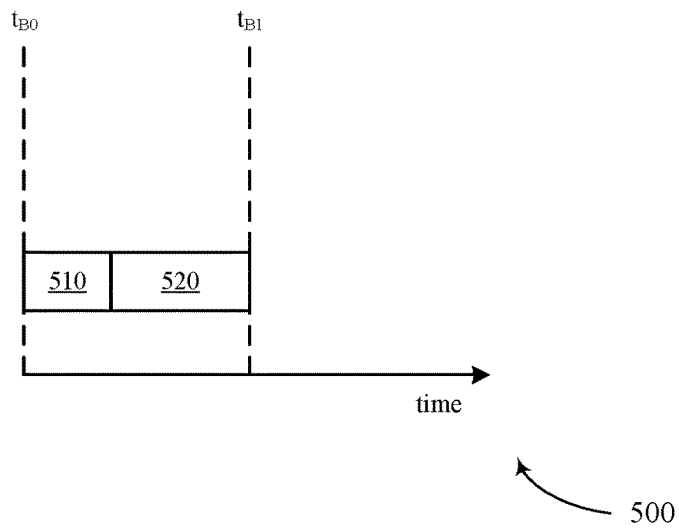


FIG. 5A

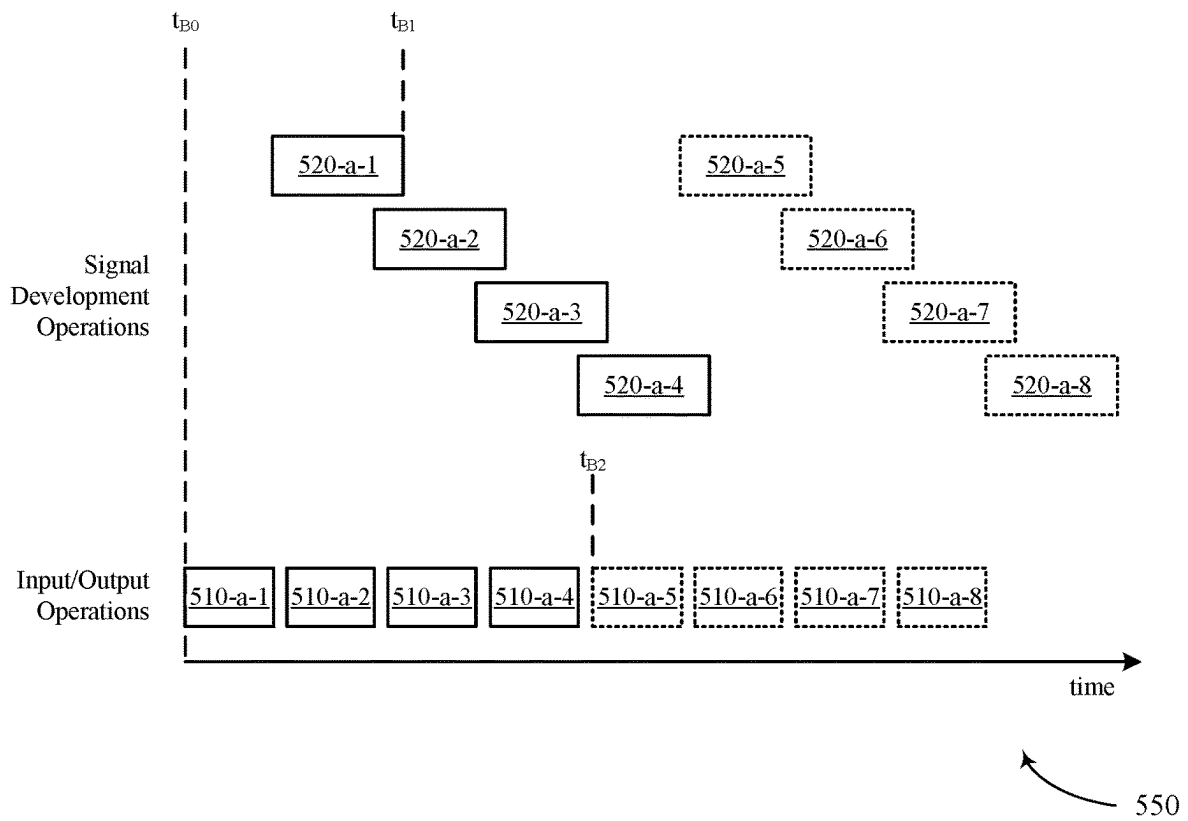


FIG. 5B

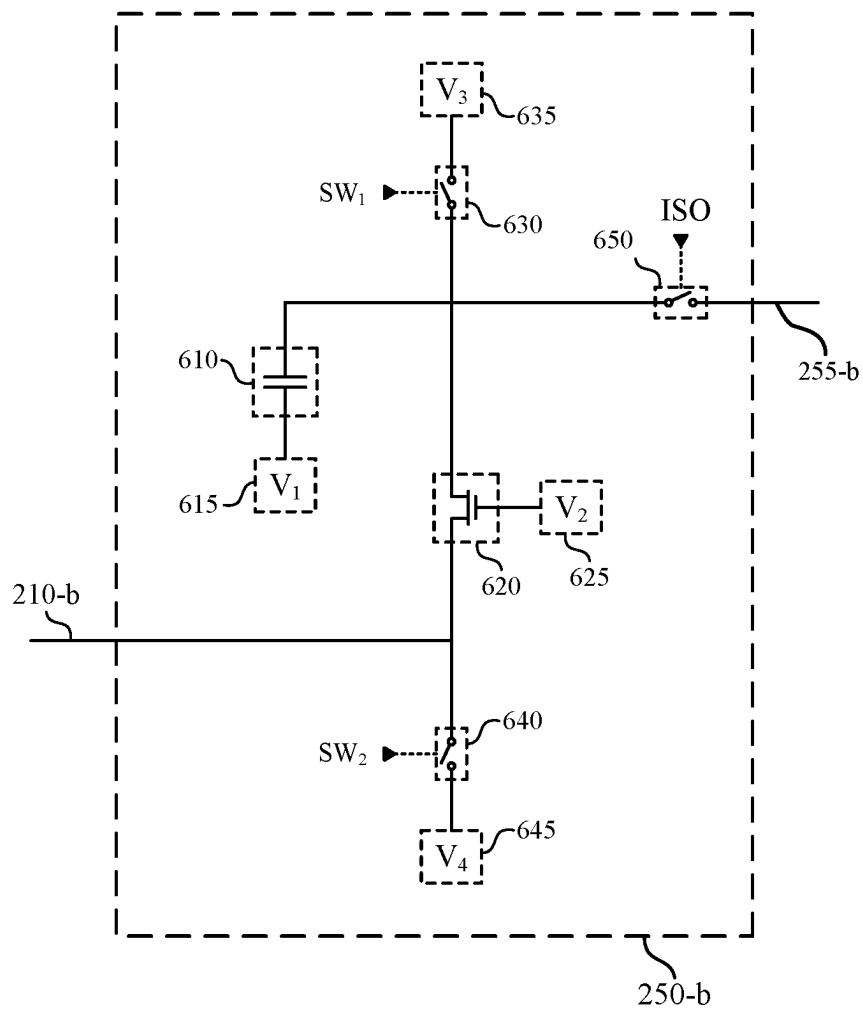


FIG. 6

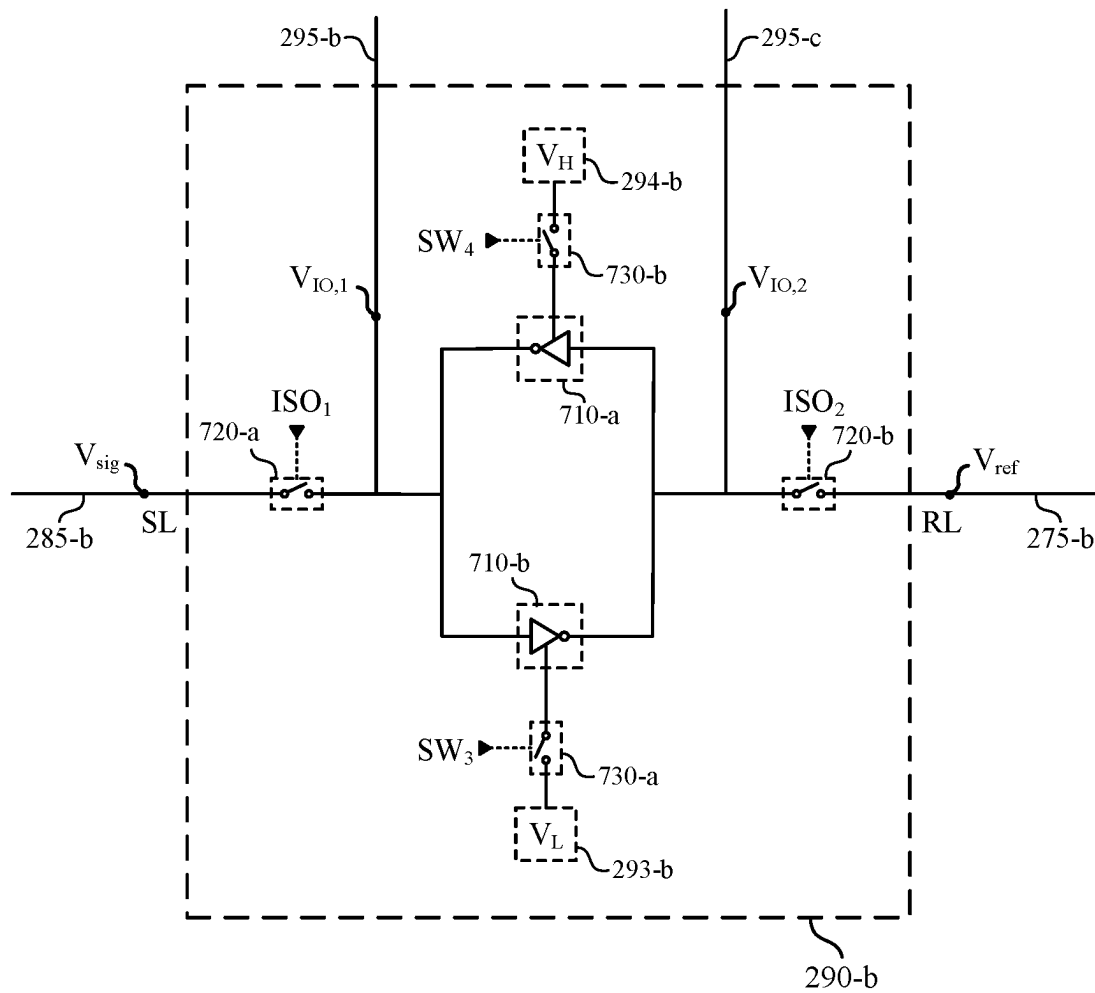


FIG. 7

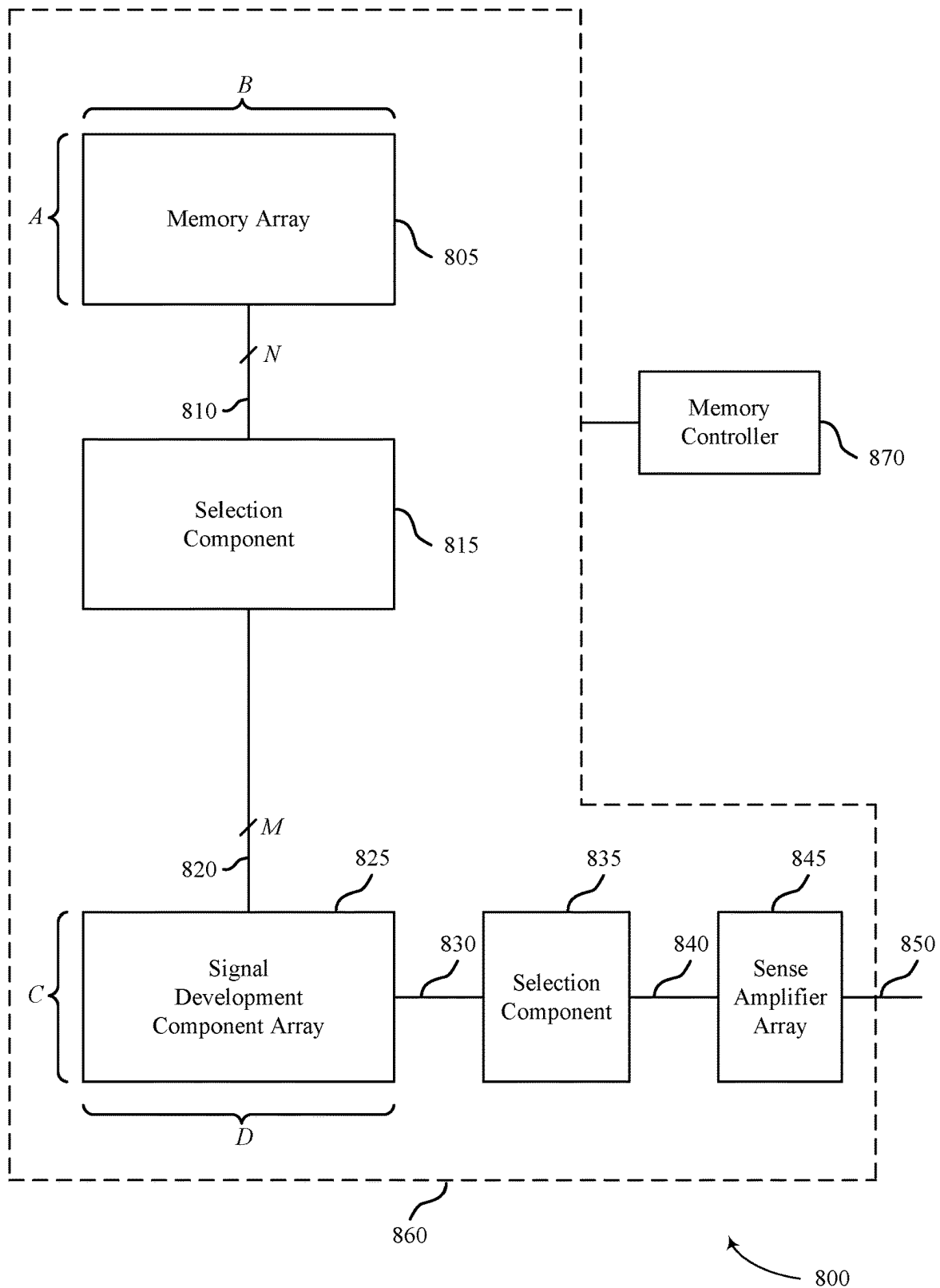


FIG. 8

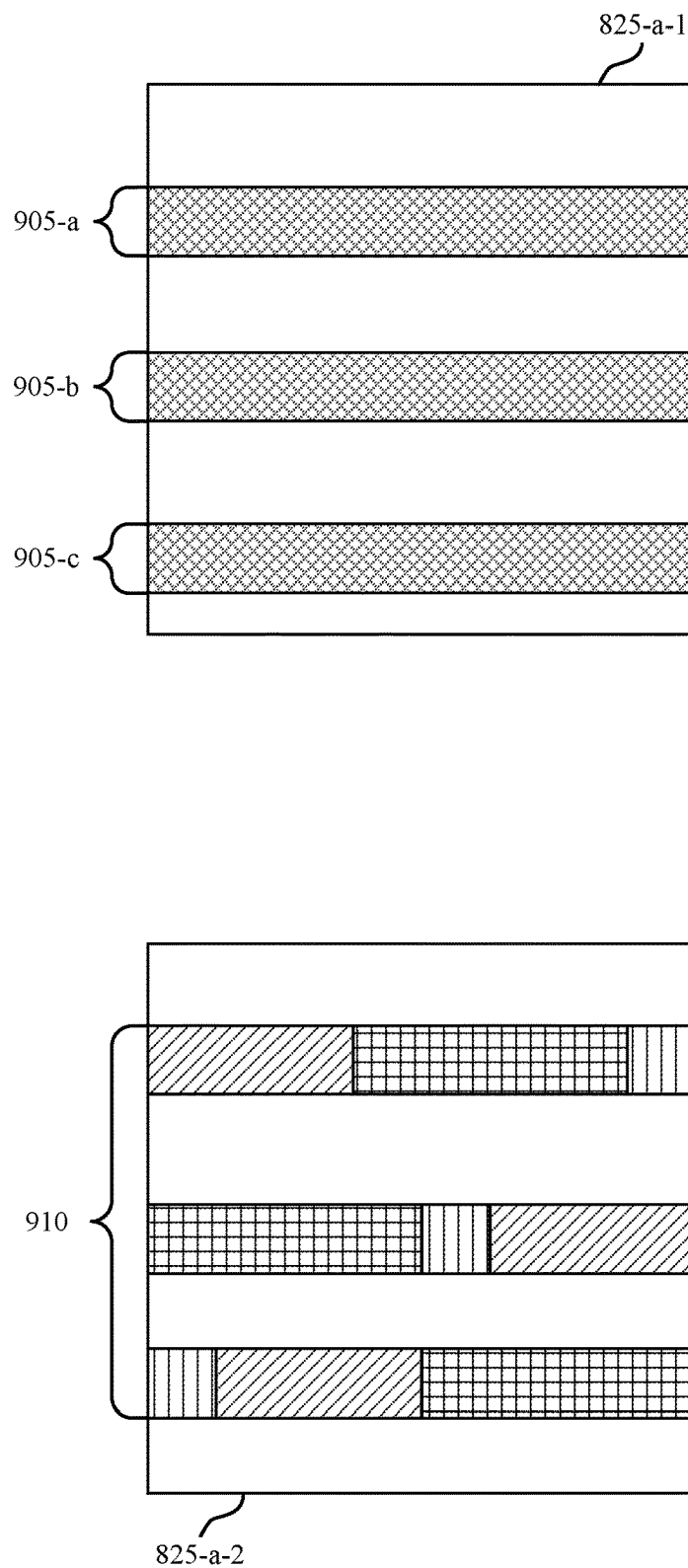


FIG. 9

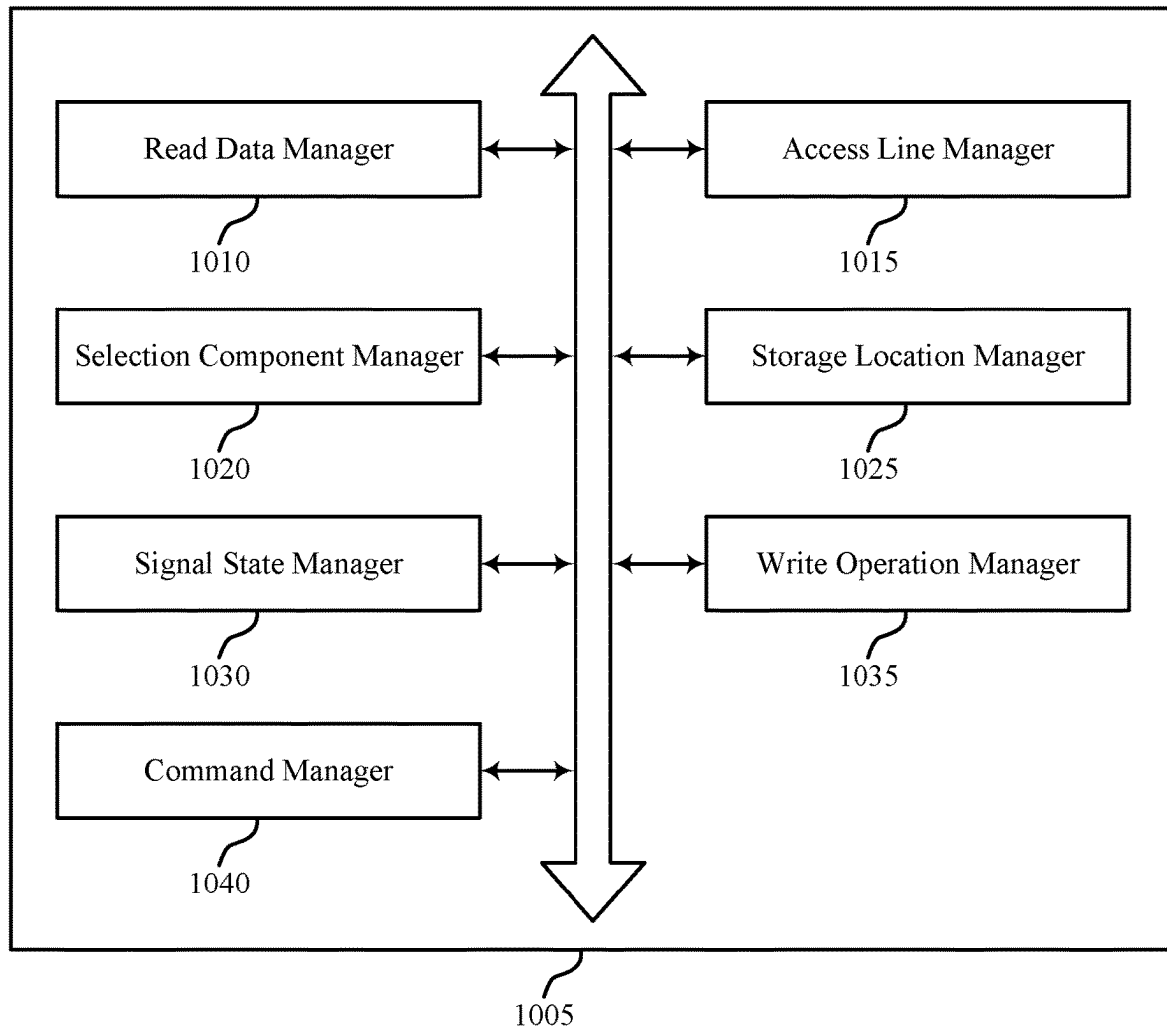


FIG. 10

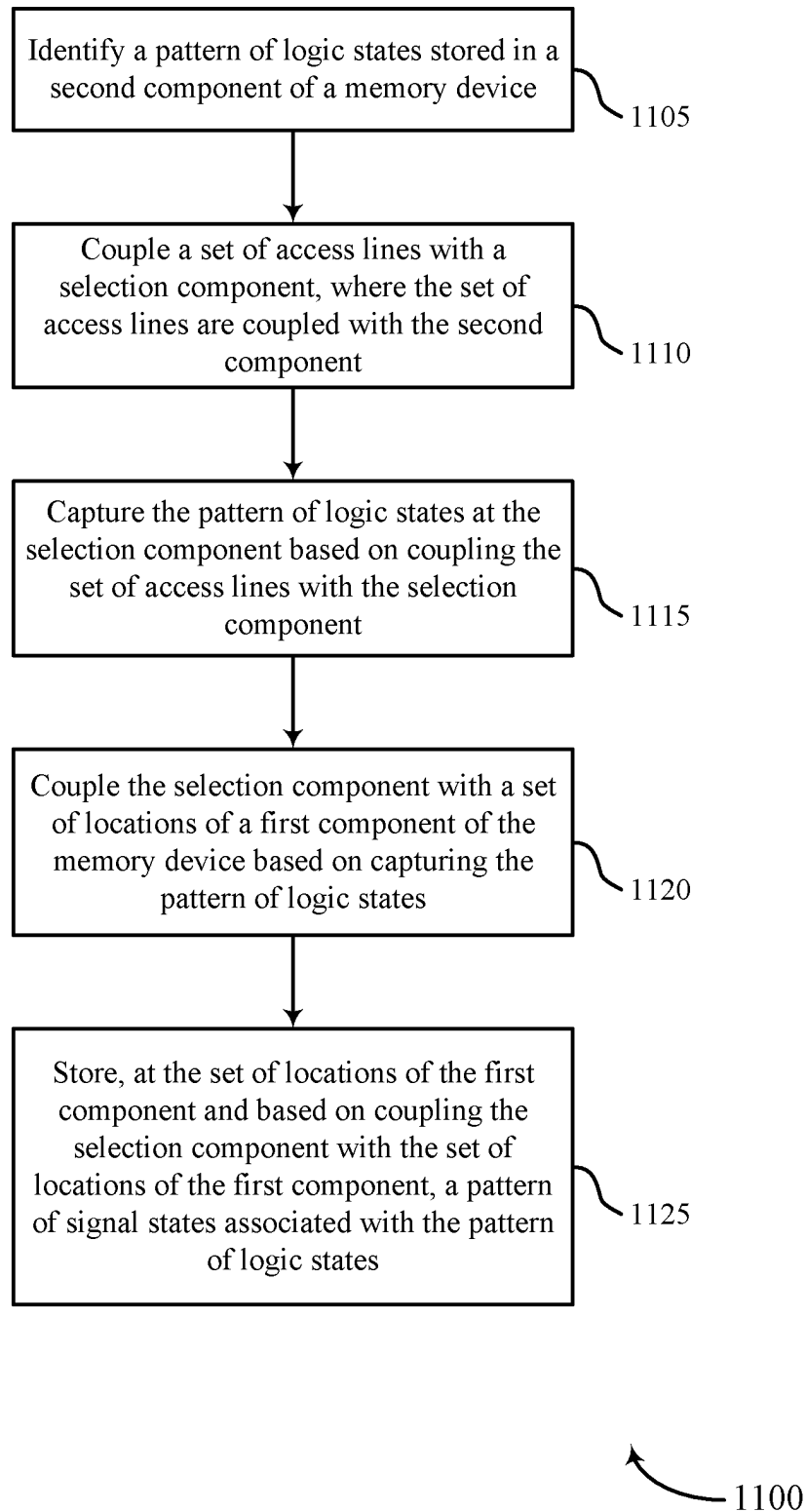
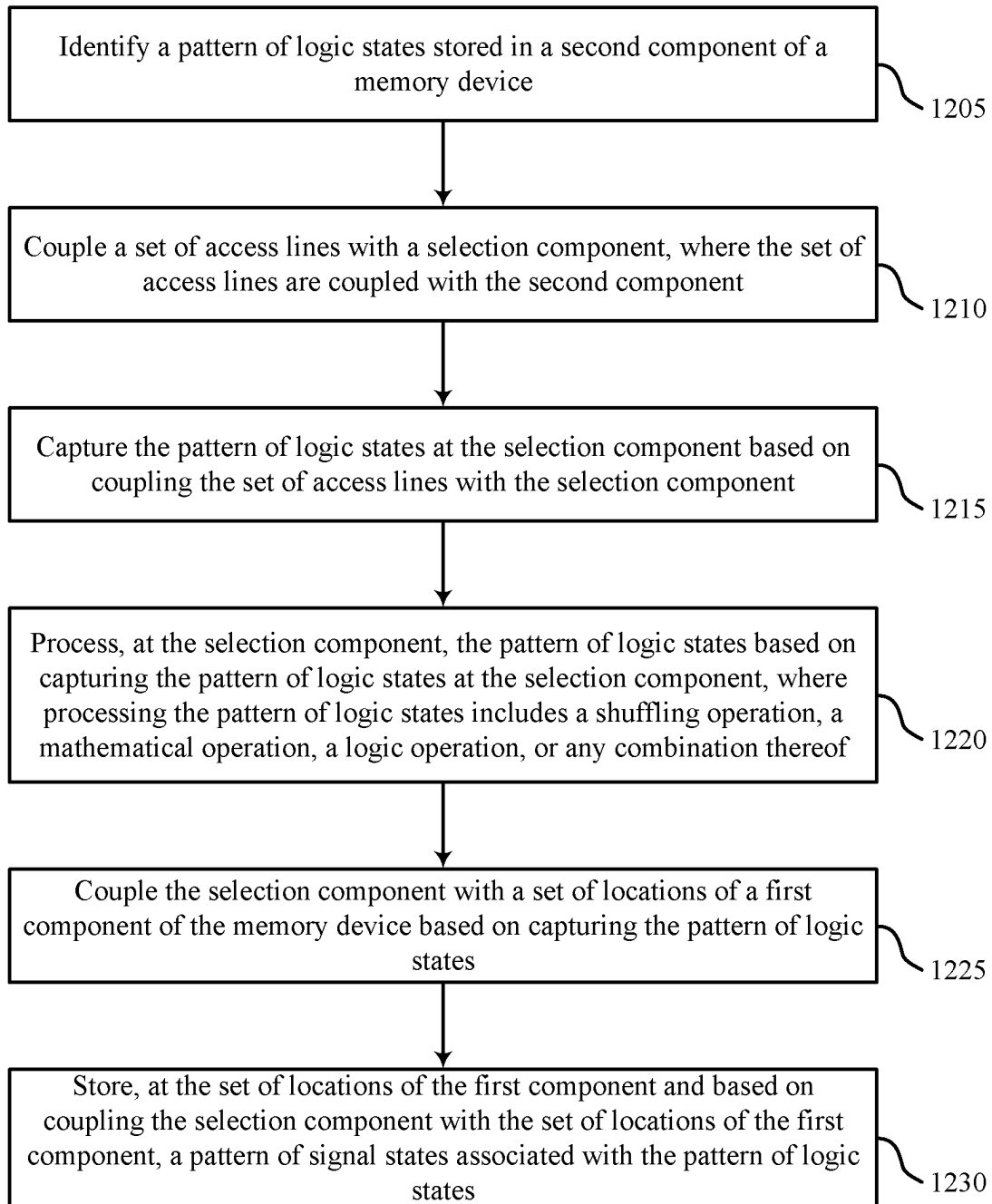


FIG. 11



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FIG. 12

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READ BROADCAST OPERATIONS ASSOCIATED WITH A MEMORY DEVICE

CROSS REFERENCE

The present Application for Patent is a continuation of U.S. patent application Ser. No. 17/846,715 by Yudanov et al., entitled “READ BROADCAST OPERATIONS ASSOCIATED WITH A MEMORY DEVICE,” filed Jun. 22, 2022, which is a continuation of U.S. patent application Ser. No. 17/414,297 by Yudanov et al., entitled “READ BROADCAST OPERATIONS ASSOCIATED WITH A MEMORY DEVICE,” filed Jun. 15, 2021, which is a 371 national phase filing of International Patent Application No. PCT/US19/67840 by Yudanov et al., entitled “READ BROADCAST OPERATIONS ASSOCIATED WITH A MEMORY DEVICE,” filed Dec. 20, 2019, and claims the benefit of U.S. Provisional Patent Application No. 62/783,388 by Yudanov et al., entitled “MULTIPLEXED SIGNAL DEVELOPMENT IN A MEMORY DEVICE” filed Dec. 21, 2018, each of which is assigned to the assignee hereof, and each of which is expressly incorporated by reference herein.

BACKGROUND

The following relates generally to memory systems and more specifically to read broadcast operations associated with a memory device.

Memory devices are widely used to store information in various electronic devices such as computers, wireless communication devices, cameras, digital displays, and the like. Information is stored by programing different states of a memory device. For example, binary memory devices have two logic states, often denoted by a logic “1” or a logic “0”. In other memory devices, more than two logic states may be stored. To access the stored information, a component of the electronic device may read, or sense, the stored logic state in the memory device. To store information, a component of the electronic device may write, or program, the logic state in the memory device.

Various types of memory devices and memory cells exist, including magnetic hard disks, random access memory (RAM), read-only memory (ROM), dynamic RAM (DRAM), synchronous dynamic RAM (SDRAM), ferroelectric RAM (FeRAM), magnetic RAM (MRAM), resistive RAM (RRAM), flash memory, phase change memory (PCM), self-selecting memory, chalcogenide memory technologies, and others. Memory cells may be volatile or non-volatile.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. 1 illustrates an example memory device that supports read broadcast operations associated with a memory device in accordance with examples as disclosed herein.

FIG. 2 illustrates an example circuit that supports read broadcast operations associated with a memory device in accordance with examples as disclosed herein.

FIG. 3 illustrates an example circuit that supports read broadcast operations associated with a memory device in accordance with examples as disclosed herein.

FIGS. 4A and 4B illustrate examples of read operations that support read broadcast operations associated with a memory device in accordance with examples as disclosed herein.

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FIGS. 5A and 5B illustrate examples of write operations that support read broadcast operations associated with a memory device in accordance with examples as disclosed herein.

FIG. 6 illustrates an example of a signal development component that supports read broadcast operations associated with a memory device in accordance with examples as disclosed herein.

FIG. 7 illustrates an example of a sense amplifier that supports read broadcast operations associated with a memory device in accordance with examples as disclosed herein.

FIG. 8 shows a block diagram of a system that supports read broadcast operations associated with a memory device in accordance with examples as disclosed herein.

FIG. 9 illustrates examples of signal development component arrays that support read broadcast operations associated with a memory device in accordance with examples as disclosed herein.

FIG. 10 shows a block diagram of a memory device that supports read broadcast operations associated with a memory device in accordance with examples as disclosed herein.

FIGS. 11 and 12 show flowcharts illustrating a method or methods that support read broadcast operations associated with a memory device in accordance with examples as disclosed herein.

DETAILED DESCRIPTION

Different latencies associated with different components used in a memory access operation, or different latencies otherwise associated with portions of a memory access operation, may cause delays in performing the memory access operation. For example, when a latency associated with developing a signal based on accessing a memory cell (e.g., an operation that includes coupling a memory cell with a signal development component) is longer in duration than a latency associated with generating an output signal at a sense amplifier (e.g., a sensing or latching operation at the sense amplifier), a memory device may be able to generate output signals more quickly than it can perform underlying signal development operations upon which the output signals are based. For a memory device that has a single signal development component for each sense amplifier (e.g., a 1:1 mapping of signal development components and sense amplifiers), the throughput of the memory device may therefore be limited by the latency or cycle duration associated with the signal development component or signal development operations, which may affect latency-sensitive applications.

In accordance with examples as disclosed herein, a memory device may include a signal development cache having a set of cache elements (e.g., signal storage elements) that may be selectively coupled with or decoupled from sense amplifiers of the memory device. For example, an array of sense amplifiers may be coupled with a selection component (e.g., a multiplexer (MUX), a transistor network, a transistor array, a switching network, a switching array), and the selection component may be coupled with a set of signal development cache elements that may each be associated with one or more memory cells of the memory device. In some examples, cell access signals (e.g., cell read signals, cell write signals) may be developed (e.g., based at least in part on a coupling with or other accessing of a respective

memory cell) at each of the signal development cache elements independently from others of the signal development cache elements.

In some examples (e.g., in a read operation), signal development cache elements may each be coupled with a respective memory cell or access line during overlapping time intervals, such that multiple cell access signals (e.g., multiple cell read signals associated with the respective memory cell or access line of each of the respective signal development components) may be generated during the overlapping time intervals. A signal development cache element may subsequently be coupled with the sense amplifier via the selection component to generate a sense or latch signal (e.g., an output signal of the sense amplifier, based on a respective cell access signal), which may be associated with a particular logic state that was stored by a respective memory cell (e.g., associated with the respective cell access signal). In examples where cell access signals have been developed at multiple signal development cache elements, the multiple signal development cache elements may be coupled with the sense amplifier in a sequential manner to generate sense or latch signals in a sequential manner.

In some examples, one or more components in a system (e.g., one or more signal development components) may enable read broadcast operations. A read broadcast may occur from one or more locations in a memory array (e.g., from one or more domains) to multiple locations of a set of signal development components (e.g., in one or more cache blocks) to a desired pattern of data, the set of locations may be concurrently coupled with one or more multiplexers to store the pattern of data from the memory array. As used herein, a “set” may include one or more elements (e.g., one element, two elements, three elements, and so on).

Features of the disclosure introduced above are further described with reference to FIGS. 1 through 3 in the context of memory arrays and memory circuits that support read broadcast operations associated with a memory device. Specific examples are then described with reference to FIGS. 4A through 5B, which illustrate particular read operations and write operations that support read broadcast operations associated with a memory device. Further examples of circuits, components, and arrangements that may support the described operations are described with reference to FIGS. 6 through 8. These and other features of the disclosure are further described with respect to FIGS. 9 through 11, which illustrate apparatus diagrams, system diagrams, and flowcharts that support read broadcast operations associated with a memory device.

FIG. 1 illustrates an example memory device 100 that supports read broadcast operations associated with a memory device in accordance with examples as disclosed herein. The memory device 100 may also be referred to as an electronic memory apparatus. The memory device 100 may include memory cells 105 that are programmable to store different states such as memory states, which may be referred to herein as logic states. In some cases, a memory cell 105 may be programmable to store two logic states, denoted a logic 0 and a logic 1. In some cases, a memory cell 105 may be programmable to store more than two logic states. Additionally or alternatively, a memory cell 105 may be programmable to store a memory state based on an analog

or stochastic operation (e.g., related to a neural network), where the memory state correspond to information other than a logic 0 or a logic 1. In some examples, the memory cells 105 may include a capacitive memory element, a ferroelectric memory element, a material memory element, a resistive element, a self-selecting memory element, a thresholding memory element, or any combination thereof.

The set of memory cells 105 may be part of a memory section 110 of the memory device 100 (e.g., including an array of memory cells 105), where in some examples a memory section 110 may refer to a contiguous tile of memory cells 105 (e.g., a contiguous set of elements of a semiconductor chip). In some examples, a memory section 110 may refer to the smallest set of memory cells 105 that may be biased in an access operation, or a smallest set of memory cells 105 that share a common node (e.g., a common plate line, a set of plate lines that are biased to a common voltage). Although a single memory section 110 of the memory device 100 is shown, various examples of a memory device in accordance with examples as disclosed herein may have a set of memory sections 110. In one illustrative example, a memory device 100, or a subsection thereof (e.g., a core of a multi-core memory device 100, a chip of a multi-chip memory device) may include 32 “banks” and each bank may include 32 sections. Thus, a memory device 100, or subsection thereof, according to the illustrative example may include 1,024 memory sections 110.

In some examples, a memory cell 105 may store an electric charge representative of the programmable logic states (e.g., storing charge in a capacitor, capacitive memory element, capacitive storage element). In one example, a charged and uncharged capacitor may represent two logic states, respectively. In another example, a positively charged and negatively charged capacitor may represent two logic states, respectively. DRAM or FeRAM architectures may use such designs, and the capacitor employed may include a dielectric material with linear or para-electric polarization properties as an insulator. In some examples, different levels of charge of a capacitor may represent different logic states (e.g., supporting more than two logic states in a respective memory cell 105). In some examples, such as FeRAM architectures, a memory cell 105 may include a ferroelectric capacitor having a ferroelectric material as an insulating (e.g., non-conductive) layer between terminals of the capacitor. Different levels of polarization of a ferroelectric capacitor may represent different logic states (e.g., supporting two or more logic states in a respective memory cell 105). In some examples, ferroelectric materials have non-linear polarization properties.

In some examples, a memory cell 105 may include a material portion, which may be referred to as a memory element, a memory storage element, a self-selecting memory element, or a self-selecting memory storage element. The material portion may have a variable and configurable electrical resistance or other characteristic that is representative of different logic states. For example, a material that can take the form of a crystalline atomic configuration or an amorphous atomic configuration (e.g., able to maintain either a crystalline state or an amorphous state over an ambient operating temperature range of the memory device 100) may have different electrical resistances depending on the atomic configuration. A more-crystalline state of the material (e.g., a single crystal, a collection of relatively large crystal grains that may be substantially crystalline) may have a relatively low electrical resistance, and may alternatively be referred to as a “SET” logic state. A more-

amorphous state of the material (e.g., an entirely amorphous state, some distribution of relatively small crystal grains that may be substantially amorphous) may have a relatively high electrical resistance, and may alternatively be referred to as a “RESET” logic state. Thus, a voltage applied to such a memory cell **105** may result in different current flow depending on whether the material portion of the memory cell **105** is in the more-crystalline or the more-amorphous state. Accordingly, the magnitude of the current resulting from applying a read voltage to the memory cell **105** may be used to determine a logic state stored by memory cell **105**.

In some examples, a memory element may be configured with various ratios of crystalline and amorphous areas (e.g., varying degrees of atomic order and disorder) that may result in intermediate resistances, which may represent different logic states (e.g., supporting two or more logic states in a respective memory cell **105**). Further, in some examples, a material or a memory element may have more than two atomic configurations, such as an amorphous configuration and two different crystalline configurations. Although described herein with reference to an electrical resistance of different atomic configurations, a memory device may use some other characteristic of a memory element to determine a stored logic state corresponding to an atomic configuration, or combination of atomic configurations.

In some cases, a memory element in a more-amorphous state may be associated with a threshold voltage. In some examples, electrical current may flow through a memory element in the more-amorphous state when a voltage greater than the threshold voltage is applied across the memory element. In some examples, electrical current may not flow through a memory element in the more-amorphous state when a voltage less than the threshold voltage is applied across the memory element. In some cases, a memory element in a more-crystalline state may not be associated with a threshold voltage (e.g., may be associated with a threshold voltage of zero). In some examples, electrical current may flow through a memory element in the more-crystalline state in response to a non-zero voltage across the memory element.

In some cases, a material in both the more-amorphous state and the more-crystalline state may be associated with threshold voltages. For example, self-selecting or thresholding memory may be based on differences in a threshold voltage of a memory cell between different programmed states (e.g., by way of different compositional distributions). The logic state of a memory cell **105** having such a memory element may be set by biasing or heating the memory element to a temperature profile over time that supports forming a particular atomic configuration, or combination of atomic configurations.

A memory device **100** may include a three-dimensional (3D) memory array, where a plurality of two-dimensional (2D) memory arrays (e.g., decks, levels) are formed on top of one another. In various examples, such arrays may be divided into a set of memory sections **110**, where each memory section **110** may be arranged within a deck or level, distributed across multiple decks or levels, or any combination thereof. Such arrangements may increase the number of memory cells **105** that may be placed or created on a single die or substrate as compared with 2D arrays, which in turn may reduce production costs or increase the performance of a memory device **100**, or both. The decks or levels may be separated by an electrically insulating material. Each deck or level may be aligned or positioned so that memory cells **105** may be approximately aligned with one another across each deck, forming a stack of memory cells **105**.

In the example of memory device **100**, each row of memory cells **105** of the memory section **110** may be coupled with one of a set of first access lines **120** (e.g., a word line (WL), such as one of WL₁ through WL_M), and each column of memory cells **105** may be coupled with one of a set of second access lines **130** (e.g., a digit line (DL), such as one of DL₁ through DL_N). In some examples, a row of memory cells **105** of a different memory section **110** (not shown) may be coupled with one of a different plurality of first access lines **120** (e.g., a word line different from WL₁ through WL_M), and a column of memory cells **105** of the different memory section **110** may be coupled with one of a different plurality of second access lines **130** (e.g., a digit line different from DL₁ through DL_N). In some cases, first access lines **120** and second access lines **130** may be substantially perpendicular to one another in the memory device **100** (e.g., when viewing a plane of a deck of the memory device **100**, as shown in FIG. 1). References to word lines and bit lines, or their analogues, are interchangeable without loss of understanding or operation.

In general, one memory cell **105** may be located at the intersection of (e.g., coupled with, coupled between) an access line **120** and an access line **130**. This intersection, or an indication of this intersection, may be referred to as an address of a memory cell **105**. A target or selected memory cell **105** may be a memory cell **105** located at the intersection of an energized or otherwise selected access line **120** and an energized or otherwise selected access line **130**. In other words, an access line **120** and an access line **130** may be energized or otherwise selected to access (e.g., read, write, rewrite, refresh) a memory cell **105** at their intersection. Other memory cells **105** that are in electronic communication with (e.g., connected to) the same access line **120** or **130** may be referred to as untargeted or non-selected memory cells **105**.

In some architectures, the logic storing component (e.g., a capacitive memory element, a ferroelectric memory element, a resistive memory element, other memory element) of a memory cell **105** may be electrically isolated from a second access line **130** by a cell selection component, which, in some examples, may be referred to as a switching component or a selector device. A first access line **120** may be coupled with the cell selection component (e.g., via a control node or terminal of the cell selection component), and may control the cell selection component of or associated with the memory cell **105**. For example, the cell selection component may be a transistor and the first access line **120** may be coupled with a gate of the transistor (e.g., where a gate node of the transistor may be a control node of the transistor). Activating the first access line **120** of a memory cell **105** may result in an electrical connection or closed circuit between the logic storing component of the memory cell **105** and its corresponding second access line **130**. The second access line **130** may then be accessed to read or write the memory cell **105**.

In some examples, memory cells **105** of the memory section **110** may also be coupled with one of a plurality of third access lines **140** (e.g., a plate line (PL), such as one of PL₁ through PL_N). Although illustrated as separate lines, in some examples, the plurality of third access lines **140** may represent or be otherwise functionally equivalent with a common plate line, a common plate, or other common node of the memory section **110** (e.g., a node common to each of the memory cells **105** in the memory section **110**), or other common node of the memory device **100**. In some examples, the plurality of third access lines **140** may couple memory cells **105** with one or more voltage sources for various

sensing and/or writing operations including those described herein. For example, when a memory cell **105** employs a capacitor for storing a logic state, a second access line **130** may provide access to a first terminal or a first plate of the capacitor, and a third access line **140** may provide access to a second terminal or a second plate of the capacitor (e.g., a terminal associated with an opposite plate of the capacitor as opposed to the first terminal of the capacitor, a terminal otherwise on the opposite side of a capacitance from the first terminal of the capacitor). In some examples, memory cells **105** of a different memory section **110** (not shown) may be coupled with one of a different plurality of third access lines **140** (e.g., a set of plate lines different from PL_1 through PL_N , a different common plate line, a different common plate, a different common node), which may be electrically isolated from the illustrated third access line **140** (e.g., plate lines PL_1 through PL_N).

The plurality of third access lines **140** may be coupled with a plate component **145**, which may control various operations such as activating one or more of the plurality of third access lines **140**, or selectively coupling one or more of the plurality of third access lines **140** with a voltage source or other circuit element. Although the plurality of third access lines **140** of the memory device **100** are shown as substantially parallel with the plurality of second access lines **130**, in other examples, a plurality of third access lines **140** may be substantially parallel with the plurality of first access lines **120**, or in any other configuration.

Although the access lines described with reference to FIG. **1** are shown as direct lines between memory cells **105** and coupled components, access lines may be associated with other circuit elements, such as capacitors, resistors, transistors, amplifiers, voltage sources, switching components, selection components, and others, which may be used to support access operations including those described herein. In some examples, an electrode may be coupled with (e.g., between) a memory cell **105** and an access line **120**, or with (e.g., between) a memory cell **105** and an access line **130**. The term electrode may refer to an electrical conductor, or other electrical interface between components, and in some cases, may be employed as an electrical contact to a memory cell **105**. An electrode may include a trace, wire, conductive line, conductive layer, conductive pad, or the like, that provides a conductive path between elements or components of memory device **100**.

Access operations such as reading, writing, rewriting, and refreshing may be performed on a memory cell **105** by activating or selecting a first access line **120**, a second access line **130**, and/or a third access line **140** coupled with the memory cell **105**, which may include applying a voltage, a charge, or a current to the respective access line. Access lines **120**, **130**, and **140** may be made of conductive materials, such as metals (e.g., copper (Cu), silver (Ag), aluminum (Al), gold (Au), tungsten (W), titanium (Ti)), metal alloys, carbon, or other conductive or semi-conductive materials, alloys, or compounds. Upon selecting a memory cell **105**, a resulting signal (e.g., a cell access signal, a cell read signal) may be used to determine the logic state stored by the memory cell **105**. For example, a memory cell **105** with a capacitive memory element storing a logic state may be selected, and the resulting flow of charge via an access line and/or resulting voltage of an access line may be detected, converted, or amplified to determine the programmed logic state stored by the memory cell **105**.

Accessing memory cells **105** may be controlled through a row component **125** (e.g., a row decoder), a column component **135** (e.g., a column decoder), or a plate component

145 (e.g., a plate driver), or a combination thereof. For example, a row component **125** may receive a row address from the memory controller **170** and select or activate the appropriate first access line **120** based on the received row address. Similarly, a column component **135** may receive a column address from the memory controller **170** and select or activate the appropriate second access line **130**. Thus, in some examples, a memory cell **105** may be accessed by selecting or activating a first access line **120** and a second access line **130**. In some examples, such access operations may be accompanied by a plate component **145** biasing one or more of the third access lines **140** (e.g., biasing one of the third access lines **140** of the memory section **110**, biasing all of the third access lines **140** of the memory section, biasing a common plate line of the memory section **110** or the memory device **100**, biasing a common node of the memory section **110** or the memory device **100**), which may be referred to as “moving the plate” of memory cells **105**, the memory section **110**, or the memory device **100**. In various examples, any one or more of the row component **125**, the column component **135**, or the plate component **145** may be referred to as, or otherwise include access line drivers or access line decoders.

In some examples, the memory controller **170** may control the operation (e.g., read operations, write operations, rewrite operations, refresh operations, discharge operations, dissipation operations, equalization operations) of memory cells **105** through the various components (e.g., row component **125**, column component **135**, plate component **145**, sense component **150**). In some cases, one or more of the row component **125**, the column component **135**, the plate component **145**, and the sense component **150** may be co-located or otherwise included with the memory controller **170**. In some examples, any one or more of a row component **125**, a column component **135**, or a plate component **145** may also be referred to as a memory controller or circuit for performing access operations of the memory device **100**. In some examples, any one or more of a row component **125**, a column component **135**, or a plate component **145** may be described as controlling or performing operations for accessing a memory device **100**, or controlling or performing operations for accessing the memory section **110** of the memory device **100**.

The memory controller **170** may generate row and column address signals to activate a desired access line **120** and access line **130**. The memory controller **170** may also generate or control various voltages or currents used during the operation of memory device **100**. Although a single memory controller **170** is shown, a memory device **100** may have more than one memory controller **170** (e.g., a memory controller **170** for each of a set of memory sections **110** of a memory device **100**, a memory controller **170** for each of a number of subsets of memory sections **110** of a memory device **100**, a memory controller **170** for each of a set of chips of a multi-chip memory device **100**, a memory controller **170** for each of a set of banks of a multi-bank memory device **100**, a memory controller **170** for each core of a multi-core memory device **100**, or any combination thereof), where different memory controllers **170** may perform the same functions and/or different functions.

Although the memory device **100** is illustrated as including a single row component **125**, a single column component **135**, and a single plate component **145**, other examples of a memory device **100** may include different configurations to accommodate a memory section **110** or a set of memory sections **110**. For example, in various memory devices **100** a row component **125** may be shared among a set of memory

sections 110 (e.g., having subcomponents common to all of the set of memory sections 110, having subcomponents dedicated to respective ones of the set of memory sections 110), or a row component 125 may be dedicated to one memory section 110 of a set of memory sections 110. Likewise, in various memory devices 100, a column component 135 may be shared among a set of memory sections 110 (e.g., having subcomponents common to all of the set of memory sections 110, having subcomponents dedicated to respective ones of the set of memory sections 110), or a column component 135 may be dedicated to one memory section 110 of a set of memory sections 110. Additionally, in various memory devices 100, a plate component 145 may be shared among a set of memory sections 110 (e.g., having subcomponents common to all of the set of memory sections 110, having subcomponents dedicated to respective ones of the set of memory sections 110), or a plate component 145 may be dedicated to one memory section 110 of a set of memory sections 110.

In general, the amplitude, shape, or duration of an applied voltage, current, or charge may be adjusted or varied, and may be different for the various operations discussed in operating the memory device 100. Further, one, multiple, or all memory cells 105 within memory device 100 may be accessed simultaneously. For example, multiple or all memory cells 105 of memory device 100 may be accessed simultaneously during a reset operation in which all memory cells 105, or a group of memory cells 105 (e.g., the memory cells 105 of a memory section 110), are set to a single logic state.

A memory cell 105 may be read (e.g., sensed) by a sense component 150 when the memory cell 105 is accessed (e.g., in cooperation with the memory controller 170) to determine a logic state stored by the memory cell 105. For example, the sense component 150 may be configured to sense a current or charge through the memory cell 105, or a voltage resulting from coupling the memory cell 105 with the sense component 150 or other intervening component (e.g., a signal development component between the memory cell 105 and the sense component 150), responsive to a read operation. The sense component 150 may provide an output signal indicative of (e.g., based at least in part on) the logic state stored by the memory cell 105 to one or more components (e.g., to the column component 135, the input/output component 160, the memory controller 170). In various memory devices 100, a sense component 150 may be shared among a set or bank of memory sections 110 (e.g., having subcomponents common to all of the set or bank of memory sections 110, having subcomponents dedicated to respective ones of the set or bank of memory sections 110), or a sense component 150 may be dedicated to one memory section 110 of a set or bank of memory sections 110.

In some examples, during or after accessing a memory cell 105, the logic storage portion of memory cell 105 may discharge, or otherwise permit electrical charge or current to flow via its corresponding access lines 120, 130, or 140. Such charge or current may result from biasing, or applying a voltage, to the memory cell 105 from one or more voltage sources or supplies (not shown) of the memory device 100, where such voltage sources or supplies may be part of a row component 125, a column component 135, a plate component 145, a sense component 150, a memory controller 170, or some other component (e.g., a biasing component). In some examples, a discharge of a memory cell 105 may cause a change in the voltage of the access line 130, which the sense component 150 may compare to a reference voltage to determine the stored state of the memory cell 105. In some

examples, a voltage may be applied to a memory cell 105 (e.g., using the corresponding access line 120 and access line 130) and the presence or magnitude of a resulting current may depend on the applied voltage and the resistance state of a memory element of the memory cell 105, which the sense component 150 may use to determine the stored state of the memory cell 105.

In some examples, when a read signal (e.g., a read pulse, a read current, a read voltage) is applied across a memory cell 105 with a material memory element storing a first logic state (e.g., a SET state, associated with a more-crystalline atomic configuration), the memory cell 105 conducts current due to the read pulse exceeding a threshold voltage of the memory cell 105. In response to, or based at least in part on this, the sense component 150 may therefore detect a current through the memory cell 105 as part of determining the logic state stored by the memory cell 105. When a read pulse is applied to the memory cell 105 with the memory element storing a second logic state (e.g., a RESET state, associated with a more-amorphous atomic configuration), which may occur before or after the application of a read pulse across a memory cell 105 with a memory element storing a first logic state, the memory cell 105 may not conduct current due to the read pulse not exceeding the threshold voltage of the memory cell 105. The sense component 150 may therefore detect little or no current through the memory cell 105 as part of determining the stored logic state.

In some examples, a threshold current may be defined for sensing the logic state stored by a memory cell 105. The threshold current may be set above a current that may pass through the memory cell 105 when the memory cell 105 does not threshold in response to the read pulse, but equal to or below an expected current through the memory cell 105 when the memory cell 105 does threshold in response to the read pulse. For example, the threshold current may be higher than a leakage current of the associated access lines 120, 130, or 140. In some examples, a logic state stored by a memory cell 105 may be determined based at least in part on a voltage (e.g., across a shunt resistance) resulting from the current driven by a read pulse. For example, the resulting voltage may be compared relative to a reference voltage, with a resulting voltage less than the reference voltage corresponding to a first logic state and a resulting voltage greater than the reference voltage corresponding to a second logic state.

In some examples, more than one voltage may be applied when reading a memory cell 105 (e.g., multiple voltages may be applied during portions of a read operation). For example, if an applied read voltage does not result in current flow, one or more other read voltages may be applied (e.g., until a current is detected by sense component 150). Based at least in part on assessing the read voltage that resulted in current flow, the stored logic state of the memory cell 105 may be determined. In some cases, a read voltage may be ramped (e.g., smoothly increasing higher in magnitude) until a current flow or other condition is detected by a sense component 150. In other cases, predetermined read voltages may be applied (e.g., a predetermined sequence of read voltages that increase higher in magnitude in a stepwise manner) until a current is detected. Likewise, a read current may be applied to a memory cell 105 and the magnitude of the voltage to create the read current may depend on the electrical resistance or the total threshold voltage of the memory cell 105.

A sense component 150 may include various switching components, selection components, multiplexers, transistors, amplifiers, capacitors, resistors, voltage sources, or

other components to detect, convert, or amplify a difference in sensing signals (e.g., a difference between a read voltage and a reference voltage, a difference between a read current and a reference current, a difference between a read charge and a reference charge), which, in some examples, may be referred to as sensing or latching or generating a sense or latch signal. In some examples, a sense component 150 may include a collection of components (e.g., circuit elements, circuitry) that are repeated for each of a set of access lines 130 connected to the sense component 150. For example, a sense component 150 may include a separate sensing circuit or circuitry (e.g., a separate sense amplifier, a separate signal development component) for each of a set of access lines 130 coupled with the sense component 150, such that a logic state may be separately detected for a respective memory cell 105 coupled with a respective one of the set of access lines 130. In some examples, a reference signal source (e.g., a reference component) or generated reference signal may be shared between components of the memory device 100 (e.g., shared among one or more sense components 150, shared among separate sensing circuits of a sense component 150, shared among access lines 120, 130, or 140 of a memory section 110).

The sense component 150 may be included in a device that includes the memory device 100. For example, the sense component 150 may be included with other read and write circuitry, decoding circuitry, or register circuitry of the memory that may be coupled with or to the memory device 100. In some examples, the detected logic state of a memory cell 105 may be output through a column component 135 or an input/output component 160 as an output. In some examples, a sense component 150 may be part of a column component 135, a row component 125, or a memory controller 170. In some examples, a sense component 150 may be connected to or otherwise in electronic communication with a column component 135, a row component 125, or memory controller 170.

Although a single sense component 150 is shown, a memory device 100 (e.g., a memory section 110 of a memory device 100) may include more than one sense component 150. For example, a first sense component 150 may be coupled with a first subset of access lines 130 and a second sense component 150 may be coupled with a second subset of access lines 130 (e.g., different from the first subset of access lines 130). In some examples, such a division of sense components 150 may support parallel (e.g., simultaneous) operation of multiple sense components 150. In some examples, such a division of sense components 150 may support matching sense components 150 having different configurations or characteristics to particular subsets of the memory cells 105 of the memory device (e.g., supporting different types of memory cells 105, supporting different characteristics of subsets of memory cells 105, supporting different characteristics of subsets of access lines 130).

Additionally or alternatively, two or more sense components 150 may be coupled (e.g., selectively coupled) with a same set of access lines 130 (e.g., for component redundancy). In some examples, such a configuration may support maintaining functionality to overcome a failure or otherwise poor or degraded operation of one of the redundant sense components 150. In some examples, such a configuration may support the ability to select one of the redundant sense components 150 for particular operational characteristics (e.g., as related to power consumption characteristics, as related to access speed characteristics for a particular sensing operation, as related to operating memory cells 105 in a volatile mode or a non-volatile mode).

In some memory architectures, accessing a memory cell 105 may degrade or destroy a logic state stored by one or more memory cells 105 of the memory section 110, and rewrite or refresh operations may be performed to return the original logic state to the memory cells 105. In DRAM or FeRAM, for example, a capacitor of a memory cell 105 may be partially or completely discharged or depolarized during a sense operation, thereby corrupting the logic state that was stored in the memory cell 105. In PCM, for example, sense operations may cause a change in the atomic configuration of a memory cell 105, thereby changing the resistance state of the memory cell 105. Thus, in some examples, the logic state stored in a memory cell 105 may be rewritten after an access operation. Further, activating a single access line 120, 130, or 140 may result in the discharge of all memory cells 105 coupled with the activated access line 120, 130, or 140. Thus, several or all memory cells 105 coupled with an access line 120, 130, or 140 associated with an access operation (e.g., all cells of an accessed row, all cells of an accessed column) may be rewritten after the access operation.

In some examples, reading a memory cell 105 may be non-destructive. That is, the logic state of the memory cell 105 may not need to be rewritten after the memory cell 105 is read. For example, in non-volatile memory such as PCM, accessing the memory cell 105 may not destroy the logic state and, thus, the memory cell 105 may not require rewriting after accessing. However, in some examples, refreshing the logic state of the memory cell 105 may or may not be needed in the absence or presence of other access operations. For example, the logic state stored by a memory cell 105 may be refreshed at periodic intervals by applying an appropriate write, refresh, or equalization pulse or bias to maintain the stored logic state. Refreshing the memory cell 105 may reduce or eliminate read disturb errors or logic state corruption due to a charge leakage or a change in an atomic configuration of a memory element over time.

A memory cell 105 may be set or written or refreshed by activating the relevant first access line 120, second access line 130, and/or third access line 140 (e.g., via a memory controller 170). In other words, a logic state may be stored in the memory cell 105 (e.g., via a cell access signal, via a cell write signal). Row component 125, column component 135, or plate component 145 may accept data, for example, via input/output component 160, to be written to the memory cells 105. In some examples, a write operation may be performed at least in part by a sense component 150, or a write operation may be configured to bypass a sense component 150.

In the case of a capacitive memory element, a memory cell 105 may be written by applying a voltage to a capacitor, and then isolating the capacitor (e.g., isolating the capacitor from a voltage source used to write the memory cell 105, floating the capacitor) to store a charge in the capacitor associated with a desired logic state. In the case of ferroelectric memory, a ferroelectric memory element (e.g., a ferroelectric capacitor) of a memory cell 105 may be written by applying a voltage with a magnitude high enough to polarize the ferroelectric memory element (e.g., applying a saturation voltage) with a polarization associated with a desired logic state, and the ferroelectric memory element may be isolated (e.g., floating), or a zero net voltage or bias may be applied across the ferroelectric memory element (e.g., grounding, virtually grounding, or equalizing a voltage across the ferroelectric memory element). In the case of PCM, a memory element may be written by applying a current with a profile that causes (e.g., by way of heating and

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cooling) the memory element to form an atomic configuration associated with a desired logic state.

The sense component 150 may include multiple signal development components that may be selectively coupled with or decoupled from respective ones of a set of the sense amplifiers. For example, a sense amplifier of the sense component 150 may be coupled with a selection component of the sense component 150, and the selection component may be coupled with a set of signal development components of the sense component 150 that may be associated with one or more memory cells 105 or one or more access lines (e.g., one or more access lines 130) of the memory device 100. In some examples, cell access signals may be developed at each of the signal development components independently from others of the signal development components.

In some examples, signal development components of the sense component 150 may each be coupled with a respective memory cell during overlapping time intervals, such that multiple cell access signals (e.g., cell read signals, cell write signals, each associated with the respective memory cell of each of the respective signal development components) may be generated during the overlapping time intervals. In examples where cell access signals have been developed at multiple signal development components (e.g., in read operations of multiple memory cells 105, in a multi-cell read operation), the multiple signal development components may be coupled with the sense amplifier (e.g., in a sequential manner, in a step-wise manner) to generate sense or latch signals of the sense amplifier based at least in part on the cell access signals (e.g., in a sequential manner, in a step-wise manner). In examples where a sequence of sense or latch signals is associated with writing or re-writing a set of memory cells 105 (e.g., in write or refresh operations of multiple memory cells 105, in a multi-cell write or refresh operation), multiple signal development components may be coupled with the sense amplifier (e.g., in a sequential manner, in a step-wise manner) to generate multiple cell access signals based at least in part on the sense or latch signals of the sense amplifier (e.g., in a sequential manner, in a step-wise manner). In some examples, the multiplexed signal development components of the sense component 150 may compensate for parts of a signal development component or portions of an access operation that are associated with different latency, which may reduce the impact of access serialization.

In some examples, signal development components of the sense component 150 may enable read broadcast operations. A read broadcast may occur from one or more locations of the memory section 110 (where each location may include or be associated with one or more memory cells 105) to multiple locations of a set of signal development components. In some examples, one or more multiplexers (e.g., a multiplexer of a selection component of the sense component 150) may act as a distributed routing delivery apparatus to transfer data from locations of the memory section 110 using access lines (e.g., access lines 120, 130, and/or 140) coupled with the locations. For example, to initialize a set of locations (e.g., one or more cache blocks) in the set of signal development components to a desired pattern of data, the set of locations may be concurrently coupled with one or more multiplexers store the pattern of data from the memory section 110.

FIG. 2 illustrates an example circuit 200 that supports read broadcast operations associated with a memory device in accordance with examples as disclosed herein. Circuit 200 may include a memory cell 105-a and a sense compo-

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nent 150-a, which may be examples of a memory cell 105 and a sense component 150 described with reference to FIG. 1. Circuit 200 may also include a word line 205, a digit line 210, and a plate line 215, which, in some examples, may correspond to a first access line 120, a second access line 130, and a third access line 140, respectively (e.g., of a memory section 110), as described with reference to FIG. 1. In some examples, the plate line 215 may be illustrative of a common plate line, a common plate, or another common node for the memory cell 105-a and another memory cell 105 (not shown) of a same memory section 110. Circuit 200 illustrates circuitry that may support the described techniques for read broadcast operations associated with a memory device.

The sense component 150-a may include a sense amplifier 290 (e.g., an amplifier component, an input/output amplifier, a "latch"), which may include a first node 291 and a second node 292. In various examples, the first node 291 and the second node 292, may be coupled with different access lines of a circuit (e.g., a signal line 285 and a reference line 275 of the circuit 200, respectively), or may be coupled with a common access line of a different circuit (not shown). In some examples, the first node 291 may be referred to as a signal node, and the second node 292 may be referred to as a reference node. The sense amplifier 290 may be associated with (e.g., coupled with, coupled to) one or more input/output (I/O) lines (e.g., I/O line 295), which may include an access line coupled with a column component 135 via input/output component 160 described with reference to FIG. 1. Although the sense amplifier 290 is illustrated as having a single I/O line 295, a sense amplifier in accordance with examples as disclosed herein may have more than one I/O line 295 (e.g., two I/O lines 295). In various examples, other configurations and nomenclature for access lines and/or reference lines are possible in accordance with examples as disclosed herein.

The memory cell 105-a may include a logic storage component (e.g., a memory element, a storage element, a memory storage element), such as a capacitor 220 that has a first plate, cell plate 221, and a second plate, cell bottom 222. The cell plate 221 and the cell bottom 222 may be capacitively coupled through a dielectric material positioned between them (e.g., in a DRAM application), or capacitively coupled through a ferroelectric material positioned between them (e.g., in a FeRAM application). The cell plate 221 may be associated with a voltage, V_{plate} , and cell bottom 222 may be associated with a voltage, V_{bottom} , as illustrated in the circuit 200. The orientation of cell plate 221 and cell bottom 222 may be different (e.g., flipped) without changing the operation of the memory cell 105-a. The cell plate 221 may be accessed via the plate line 215 and cell bottom 222 may be accessed via the digit line 210. As described herein, various logic states may be stored by charging, discharging, or polarizing the capacitor 220.

The capacitor 220 may be in electronic communication with the digit line 210, and the stored logic state of the capacitor 220 may be read or sensed by operating various elements represented in circuit 200. For example, the memory cell 105-a may also include a cell selection component 225 which, in some examples, may be referred to as a switching component or a selector device coupled with or between an access line (e.g., the digit line 210) and the capacitor 220. In some examples, a cell selection component 225 may be considered to be outside the illustrative boundary of the memory cell 105-a, and the cell selection component 225 may be referred to as a switching component or

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selector device coupled with or between an access line (e.g., the digit line **210**) and the memory cell **105-a**.

The capacitor **220** may be selectively coupled with the digit line **210** when the cell selection component **225** is activated (e.g., by way of an activating logical signal or voltage), and the capacitor **220** can be selectively isolated or decoupled from the digit line **210** when the cell selection component **225** is deactivated (e.g., by way of a deactivating logical signal or voltage). A logical signal or other selection signal or voltage may be applied to a control node **226** (e.g., a control node, a control terminal, a selection node, a selection terminal) of the cell selection component **225** (e.g., via the word line **205**). In other words, the cell selection component **225** may be configured to selectively couple or decouple the capacitor **220** (e.g., a logic storage component) and the digit line **210** based on a logical signal or voltage applied via the word line **205** to the control node **226**.

Activating the cell selection component **225** may be referred to as selecting the memory cell **105-a** in some examples, and deactivating the cell selection component **225** may be referred to as deselecting the memory cell **105-a** in some examples. In some examples, the cell selection component **225** is a transistor (e.g., an n-type transistor) and its operation may be controlled by applying an activation or selection voltage to the transistor gate (e.g., a control or selection node or terminal). The voltage for activating the transistor (e.g., the voltage between the transistor gate terminal and the transistor source terminal) may be a voltage greater than the threshold voltage magnitude of the transistor (e.g., a positive activation or selection voltage). The voltage for deactivating the transistor may be a voltage less than the threshold voltage magnitude of the transistor (e.g., a ground or negative deactivation or deselection voltage).

The word line **205** may be used (e.g., by a row component **125**) to activate or deactivate the cell selection component **225**. For example, a selection voltage applied to the word line **205** (e.g., a word line logical signal or a word line voltage) may be applied to the gate of a transistor of cell selection component **225**, which may selectively connect or couple the capacitor **220** with the digit line **210** (e.g., providing a conductive path between the capacitor **220** and the digit line **210**). A deselection or deactivation voltage applied to the word line **205** may be applied to the gate of the transistor of cell selection component **225**, which may selectively disconnect, decouple, or isolate the capacitor **220** from the digit line **210**. In some examples, activating the cell selection component **225** may be referred to as selectively coupling the memory cell **105-a** with the digit line **210**, and deactivating the cell selection component **225** may be referred to as selectively decoupling or isolating the memory cell **105-a** from the digit line **210**.

In other examples, the positions of the cell selection component **225** and the capacitor **220** in the memory cell **105-a** may be switched, such that cell selection component **225** may be coupled with or between the plate line **215** and the cell plate **221**, and the capacitor **220** may be coupled with or between the digit line **210** and the other terminal of the cell selection component **225**. In such an example, the cell selection component **225** may remain connected (e.g., in electronic communication) with the digit line **210** through the capacitor **220**. This configuration may be associated with alternative timing and biasing for access operations.

In examples that employ a ferroelectric capacitor **220**, the capacitor **220** may or may not fully discharge upon connection to or coupling with the digit line **210**. In various schemes, to sense the logic state stored by a ferroelectric capacitor **220**, a voltage may be applied to the plate line **215**

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and/or the digit line **210**, and the word line **205** may be biased (e.g., by activating the word line **205**) to select the memory cell **105-a**. In some cases, the plate line **215** and/or the digit line **210** may be virtually grounded and then isolated from the virtual ground, which may be referred to as a floating condition, an idle condition, or a standby condition, prior to activating the word line **205**.

Operation of the memory cell **105-a** by varying the voltage of the cell plate **221** (e.g., via the plate line **215**) may be referred to as "moving the cell plate." Biasing the plate line **215** and/or the digit line **210** may result in a voltage difference (e.g., the voltage of the digit line **210** minus the voltage of the plate line **215**) across the capacitor **220**. The voltage difference may accompany a change in the stored charge on capacitor **220**, where the magnitude of the change in stored charge may depend on the initial state of the capacitor **220** (e.g., whether the initial logic state stored a logic 1 or a logic 0). In some schemes, the change in the stored charge of the capacitor **220**, or some portion of such a charge, may be used by the sense component **150-a** to determine the logic state stored by the memory cell **105-a** (e.g., in a charge transfer sensing scheme). In some schemes, the change in the stored charge of the capacitor **220** may cause a change in the voltage of the digit line **210**, which may be used by the sense component **150-a** to determine the logic state stored by the memory cell **105-a**. A cell access signal may refer to a signal generated while the memory cell **105-a** is selected or activated (e.g., while coupled with the signal development component), which may include a cell read signal in a read operation of the memory cell **105-a**, or a cell write signal in a write operation, a rewrite operation, or a refresh operation of the memory cell **105-a**. In various examples, a cell access signal may be referred to as a cell coupling signal or a cell charge sharing signal.

In some examples, the digit line **210** may be coupled with additional memory cells **105** (not shown), which each may be coupled with different word lines **205** (not shown). In other words, different memory cells **105** that are coupled with the digit line **210** may, in some examples, be selected or activated based at least in part on different word line logical signals.

The digit line **210** may have properties that result in an intrinsic capacitance **230** (e.g., on the order of picofarads (pF), which may in some cases be non-negligible), which may couple the digit line **210** with a voltage source **240-a** having a voltage V_0 . The voltage source **240-a** may represent a common ground or virtual ground voltage, or the voltage of an adjacent access line of the circuit **200** (not shown). Although illustrated as a separate element in FIG. 2, the intrinsic capacitance **230** may be associated with properties distributed throughout the digit line **210** or another part of the circuit **200**.

In some examples, the intrinsic capacitance **230** may depend on physical characteristics of the digit line **210**, including conductor dimensions (e.g., length, width, thickness) of the digit line **210**. The intrinsic capacitance **230** may also depend on characteristics of adjacent access lines or circuit components, proximity to such adjacent access lines or circuit components, or insulation characteristics between the digit line **210** and such access lines or circuit components. Thus, a change in voltage of digit line **210** after selecting or activating the memory cell **105-a** may depend on the net capacitance of (e.g., associated with) the digit line **210**. In other words, as charge flows along the digit line **210** (e.g., to the digit line **210**, from the digit line **210**), some finite charge may be stored along the digit line **210** (e.g., in the intrinsic capacitance **230**, in another capacitance coupled

with the digit line 210), and the resulting voltage of the digit line 210 may depend on the net capacitance of the digit line 210.

The circuit 200 (e.g., the sense component 150-a) may include a signal development component 250, which may be an example of a signal development component or signal development circuit coupled with or between the memory cell 105-a and the sense amplifier 290. In some examples, an access line associated with a signal development component 250 (e.g., an access line coupled with an input/output of the signal development component 250, an access line coupled with or between the signal development component 250 and the sense amplifier 290) may be referred to as a signal development line (SDL) (e.g., signal development line 255, a “cacheline” (CL)). The signal development component 250 may amplify or otherwise convert signals (e.g., cell access signals) of the digit line 210 and the signal development line 255. For example, for a read operation, the signal development component 250 may generate or be otherwise associated with generating a cell read signal based at least in part on being coupled with the capacitor 220 (e.g., prior to a sensing operation of the sense amplifier 290), which may include a charge sharing between the signal development component 250 and the capacitor 220. In another example, for a write operation, a rewrite operation, or a refresh operation, the signal development component 250 may generate or be otherwise associated with generating a cell write signal for the capacitor 220 (e.g., based at least in part on being coupled with the sense amplifier 290, in response to a write command, a refresh command, a rewrite command, or a read command), which may include a charge sharing between the signal development component 250 and the capacitor 220.

In some examples, the signal development component 250 may include a signal storage element such as capacitor (e.g., a signal development cache element, an integrator capacitor, an amplifier capacitor (AMPCap), which may in some cases alternatively be referred to as a “fast cap”) or another type of charge storage element configured to store a signal or signal state different than a logic state stored at a memory cell 105 (e.g., different than a logic state stored at the memory cell 105-a). Additionally or alternatively, the signal development component 250 may include, a transistor, an amplifier, a cascode, or any other charge or voltage conversion or amplification component. For example, the signal development component 250 may include a charge transfer sensing amplifier (CTSA), which in some examples may include a transistor having a gate terminal coupled with a voltage source.

Although the sense component 150-a is illustrated with a single signal development component 250, the sense component 150-a may include one or more additional signal development components 250 (not shown) to form a set of signal development components 250 (e.g., a signal development cache) in accordance with examples as disclosed herein. Each of the set of signal development components 250 of the sense component 150-a may be associated with (e.g., configured to be selectively coupled with or decoupled from, configured to develop cell access signals for) one or more memory cells 105 or one or more digit lines 210, which may or may not include the memory cell 105-a or the digit line 210. For example, each signal development component 250 of the set of signal development components 250 may be selectively coupled with or decoupled from one or more digit lines 210 of a memory section 110 of a memory array. In examples where a respective one of the signal development components 250 is coupled with more than one

memory cell 105 or more than one digit line 210, any of the memory cells 105 or digit lines 210 may be selectively coupled with or decoupled from the respective signal development component 250 by a selection component (e.g., a digit line selection component, a multiplexer, a transistor network, a transistor array, a switching network, a switching array, not shown) between the respective signal development component 250 and the associated memory cells 105 or digit lines 210.

The sense component 150-a may also include a selection component 280 (e.g., a signal development component selection component, a multiplexer, a transistor network, a transistor array, a switching network, a switching array) coupled with or between a set of signal development components 250 (e.g., with or between a set of signal development lines 255) and the sense amplifier 290. The selection component 280 may be configured to selectively couple or decouple any of the set of signal development components 250 or signal development lines 255 with the sense amplifier 290. The selection component 280 may be associated with an access line, such as the signal line 285, for conveying signals (e.g., voltage, charge, current) between the selection component 280 and the sense amplifier 290. The output of the selection component 280 (e.g., in a read operation), for example, may be an output signal (e.g., a signal conveyed via the signal line 285) that is based at least in part on an input signal (e.g., a signal conveyed from a signal development component 250 selected by the selection component 280, a signal conveyed by a signal development line 255 selected by the selection component 280). In some examples, the output signal of the selection component 280 may be equal to, or substantially equal to the input signal of the selection component 280 (e.g., where $V_{sig} = V_{SDL}$). Although described in the context of an input signal via a signal development line 255 and an output signal via a signal line 285, the interpretation of input and output may be reversed in certain access operations that employ the circuit 200 (e.g., in a write operation, a rewrite operation, a refresh operation).

In a read operation, the voltage of the signal line 285 after selecting the memory cell 105-a (e.g., a cell read signal, after coupling the memory cell 105-a or the digit line 210 with the signal development component 250, after selecting the signal development component 250 at the selection component 280) may be compared to a reference (e.g., a voltage of the reference line 275) by the sense component 150-b to determine the logic state that was stored in the memory cell 105-a (e.g., to generate a sense or latch signal). In some examples, a voltage of the reference line 275 may be provided by a reference component 270. In other examples, the reference component 270 may be omitted and a reference voltage may be provided, for example, by accessing the memory cell 105-a or the digit line 210 to generate the reference voltage (e.g., in a self-referencing access operation). Other operations may be used to support selecting and/or sensing the memory cell 105-a.

In some examples, the circuit 200 may include a bypass line 260 that may permit bypassing (e.g., selectively bypassing) the signal development component 250 or some other portion of a circuit between the memory cell 105-a and the sense amplifier 290. In some examples, the bypass line 260 may be selectively enabled or disabled by way of a switching component 265. In other words, when the switching component 265 is activated, the digit line 210 may be coupled with the signal development line 255 or the selection component 280 via the bypass line 260 (e.g., coupling

the memory cell **105-a** with the selection component **280** or some other portion of a circuit between the memory cell and the sense amplifier **290**).

In some examples, when the switching component **265** is activated, the signal development component **250** may be selectively isolated or decoupled from one or both of the digit line **210** or the signal development line **255** (e.g., by another switching component or selection component, not shown). When the switching component **265** is deactivated, the digit line **210** may be selectively coupled with the signal development line **255** or the selection component **280** via the signal development component **250**. In other examples, one or more additional selection components (not shown) may be used to selectively couple the memory cell **105-a** (e.g., the digit line **210**) with one of the signal development component **250** (e.g., via the signal development line **255**) or the bypass line **260**.

Additionally or alternatively, in some examples, a switching or selection component may be used to selectively couple the selection component **280** with one of the signal development component **250** (e.g., via the signal development line **255**) or the bypass line **260**. In some examples, a selectable bypass line **260** may support generating a cell access signal (e.g., a cell read signal) for detecting a logic state of the memory cell **105-a** by using the signal development component **250**, and generating a cell access signal (e.g., a cell write signal) to write a logic state to the memory cell **105-a** that bypasses the signal development component **250**.

Some examples of a memory device that supports read broadcast operations may share a common access line (not shown) between a memory cell **105** and a sense amplifier **290** to support generating a sense signal and a reference signal from the same memory cell **105**. In one example, a common access line between a signal development component **250** and a sense amplifier **290** may be referred to as a "common line," and the common access line may take the place of the signal line **285** and the reference line **275** illustrated in circuit **200**.

In such examples, the common access line may be connected to the sense amplifier **290** at two different nodes (e.g., a first node **291** and a second node **292**, as described herein). In some examples, a common access line may permit a self-referencing read operation to share, in both a signal generating operation and a reference generating operation, components that may exist between the sense amplifier **290** and a memory cell **105** being accessed. Such a configuration may reduce the sensitivity of the sense amplifier **290** to operational variations of various components in a memory device, such as memory cells **105**, access lines (e.g., a word line **205**, a digit line **210**, a plate line **215**), signal development circuits (e.g., signal development component **250**), transistors, voltage sources **293** and **294**, and others.

Although the digit line **210**, the signal development line **255**, and the signal line **285** are identified as separate lines, the digit line **210**, the signal development line **255**, the signal line **285**, and any other lines connecting a memory cell **105** with a sense amplifier **290** may be referred to as a single access line in accordance with examples as disclosed herein. Constituent portions of such an access line may be identified separately for the purposes of illustrating intervening components and intervening signals in various example configurations.

The sense amplifier **290** may include various transistors or amplifiers to detect, convert, or amplify a difference in signals, which may include or otherwise be referred to as generating a sense signal or a latch signal. For example, the

sense amplifier **290** may include circuit elements that receive and compare a sense signal voltage (e.g., a cell read signal, V_{sig}) at the first node **291** with a reference signal voltage (e.g., V_{ref}) at the second node **292**. An output of the sense amplifier **290** (e.g., a sense or latch signal) may be driven to a higher (e.g., a positive voltage) or a lower voltage (e.g., a negative voltage, a ground voltage) based on the comparison at the sense amplifier **290**.

For example, if the first node **291** has a lower voltage than the second node **292**, the output of the sense amplifier **290** may be driven to a relatively lower voltage of a low voltage source **293** (e.g., a voltage of V_L , which may be a ground voltage substantially equal to V_0 or a negative voltage). A sense component **150** that includes the sense amplifier **290**, or an I/O component **160** that is coupled with such a sense component **150**, may latch the output of the sense amplifier **290** to determine the logic state stored in the memory cell **105-a** (e.g., detecting a logic 0 when the first node **291** has a lower voltage than the second node **292**).

If the first node **291** has a higher voltage than the second node **292**, the output of the sense amplifier **290** may be driven to the voltage of a high voltage source **294** (e.g., a voltage of V_H). A sense component **150** that includes the sense amplifier **290**, or an I/O component **160** that is coupled with such a sense component **150**, may latch the output of the sense amplifier **290** to determine the logic state stored in the memory cell **105-a** (e.g., detecting a logic 1 when the first node **291** has a higher voltage than the second node **292**). The latched output of the sense amplifier **290**, corresponding to the detected logic state of memory cell **105-a**, may then be output via one or more input/output (I/O) lines (e.g., I/O line **295**).

To perform a write operation, rewrite operation, or refresh operation on the memory cell **105-a**, a voltage (e.g., a cell write signal) may be applied across the capacitor **220**. Various methods may be used. In one example, the cell selection component **225** may be selected or activated through the word line **205** (e.g., by selecting or activating the word line **205**) to electrically connect the capacitor **220** to the digit line **210**. A voltage may be applied across capacitor **220** by controlling the voltage of the cell plate **221** (e.g., through the plate line **215**) and the cell bottom **222** (e.g., through the digit line **210**). In some examples, write operations, rewrite operations, or refresh operations may be based at least in part on a sense or latch signal at the sense amplifier **290**, which may be based on a signal received via the I/O line **295** (e.g., a write signal, a refresh signal) or based on a signal generated at the sense amplifier **290** (e.g., a rewrite signal).

For example, to write a logic 0, the cell plate **221** may be taken high (e.g., applying a positive voltage to the plate line **215**), and the cell bottom **222** may be taken low (e.g., grounding the digit line **210**, virtually grounding the digit line **210**, applying a negative voltage to the digit line **210**). The opposite process may be performed to write a logic 1, where the cell plate **221** is taken low and the cell bottom **222** is taken high. In some cases, the voltage applied across the capacitor **220** during a write operation may have a magnitude equal to or greater than a saturation voltage of a ferroelectric material in the capacitor **220**, such that the capacitor **220** is polarized, and thus maintains a charge even when the magnitude of applied voltage is reduced, or if a zero net voltage is applied across the capacitor **220**. In some examples, the sense amplifier **290** or the signal development component **250** may be used to perform the write operations, which may include coupling the low voltage source **293** or the high voltage source **294** with the digit line. When the

sense amplifier **290** is used to perform the write operations, the signal development component **250** may or may not be bypassed (e.g., by applying a write signal via the bypass line **260**).

The circuit **200**, including the sense component **150-a**, the cell selection component **225**, the signal development component **250**, the switching component **265**, the reference component **270**, the selection component **280**, or the sense amplifier **290** may include various types of transistors. For example, the circuit **200** may include n-type transistors, where applying a relative positive voltage to the gate of the n-type transistor that is above a threshold voltage for the n-type transistor (e.g., an applied voltage having a positive magnitude, relative to a source terminal, that is greater than a threshold voltage) enables a conductive path between the other terminals of the n-type transistor (e.g., the source terminal and a drain terminal).

In some examples, an n-type transistor may act as a switching component, where the applied voltage is a logical signal that is used to selectively enable conductivity through the transistor by applying a relatively high logical signal voltage (e.g., a voltage corresponding to a logic 1 state, which may be associated with a positive logical signal voltage supply), or to selectively disable conductivity through the transistor by applying a relatively low logical signal voltage (e.g., a voltage corresponding to a logic 0 state, which may be associated with a ground or virtual ground voltage, or a negative voltage). In some examples where a n-type transistor is employed as a switching component, the voltage of a logical signal applied to the gate terminal may be selected to operate the transistor at a particular working point (e.g., in a saturation region or in an active region).

In some examples, the behavior of an n-type transistor may be different (e.g., more complex) than a logical switching, and selective conductivity across the transistor may also be a function of varying source and drain voltages. For example, the applied voltage at the gate terminal may have a particular voltage level (e.g., a clamping voltage, a control voltage) that is used to enable conductivity between the source terminal and the drain terminal when the source terminal voltage is below a certain level (e.g., below the gate terminal voltage minus the threshold voltage). When the source terminal voltage or drain terminal voltage rises above the certain level, the n-type transistor may be deactivated such that the conductive path between the source terminal and drain terminal is opened.

Additionally or alternatively, the circuit **200** may include p-type transistors, where applying a relative negative voltage to the gate of the p-type transistor that is above a threshold voltage for the p-type transistor (e.g., an applied voltage having a negative magnitude, relative to a source terminal, that is greater than a threshold voltage) enables a conductive path between the other terminals of the p-type transistor (e.g., the source terminal and a drain terminal).

In some examples, an p-type transistor may act as a switching component, where the applied voltage is a logical signal that is used to selectively enable conductivity by applying a relatively low logical signal voltage (e.g., a voltage corresponding to a logical "1" state, which may be associated with a negative logical signal voltage supply), or to selectively disable conductivity by applying a relatively high logical signal voltage (e.g., a voltage corresponding to a logical "0" state, which may be associated with a ground or virtual ground voltage, or a positive voltage). In some examples where a p-type transistor is employed as a switching component, the voltage of a logical signal applied to the

gate terminal may be selected to operate the transistor at a particular working point (e.g., in a saturation region or in an active region).

In some examples, the behavior of a p-type transistor may be different (e.g., more complex) than a logical switching by the gate voltage, and selective conductivity across the transistor may also be a function of varying source and drain voltages. For example, the applied voltage at the gate terminal may have a particular voltage level that is used to enable conductivity between the source terminal and the drain terminal so long as the source terminal voltage is above a certain level (e.g., above the gate terminal voltage plus the threshold voltage). When the voltage of the source terminal voltage falls below the certain level, the p-type transistor may be deactivated such that the conductive path between the source terminal and drain terminal is opened.

A transistor of the circuit **200** may be a field-effect transistor (FET), including a metal oxide semiconductor FET, which may be referred to as a MOSFET. These, and other types of transistors may be formed by doped regions of material on a substrate. In some examples, the transistor(s) may be formed on a substrate that is dedicated to a particular component of the circuit **200** (e.g., a substrate for the sense amplifier **290**, a substrate for the signal development component **250**, a substrate for the memory cell **105-a**), or the transistor(s) may be formed on a substrate that is common for particular components of the circuit **200** (e.g., a substrate that is common for the sense amplifier **290**, the signal development component **250**, and the memory cell **105-a**). Some FETs may have a metal portion including aluminum or other metal, but some FETs may implement other non-metal materials such as polycrystalline silicon, including those FETs that may be referred to as a MOSFET. Further, although an oxide portion may be used as a dielectric portion of a FET, other non-oxide materials may be used in a dielectric material in a FET, including those FETs that may be referred to as a MOSFET.

In some examples, different portions of the circuit **200**, or different operations that use portions of the circuit **200**, may be associated with different latencies. For example, in one portion of an access operation (e.g., a first sub-operation, a first set of sub-operations), a cell access signal may be developed by coupling the memory cell **105-a** with the signal development component **250** (e.g., based at least in part on activating or selecting the cell selection component **225**, based at least in part on activating another switching component, isolation component, or selection component between the memory cell **105-a** and the signal development component **250**). In some examples, the cell access signal may be developed based at least in part on, or may be otherwise associated with a charge sharing between the memory cell **105-a** (e.g., the capacitor **220**) and the signal development component **250** (e.g., charge flowing from the capacitor **220** to the signal development component **250**, charge flowing from the signal development component **250** to the capacitor **220**). In some examples (e.g., in a read operation), the developed cell access signal (e.g., a cell read signal) or charge sharing may be based at least in part on a logic state stored by the memory cell **105-a**. In some examples (e.g., in a write operation, a rewrite operation, a refresh operation), the developed cell access signal (e.g., a cell write signal) or charge sharing may be based at least in part on a developed sense or latch signal (e.g., at the sense amplifier **290**, at the signal line **285**). As disclosed herein, the charge sharing between the memory cell **105-a** and the signal development component **250** may be associated with

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a change in voltage of the digit line **210**, or a change in voltage of the signal development line **255**, or both.

The development of a cell access signal for an access operation may be associated with a latency, which may refer to an amount of time (e.g., a duration) for developing the cell access signal, a delay between initiating a cell access signal development operation and a cell access signal reaching a threshold level suitable for subsequent portions of the access operation (e.g., in a read operation), or a delay between initiating a cell access signal development operation and a memory cell **105** being written with a logical value (e.g., in a write operation, a rewrite operation, or a refresh operation). In some examples (e.g., in a read operation) the duration or latency may be referred to as a "row-to-column address delay," and in some examples (e.g., in a write operation) the duration or latency may be referred to as a "row precharge delay," which may be longer or shorter than a row-to-column address delay.

In some examples, the sharing of charge between the memory cell **105-a**, the digit line **210** (e.g., intrinsic capacitance **230**) and the signal development component **250** may be associated with a time constant behavior (e.g., a time constant behavior of a change in voltage V_{DL} , a time constant behavior of a change in voltage V_{SDL}), or otherwise include a logarithmic or exponential behavior. The duration or latency for developing the cell access signal may refer to a duration between a coupling or activation operation (e.g., a selection or activation of the cell selection component **225**, a selection or activation of another component configured to selectively couple the memory cell **105-a** and the signal development component **250**) and the digit line **210** or signal development line **255** reaching a steady state voltage, or the digit line **210** or signal development line **255** reaching a threshold proportion of a steady state voltage (e.g., 95% of a steady state voltage, 99% of a steady state voltage).

In some examples, the duration or latency for developing a cell access signal may be expressed as a time constant (e.g., a duration of time for reaching 63% of a change between initial voltage and steady state voltage), or expressed as a multiple of time constants. For example, the duration or latency for developing the cell access signal may be expressed as a duration of 3 time constants, or a duration otherwise associated with the cell access signal being within 5% of a steady state value. In another example, the duration or latency for developing the cell access signal may be expressed as a duration of 5 time constants, or a duration otherwise associated with the cell access signal being within 1% of a steady state value.

In some examples, charge sharing behavior and associated time constants or other latency may be based at least in part on a capacitance of the memory cell **105-a**, the signal development component **250**, or other capacitance between the memory cell **105-a** and the signal development component **250** (e.g., intrinsic capacitance, such as intrinsic capacitance **230**). For example, a relatively high capacitance of the digit line **210** (e.g., a relatively high intrinsic capacitance **230**) may be associated with a relatively high latency (e.g., a relatively long duration to develop a cell read signal), and a relatively low capacitance of the digit line **210** may be associated with a relatively low latency (e.g., a relatively short duration to develop a cell read signal). In another example, a relatively high capacitance of memory cell **105-a** (e.g., capacitor **220**) may be associated with a relatively low latency (e.g., a relatively short duration to develop a cell read signal), and a relatively low capacitance of the memory cell **105-a** may be associated with a relatively high latency (e.g., a relatively long duration to develop a cell read signal).

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Although described with reference to time constant behavior, a duration or latency associated with developing a cell access signal may additionally or alternatively include other behaviors such as ramped, stepped, or oscillating (e.g., underdamped) behaviors. In some examples, developing a cell access signal may include a set of operations, such as a set of coupling, isolating, activating, deactivating, selecting, or deselecting operations, and a duration or latency associated with developing the cell access signal may include the associated circuit behaviors of each of the set of operations. For example, developing a cell access signal may include activating switching or selection components along the digit line **210** or signal development line **255**, activating switching or selection components between the digit line or signal development line and another component (e.g., selectively coupling a voltage source (not shown) with the digit line **210** or the signal development line **255**), or other operations or combinations of operations.

In another portion of the access operation (e.g., a second sub-operation, a second set of sub-operations), a sense signal (e.g., a latch signal, an output signal, an input/output signal) may be developed by activating the sense amplifier **290** (e.g., based at least in part on selectively coupling the signal development component **250** with the sense amplifier **290**, based at least in part on selectively coupling the sense amplifier with one or both of the low voltage source **293** or the high voltage source **294**). In some examples, the sense signal may be developed based at least in part on, or may be otherwise associated with a charge sharing between the signal development component **250** and the sense amplifier **290**. In some examples (e.g., in a read operation), the sense signal or charge sharing may be based at least in part on the developed cell access signal (e.g., at the signal development component **250**, at the signal development line **255**). As described herein, the charge sharing between the signal development component **250** and the sense amplifier **290** may be associated with a change in voltage of the I/O line **295**, which may be based at least in part on a comparison between voltage V_{sig} and voltage V_{ref} (e.g., an output of V_L when V_{sig} is less than V_{ref} , an output of V_H when V_{sig} is greater than V_{ref}).

The development of a sense or latch signal for an access operation may also be associated with a latency, which may refer to an amount of time for developing the sense or latch signal, or a delay between initiating a sense or latch signal generation operation and a sense or latch signal reaching a threshold level suitable for subsequent portions of the access operation (e.g., an output indicative of a logic state stored by the memory cell **105-a**). For example, the sharing of charge between the signal development component **250** and the sense amplifier **290** may also be associated with a time constant behavior (e.g., a time constant behavior of a change in voltage of the I/O line **295**), or other logarithmic or exponential behavior. The duration or latency for developing the sense or latch signal may refer to a duration between a coupling or activation operation (e.g., a selection or activation of a switching component or selection component, such as the selection component **280**, configured to selectively couple the signal development component **250** with the sense amplifier **290**, a coupling of the sense amplifier **290** with one or both of the low voltage source **293** or the high voltage source **294**) and the I/O line **295** reaching a steady state voltage, or the I/O line **295** reaching a threshold proportion of a steady state voltage (e.g., 90% of a steady state voltage, 95% of a steady state voltage).

The duration or latency for developing a sense or latch signal may also be expressed as a time constant, or as a

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multiple of time constants. Although described with reference to time constant behavior, a duration or latency associated with developing a sense or latch signal may additionally or alternatively include other behaviors such as ramped, stepped, or oscillating (e.g., underdamped) behaviors. In some examples, developing a sense or latch signal may include a set of operations, such as a set of coupling, isolating, activating, deactivating, selecting, or deselecting operations, and a duration or latency associated with developing the sense or latch signal may include the associated circuit behaviors of each of the set of operations.

In some examples of the circuit 200, a latency associated with developing a cell access signal may be longer in duration than a latency associated with generating a sense or latch signal. For example, a charge sharing between the signal development component 250 and the memory cell 105-a may be associated with a different amount of charge, or a slower transfer of charge, than a charge sharing between the signal development component 250 and the sense amplifier 290. In other words, the signal development component 250 or the memory cell 105-a may be associated with or be otherwise considered as relatively high latency portions of the circuit 200 and the sense amplifier 290 may be associated with or considered as a relatively low latency portion of the circuit 200. In such examples, the circuit 200 may support performing input or output operations more quickly than performing signal development operations.

In accordance with examples as disclosed herein, a memory device 100 that includes the circuit 200 may couple each of a set of signal development components 250 with a respective memory cell 105 during overlapping time intervals, such that multiple cell access signals (e.g., associated with the respective memory cell 105 of each of the respective signal development components 250) may be generated during the overlapping time intervals. Each of the set of signal development components 250 may be selectively coupled with the sense amplifier 290 via the selection component 280 (e.g., in a sequential order) to generate a sequence of sense or latch signals at the sense amplifier 290, or vice versa. For example, in a read operation or set of read operations, the sequence of sense or latch signals generated at the sense amplifier 290 may be based on respective cell access signals (e.g., cell read signals) developed during overlapping time intervals at the set of signal development components 250, which may be associated with particular logic states stored by respective memory cells 105. Thus, as disclosed herein, a memory device 100 that includes the circuit 200 may include signal development components 250 that are multiplexed via the selection component 280, which in some examples may compensate for portions of an access operation that are associated with different latencies.

In some examples, one or more signal development components 250 of the circuit 200 may enable read broadcast operations. A read broadcast may occur from one or more locations of a memory array (e.g., a memory array including one or more memory sections 110) to multiple locations of a set of signal development components 250. In some examples, one or more multiplexers (e.g., of the selection component 280) may act as a distributed routing delivery apparatus to transfer data from locations of the memory array using access lines (e.g., word lines 205, digit lines 210, plate lines 215, etc.) coupled with the locations. For example, to initialize a set of locations in the set of signal development components 250 (e.g., in one or more cache blocks) to a desired pattern of data, the set of locations

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may be concurrently coupled with one or more of the multiplexers to store the pattern of data from the memory array.

FIG. 3 illustrates an example circuit 300 that supports read broadcast operations associated with a memory device in accordance with examples as disclosed herein. It is to be understood that circuit 300 is merely one illustrative example, and that many implementations, including other specific circuits and topologies, are possible while adhering to the principles and techniques disclosed herein, as will be appreciated by one of ordinary skill in the art.

Circuit 300 includes a set of memory cells 105-b (e.g., memory cells 105-b-111 through 105-b-srm) and a sense component 150-b. Although the memory cells 105-b are illustrated as including a capacitor and a cell selection component, memory cells 105-b in accordance with examples as disclosed herein may include various configurations (e.g., with or without cell selection components) and various types of logic storage elements (e.g., a capacitive memory element, a ferroelectric memory element, a material memory element, a resistive memory element, a thresholding memory element, other memory element) to support various types of memory devices (e.g., DRAM memory devices, FeRAM memory devices, PCM devices, chalcogenide memory devices). Circuit 300 illustrates circuitry that may support the described techniques for read broadcast operations associated with a memory device.

The sense component 150-b may include a set of signal development components 250-a (e.g., signal development components 250-a-1 through 250-a-s), each associated with one or more of the memory cells 105-b. The sense component 150-b may also include a selection component 280-a (e.g., a signal development component selection component, a MUX, a transistor network, a transistor array, a switching network, a switching array) that is coupled with the set of signal development components 250-a (e.g., via signal development lines 255-a-1 through 255-a-s). The selection component 280-a may be configured to selectively couple a selected one of the signal development components 250-a (e.g., a selected one of the signal development lines 255-a) with a sense amplifier 290-a of the sense component 150-b (e.g., via signal line 285-a, in response to a logical or selection signal, such as a signal development component multiplexing (SDCM) signal). The sense amplifier 290-a may exchange (e.g., communicate, receive, transmit) input or output signals with other components of a memory device (e.g., an input/output component 160) via the I/O line 295-a.

In the example of circuit 300, the memory cells 105-b may be arranged according to a set of domains 310-a (e.g., domains 310-a-1 through 310-a-s). In other words, the circuit 300 may illustrate an example of a set of memory cells 105-b that are divided across or otherwise associated with s domains. In the example of circuit 300, each of the domains 310-a may be associated with (e.g., coupled with) one of the signal development components 250-a (e.g., domain 310-a-1 being associated with signal development component 250-a-1). However, in various examples of circuitry that supports the described techniques, a domain 310 may be associated with more than one signal development component 250, or a signal development component 250 may be associated with more than one domain 310, or both.

Although the example domains 310-a of circuit 300 are described with reference to certain characteristics, alternative definitions or organizations of domains may also be utilized in support of the described techniques. As one such example, memory cells 105 or access lines (e.g., word lines 205, digit lines 210, plate lines 215) of a domain may be

organized or subdivided in a different manner than the domains **310-a** illustrated in the circuit **300**, or a domain may be defined in a different manner than the domains **310-a** illustrated in the circuit **300** (e.g., which components are included within an illustrative boundary of a domain), or domains may be coupled with signal development components **250** or sense amplifiers **290** in a different manner than the domains **310-a** illustrated in the circuit **300** (e.g., with different multiplexing organizations or schemes, different selection components).

In the example of circuit **300**, each of the domains **310-a** may include memory cells **105-b** that are coupled with or between one of a set of digit lines **210-a** and one of a set of plate lines **215-a**. For example, for domain **310-a-1**, each of the set of memory cells **105-b** (e.g., each of memory cells **105-b-111** through **105-b-1rm**) may be coupled with one of the digit lines **210-a-11** through **210-a-1r** and may be coupled with one of the plate lines **215-a-11** through **215-a-1r**. In other words, the domains **310-a** may illustrate an arrangement of memory cells **105-b** that are divided across or otherwise associated with r digit lines **210-a** or “columns.” Although the example circuit **300** is illustrated as having separate plate lines **215-a**, in some examples, a set of plate lines **215-a** (e.g., a set of two or more of the plate lines **215-a-11** through **215-a-1r**) may represent or be otherwise functionally equivalent with a common plate line of a domain **310-a** (e.g., domain **310-a-1**), or may represent or be otherwise functionally equivalent with a common plate line of a portion of a domain **310-a** (e.g., a “sub-domain”), or a different set of plate lines **215-a** (e.g., a set of two or more of the plate lines **215-a-11** through **215-a-sr**) may represent or be otherwise functionally equivalent with a common plate line of a set of domains **310-a** (e.g., a set of domains **310-a-1** through **310-a-s**).

Domains **310-a** may also illustrate an arrangement of memory cells **105-b** that are divided across or otherwise associated with m word lines **205-a** or “rows.” For example, domain **310-a-1** may include respective sets of m memory cells **105-b** that are coupled with or between each of the digit lines **210-a** of the domain **310-a** and the plate lines **215-a** of the domain (e.g., a set of memory cells **105-b-111** through **105-b-11m** coupled with or between the digit line **210-a-11** and the plate line **215-a-11**). For a set of memory cells **105-b** coupled with a same digit line **210-a** and a same plate line **215-a**, each of the set may be individually selected or accessed based at least in part on an associated logical signal WL (e.g., for domain **310-a**, one of logical signals WL_{11} through WL_{1m}). Although illustrated as sharing a common set of word lines **205-a** in a domain **310-a** (e.g., word lines **205-a-11** through **205-a-1m** shared across each of the columns of domain **310-a-1**), other examples of a memory device may have a different arrangement of word lines **205** in a domain **310**.

In the example of circuit **300**, each of the domains **310-a** may also include or be otherwise associated with a selection component **320-a** (e.g., a digit line selection component, a MUX, a transistor network, a transistor array, a switching network, a switching array) that is coupled with each of the set of digit lines **210-a** of the domain **310-a**. For example, the domain **310-a-1** may include a selection component **320-a-1** that is coupled with each of the digit lines **210-a-11** through **210-a-1r**. The selection component **320-a-1**, for example, may be configured to selectively couple a selected one of the digit lines **210-a-11** through **210-a-1r**, or one of the memory cells **105-b-111** through **105-b-11m**, with the signal development component **250-a-1** (e.g., in response to a logical or selection signal, such as a digit line multiplexing

(DLM) signal DLM_1). Accordingly, each of the selection components **320-a-1** through **320-a-s** may be associated with a respective one of the signal development components **250-a-1** through **250-a-s**.

In the example of circuit **300**, each of the signal development components **250-a** may be associated with a respective set of memory cells **105-b** or a respective set of digit lines **210-a**. In some examples, the selection components **320-a-1** through **320-a-s** may be an example of a plurality of second selection components, where each second selection component of the plurality of second selection components is associated with a respective signal development component **250**, and is configured to selectively couple any one memory cell **105-b** or digit line **210-a** of the set with the respective signal development component **250**.

In an illustrative example, each of the domains **310-a** may include 1,048,576 memory cells **105-b** arranged in 1,024 uniquely addressed rows and 1,024 columns (e.g., where $m=1024$ and $r=1024$). According to the illustrative example of circuit **300**, one signal development component **250-a** may be mapped to a particular domain **310-a**, but in other examples a set of more than one signal development component **250-a** may be mapped to a particular domain **310-a** (e.g., to respective sets of digit lines **210-a** of a domain **310-a**). In some examples, such a mapping may be fixed (e.g., where respective sets of digit lines **210-a** are mapped to a respective signal development component **250-a** within each domain **310-a**) which, in some examples, may reduce multiplexing or selection circuit complexity. In various other examples (not shown), a signal development component **250** may be mapped to more than one domain **310**, more than one set of digit lines **210** (e.g., of a domain), or other configurations. Additionally or alternatively, a domain **310** or a set of digit lines **210** may be mapped to more than one signal development component **250**. In other words, a memory device may include various configurations of signal development components **250** to support examples of the read broadcast operations described herein.

In the example of circuit **300**, each of the digit lines **210-a** is associated with (e.g., configured to be selectively coupled with) a single one of the signal development components (e.g., via a respective one of the selection components **320-a-1**). For example, the digit line **210-a-11** may be associated with signal development component **250-a-1**, but not signal development component **250-a-s**. However, in various examples of circuitry that supports the described techniques for read broadcast operations associated with a memory device, a particular digit line **210-a** may be associated with (e.g., configured to be selectively coupled with) more than one signal development component **250-a**, which may include a selection component different from the set of selection components **320-a-1** through **320-a-s** illustrated in circuit **300**. For example, the digit line **210-a-11** may be associated with (e.g., configured to be selectively coupled with) either the signal development component **250-a-1** or the signal development component **250-a-s**, or any other signal development components **250-a** of the circuit **300**.

In another illustrative example that supports the described techniques for read broadcast operations, another circuit may include several domains each with 1,048,576 memory cells **105** arranged in 1,024 uniquely addressed rows and 1,024 columns, which may refer to an organization of components that is different than the circuit **300**. Each of the domains of the other circuit may be arranged such that $m=1024$ and $r=1024$, and the digit lines **210** of a respective domain of this other circuit may collectively be mapped to an array of 64 signal development components **250** (e.g.,

according to a many to-one mapping, according to a many-to-many mapping). In one example of the other circuit, each of the signal development components **250** may be mapped to a respective subset of the digit lines **210** of the domain (e.g., one signal development component **250** may be mapped to $1024/64=16$ digit lines **210** within each domain). In some examples, such a mapping may be fixed (e.g., where groups or subsets of 16 digit lines **210** are mapped to a respective signal development component **250** within each domain) which, in some examples, may reduce multiplexing or selection circuit complexity.

In this other example, a row of 1024 memory cells **105** (e.g., spanning one domain of the other circuit) may be selected by a single word line **205** in each domain. In other words, with 64 signal development components **250** per domain and $r=1024$, the activation of a word line in one domain and the activation of another word line in another domain (e.g., including other independent word lines in other domains) may select memory cells **105** associated with the respective row. With 64 signal development components **250** per domain of such a circuit, 64 of the set of 1,024 memory cells **105** may be accessed at a time in each domain (e.g., by selectively coupling a respective digit line **210** with each of the 64 signal development components **250** via a respective selection component). During such accessing, other digit lines **210** may be selectively isolated from the respective signal development component **250** and other signal development components **250** interfacing the same domain. Further, the other digit lines **210** may be shunted or masked as described herein.

Thus, examples in accordance with the techniques disclosed herein may include examples in which word lines **205** within a domain, or word lines **205** across multiple domains, or some combination thereof, are independent (e.g., selectable independently of one another). Examples in accordance with the techniques disclosed herein may also include examples in which word lines **205** within a domain, or word lines **205** across multiple domains, or some combination thereof, are locked (e.g., hard-wired) to be selected together (jointly). It is to be understood that in examples in which word lines **205** are independently selectable, such word lines **205** may nevertheless be operated synchronously (e.g., as though locked), at least at certain times or under certain conditions. Further, examples in accordance with the techniques disclosed herein may include examples in which many digit lines **210** are mapped to many signal development components **250** within a domain, as well as examples where many digit lines **210** are mapped to one signal development component **250** within a domain (e.g., a selection component **280** may have many-to-one or many-to-many functionality). Aspects of these and other example variations are described throughout the disclosure, including with reference to FIG. 8.

In some examples, operations associated with word line selection may be time-bounded to prevent loss or corruption of data, which may involve waiting for completion of operations that are in progress with accessed cells. For example, when switching from a first word line **205-a** of a domain **310-a** to a second word line **205-a** of the same domain **310-a**, such a switching may need to wait for cell access signal development of the domain **310-a** (e.g., of the signal development component **250-a**) to be completed before the switching takes place. In examples where a word line **205-a** is shared across domains (e.g., a word line **205-a** that is shared between domain **310-a-1** and **310-a-s**, word line **205-a-11** being functionally equivalent to word line **205-a-s1**), when switching from a first shared word line

205-a to a second shared word line **205-a**, such a switching may need to wait for cell access signal development of each of the domains **310-a-1** and **310-a-s** (e.g., each of the signal development components **250-a-1** and **250-a-s**) to be completed before the switching takes place

In the example of circuit **300**, each of the domains **310-a** may also include or be otherwise associated with a set of shunts **330-a** (e.g., digit line shunts, digit-to-plate shunts). For example, domain **310-a-1** may include a set of shunts **330-a-11** through **330-a-1r**. Each of the shunts **330-a** may be coupled with or between a digit line **210-a** and plate line **215-a**. For example, for domain **310-a-1**, shunt **330-a-11** may be coupled with or between the digit line **210-a-11** and the plate line **215-a-11**. The shunt **330-a-11**, for example, may be configured to selectively couple the digit line **210-a-11** with the plate line **215-a-11** (e.g., in response to a logical or switching signal DLS_{11}). In some examples, a shunt **330-a** may be configured to selectively equalize a bias between a digit line **210-a** and a plate line **215-a**, or equalize one or more memory cells **105-b** that are coupled with or between a digit line **210-a** and a plate line **215-a**. In some examples, a shunt **330-a** may be configured to selectively discharge one or more memory cells **105-b** that are coupled with or between a digit line **210-a** and a plate line **215-a**.

In some examples, the circuit **300** may be operated according to a shunt mask. For example, when multiplexing is performed on a domain **310-a** (e.g., using selection components **320-2**), a shunt **330-a** of a masked digit line **210-a** (e.g., a digit line **210-a** that is not associated with an access operation that is being performed) may support a selective coupling with a plate line **215-a** to prevent or reduce data loss (e.g., charge leakage) of memory cells **105-b** that are associated with the masked digit line **210-a**. In other words, a shunt **330-a** may turn off bit transfer on masked digit lines **210-a** that are not associated with an access operation that is being performed.

The selection component **280-a** and the selection components **320-a** may include various configurations of components, and each may be referred to as a multiplexer, a transistor network, a transistor array, a switching network, or a switching array. In one example, the selection component **280-a** may include a set of transistors that are each coupled with the sense amplifier **290-a** (e.g., each coupled with the signal line **285-a**). Each of the set of transistors may also be coupled with a respective one of the signal development components **250-a** (e.g., a respective one of the signal development lines **255-a-1** through **255-a-s**). Each of the set of transistors may be configured to selectively couple the respective one of the signal development components **250-a** with the sense amplifier **290-a**, responsive to one of a set of switching or logical signals provided to a gate of the transistor.

In some examples, a selection component **280-a** or a selection component **320-a** may include decoder or other logical or selection signal conversion component. A decoder of the selection component **280-a**, for example, may receive a logical or selection signal (e.g., signal SDCM), which may be a digital signal (e.g., a signal having or otherwise representing multiple bits) received over a signal bus. In some examples, the decoder may receive the digital signal as an input to generate a set of binary signals (e.g., switching or logical signals) that may be applied to the gates of a set of transistors configured in a switching arrangement. For example, the decoder of the selection component **280-a** may receive a selection signal SDCM as a 4-bit digital input signal, and generate 16 binary (e.g., on/off) switching sig-

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nals, each applied to the gate of one of a set of 16 transistors configured in a switching arrangement.

In various examples, the selection component **280-a** may be configured such that one of the signal development components **250-a-1** through **250-a-s** is coupled with (e.g., selectively coupled with) the sense amplifier **290-a** at a time, and others of the signal development components **250-a-1** through **250-a-s** may be decoupled from (e.g., selectively decoupled from) the sense amplifier **290-a** at that time (e.g., the time when the one of the signal development components **250-a-1** through **250-a-s** is selectively coupled with the sense amplifier **290-a**). In some examples, the selection component **280-a** may also be configured to support operations where none of the signal development components **250-a-1** through **250-a-s** are coupled with the sense amplifier **290-a** at a particular time (e.g., where each of the signal development components **250-a-1** through **250-a-s** are selectively isolated from the sense amplifier **290-a**). In various examples of the circuit **300**, the selection components **320-a** may include similar features or sets of features as a selection component **280-a**, or the selection components **320-a** may include different features or sets of features as a selection component **280-a**.

In some examples of the circuit **300**, the signal development components **250-a** or the memory cells **105-b** may be associated with or be otherwise considered as relatively high latency portions of the circuit **300**, and the sense amplifier **290-a** may be associated with or considered as a relatively low latency portion of the circuit **300**. In accordance with examples as disclosed herein, the sense component **150-b** may illustrate an example of dividing memory cell access circuitry into high-latency parts (e.g., signal development components **250-a**) and low-latency parts (e.g., sense amplifier **290-a**), and coupling a set of high-latency parts with a set of low-latency parts through a multiplexer (e.g., selection component **280-a**).

In the example of circuit **300**, the selection component **280-a** may provide a first degree of data pipelining, which may reduce the impact of data access serialization due to row buffer conflicts. For example, the selection component **280-a** may support overlapping data transfers on different sets of digit lines **210-a** (e.g., different domains **310-a**). Thus, the sense amplifier **290-a** may be free to support read, write, rewrite, or refresh operations (e.g., while coupled with one of the signal development components **250-a**) while other signal development components **250-a** are involved in data transfer (e.g., while other signal development components **250-a** are coupled with digit lines **210-a** or memory cells **105-b**).

The set of signal development components **250-a** may be considered to be a small, fast local cache (e.g., a signal development cache), where the respective signal development components **250-a** may be configured to store a signal state, different than logic states stored at the memory cells **105-b**. Such a configuration may be used to support reducing a rate of row buffer conflicts, increasing internal bandwidth, or other benefits. In some examples, the selection components **320-a** may provide further gains by providing a second degree of data pipelining via multiplexed digit lines **210-a**. Thus, in accordance with examples as disclosed herein, a memory device **100** that includes the circuit **300** may include signal development components **250-a** that are multiplexed via the selection component **280-a**, or digit lines **210-a** that are multiplexed via one or more selection components **320-a**, which may compensate for portions of an access operation or portions of access circuitry that are associated with different latencies.

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Various memory devices (e.g., memory device **100**) may include various arrangements of the circuit **300**. For example, a memory device **100** may include a set of sense components **150-b**, or a sense component **150** may otherwise include a set of sense amplifiers **290-a** and corresponding sets of multiplexed signal development components **250-a**. In one example, a memory device **100**, or portion thereof, may include 16 sense amplifiers **290-a** that are multiplexed with 1024 digit lines **210-a**, which may or may not include multiplexing via selection components **320-a**. In some examples, a set of sense amplifiers **290-a** may be included in a composite array where the set of sense amplifiers **290-a** are accessed as a single “row” of sense amplifiers of the composite array. In various examples, multiplexed digit lines **210-a** may be in the same domain **310-a** or different domains **310**. In some examples, each of the domains **310-a** may be independently controllable, and may be accessed via the same row component **125** or different row components **125**.

In some examples, one or more signal development components **250-a** of the circuit **300** may enable read broadcast operations. A read broadcast may occur from one or more locations of a memory array (e.g., in one or more domains **310-a**) to multiple locations of a set of signal development components **250-a**. In some examples, one or more multiplexers (e.g., of the selection component **280-a**) may act as a distributed routing delivery apparatus to transfer data from locations of the memory array using access lines (e.g., word lines **205-a**, digit lines **210-a**, plate lines **215-a**, etc.) coupled with the locations. For example, to initialize a set of locations in the set of signal development components **250-a** (e.g., in one or more cache blocks) to a desired pattern of data, the set of locations may be concurrently coupled with one or more multiplexers to store the pattern of data from the memory array.

FIG. 4A illustrates an example of a read operation **400** that supports read broadcast operations associated with a memory device in accordance with examples as disclosed herein. The read operation **400** may illustrate portions (e.g., time intervals) of an access operation that are associated with generating cell access signals (e.g., cell read signals, cell write signals) and latch signals when accessing a memory cell **105**. For example, the read operation **400** may be divided into a read signal development portion **410** (e.g., a cell read portion), a latch signal generation portion **420**, and a rewrite signal development portion **430** (e.g., a cell rewrite portion). The read operation **400** may employ circuitry that supports read broadcast operations, such as the circuit **300** described with reference to FIG. 3. As an illustrative example, the read operation **400** is described with reference to reading a logic state stored by the memory cell **105-b-111** of the circuit **300**, but the read operation **400** may be illustrative of operations that may be performed on any one or more of the memory cells **105-b** of the circuit **300**.

The read signal development portion **410** may be associated with a charge sharing between the memory cell **105-b-111** (e.g., a capacitive storage element of the memory cell **105-b-111**, a linear capacitor or a ferroelectric capacitor), the digit line **210-a-11** (e.g., an intrinsic capacitance **230**), and the signal development component **250-a-1**. The read signal development portion **410** may be an example of developing a signal (e.g., a signal state, a cache signal) at the signal development component **250-a-1** based at least in part on selectively coupling the signal development component **250-a-1** with the memory cell **105-b-111**. In some examples, developing the read signal at the signal development com-

ponent **250-a-1** is associated with a first latency (e.g., a relatively high latency or long duration). During the read signal development portion **410**, the signal development component **250-a-1** may be selectively decoupled from the sense amplifier **290-a**.

In some examples of the read signal development portion **410**, an access line of the signal development component **250-a-1** (e.g., the signal development line **255-a-1**) may be biased with a relatively high voltage, which may be associated with storing a relatively high voltage charge at the signal development component **250-a-1** (e.g., in a signal storage component of the signal development component **250-a-1**, such as an integrator capacitor). In some examples, such a biasing may be associated with a “plate-low” read operation where, during the read signal development portion **410**, the plate line **215-a-11** associated with the memory cell **105-b-111** being accessed is biased at a lower voltage (e.g., a ground voltage) than the digit line **210-a-1** associated with the memory cell **105-b-111**.

The read signal development portion **410** may also include selectively coupling the memory cell **105-b-111** with the signal development component **250-a-1**. In some examples, the read signal development portion **410** may include activating the word line **205-a-11** that is associated with the memory cell **105-b-111** that is being read (e.g., activating the logical signal WL_1), which may selectively couple a memory storage element (e.g., a capacitor **220**) with the respective digit line **210-a-11** (e.g., via a cell selection component **225** of the memory cell **105-b-111**). In some examples, the read signal development portion **410** may include selectively coupling the respective digit line **210-a-11** with the signal development component **250-a-1** (e.g., via selection component **320-a-1**, based on a selection signal DLM_1 , or some other switching component). Charge may accordingly be shared between the memory cell **105-b-111** and the signal development component **250-a-1**, and may settle after some time (e.g., according to a time constant behavior), with changes in voltage of the digit line **210-a-11** and the signal development line **255-a-1** that are based at least in part on the logic state stored by the memory cell **105-b-111**.

In some examples, a read signal development portion **410** may include a delay (e.g., a delay portion, a delay duration) between developing a read signal (e.g., a read signal at a signal development component **250** reaching a steady state, a read signal reaching a maximum value at a signal development component **250**) and providing the developed read signal (e.g., as maintained by the signal development component **250**) to a sense amplifier **290**. In other words, there may be a delay or inactivity period during read signal development portion **410** before initiating a latch signal generation portion **420**, which in some examples may include a decay of a developed read signal (e.g., a decay of a maintained read signal). In some examples, a circuit **300** may be configured such that a duration of such a delay or inactivity period, or an amount of decay of a developed read signal, can be tolerated while still reliably detecting a logic state stored by a memory cell **105**. In some examples, such functionality of the circuit **300** may be supported by refreshing operations of signal development components **250** that mitigate decay of developed read signals (e.g., maintaining cache signals at the signal development components **250**). These and other configurations may support signal development components **250** performing a caching function (e.g., a caching of a developed read signal or cache signal for some amount of time) in the circuit **300**.

In some examples, the charge sharing of the read signal development portion **410** may be associated with a destructive read operation (e.g., where the originally-stored logic state of the memory cell **105-b-111** is lost or otherwise degraded at the memory cell **105-b-111**), and therefore may be followed by rewrite operations (e.g., the rewrite signal development portion **430**). In some examples, a rewrite operation may not immediately follow a read signal development portion **410**, such as when stored data is transferred to a signal development component **250**, where it may be stored and further read, written, or modified. In various examples, data may be returned to a same memory cell **105** or a different memory cell **105**, which may be associated with operations that make the signal development component **250** available for other operations. In some examples, the charge sharing of the read signal development portion **410** may be associated with a non-destructive read operation (e.g., where the originally-stored logic state of the memory cell **105-b-111** is maintained at the memory cell **105-b-111**), and therefore may not be followed by rewrite operations (e.g., rewrite signal development portion **430** may be omitted).

The charge sharing of the read signal development portion **410** may be associated with a delay or latency known as a row-to-column address delay. In a DRAM application, data may be stored at a memory cell **105** as electrode charge, and may be relatively fast to respond (e.g., having a relatively low latency). In an FeRAM application, data may be stored at a memory cell **105** as a cell state in form of dipole orientation or polarization. The kinetics of such dipoles may be relatively slow (e.g., having a relatively high latency), which may lead to a longer sense time for FeRAM applications (e.g., longer than DRAM applications). Thus, in some examples (e.g., in an FeRAM application), the read signal development portion **410** may be associated with a relatively high latency or long duration (e.g., in comparison with a latch signal generation portion **420**). In some FeRAM applications, for example, the latency associated with the operations of the read signal development portion **410** may be approximately 50 nanoseconds.

In some examples of the read signal development portion **410**, the shunts **330-a** associated with other memory cells **105-b** of the domain **310-a-1**, such as shunts **330-a-12** (not shown, which may be associated with a digit line **210-a-12** or a plate line **215-a-12**) through **330-a-1r**, may be selected or activated, which may equalize a bias across memory cells **105-b** that are not being accessed (e.g., equalizing a bias between a digit line **210-a-12** and a plate line **215-a-12**, equalizing a bias between a digit line **210-a-1r** and a plate line **215-a-1r**, and so on). In FeRAM applications, for example, such an equalization of bias may prevent or reduce a loss of data (e.g., due to charge leakage) of memory cells **105-b** other than the memory cell **105-b-111** that is being accessed during the read signal development portion **410**.

The latch signal generation portion **420** may be associated with a charge sharing between the signal development component **250-a-1** and the sense amplifier **290-a**. The latch signal generation portion **420** may be an example of generating an output signal of the sense amplifier **290-a** (e.g., an amplifier component) based at least in part on the developed signal at the signal development component **250-a-1** (e.g., the cell read signal). In some examples, generating the latch signal at the sense amplifier **290-a** is associated with a second latency (e.g., a relatively low latency or short duration). The transition from the read signal development portion **410** to the latch signal generation portion **420** may

include selectively coupling the signal development component **250-a-1** with the sense amplifier **290-a**.

In some examples, selectively coupling the signal development component **250-a-1** with the sense amplifier **290-a** may include a selection via the selection component **280-a**, based on a logical selection signal SDCM. In some examples, selectively coupling the signal development component **250-a-1** with the sense amplifier **290-a** may include a selective coupling via some other switching component (e.g., an isolation switching component) between the signal development component **250-a-1** and the sense amplifier **290-a**. In some examples, the charge sharing of the latch signal generation portion **420** may be relatively rapid, and may take some fraction of the amount of time involved for the charge sharing between the memory cell **105-b-11** and the signal development component **250-a-1**. In other words, the latch signal generation portion **420** may be shorter in duration than the read signal development portion **410**. In some FeRAM applications, for example, the latency associated with the operations of the latch signal generation portion **420** may be approximately 5 to 10 nanoseconds.

In some examples, the latch signal generation portion **420** may include “firing” the sense amplifier **290-a**, which may include selectively coupling one or more voltage sources with the sense amplifier **290-a** (e.g., a low voltage source **293**, a high voltage source **294**). Thus, an output signal may be generated at the sense amplifier **290-a** that is based at least in part on the cell read signal (e.g., based at least in part on the logic state stored by the memory cell **105-b-111**). The output signal may be passed from the sense amplifier **290-a** to another component of a memory device (e.g., an input/output component **160**) via the I/O line **295** to provide an indication of the data stored by the memory cell **105-b-111**. In some examples, the output signal or some other signal associated with the generated latch signal may also be passed back to, or otherwise shared with the signal development component **250-a-1**, which in some examples may support a rewrite operation (e.g., following a destructive read operation). For example, based on the generated latch signal or output signal (e.g., based on whether the memory cell **105-b-111** stored a logic 0 or a logic 1), a rewrite signal may be passed or otherwise shared or generated with the signal development component **250-a-1** (e.g., via the signal development line **255-a-1**) as part of the latch signal generation portion **420**. In some examples, the generated latch signal or output signal may be passed back to the signal development component **250-a-1** to reinforce a charge or other signal maintained at the signal development component **250-a-1**, which may support a rewrite operation on the memory cell **105-b-111**.

In some examples of the latch signal generation portion **420**, the shunts **330-a** associated with other memory cells **105-b** of the domain **310-a-1**, such as shunts **330-a-12** (not shown, which may be associated with a digit line **210-a-12** or a plate line **215-a-12**) through **330-a-1r**, may be selected or activated, which may equalize a bias across memory cells **105-b** that are not being accessed (e.g., equalizing a bias between a digit line **210-a-12** and a plate line **215-a-12**, equalizing a bias between a digit line **210-a-1r** and a plate line **215-a-1r**, and so on). In FeRAM applications, for example, such an equalization of bias may prevent or reduce a loss of data (e.g., due to charge leakage) of memory cells **105-b** other than the memory cell **105-b-111** that is being accessed during the latch signal generation portion **420**.

The rewrite signal development portion **430** may be associated with a charge sharing between the memory cell **105-b-111**, the digit line **210-a-11**, and the signal develop-

ment component **250-a-1**. The rewrite signal development portion **430** may be an example of developing a cell access signal (e.g., a cell write signal, a cell rewrite signal) at or using the signal development component **250-a-1**. In some cases, developing a cell access signal (e.g., a cell write signal, a cell rewrite signal) at or using the signal development component **250-a-1** may be based at least in part on a latch signal of the sense amplifier **290-a** (e.g., as generated during the latch signal generation portion **420**). In some examples, a cell access signal (e.g., a cell write signal, a cell rewrite signal) at or using the signal development component **250-a-1** may be based on a charge or voltage maintained at the signal development component **250-a-1** (e.g., based at least in part on the read signal development portion **410**), where the charge or voltage maintained at the signal development component **250-a-1** may be indicative of the logic state originally stored by the memory cell **105-b-111**. In some examples, the charge or voltage maintained at the signal development component **250-a-1** may be independent of the latch signal at the sense amplifier **290-a**, or may be reinforced by the latch signal at the sense amplifier **290-a** (e.g., as reinforced during the latch signal generation portion **420**).

In some examples, developing the rewrite signal at the signal development component **250-a-1** is associated with a third latency (e.g., a relatively high latency or long duration), which may or may not be equal to the first latency. The transition from the latch signal generation portion **420** to the rewrite signal development portion **430** may include selectively decoupling or isolating the signal development component **250-a-1** from the sense amplifier **290-a** (e.g., via the selection component **280-a** or an isolation switching component). Although the rewrite signal development portion **430** may support rewriting a logic state to a memory cell **105** that has been discharged, depolarized, or otherwise destroyed or degraded in a read operation, in examples of non-destructive read operations (e.g., when **105-b-111** maintains a stored logic state after the read signal development portion **410**), the rewrite signal development portion **430** may be omitted, and the latch signal generation portion **420** may be followed by another access operation (e.g., a read operation, a write operation, a refresh operation).

In various examples, a rewrite of the memory cell **105-b-111** during the rewrite signal development portion **430** may be performed or modified based on whether a rewrite signal is generated or otherwise provided by the sense amplifier **290-a**, or based on whether a rewrite signal is generated or otherwise provided by a signal development component **250-a**. For example, a rewrite operation of the rewrite signal development portion **430** may be performed without relying on a rewrite signal of the sense amplifier **290-a**, such as when a signal development component **250-a** is configured to locally maintain a charge or other state (e.g., cache state, signal state) associated with the originally-stored logic state of the memory cell **105-b-111** until it is transferred back to the memory cell **105-b-111** (e.g., providing a local caching function as related to rewrite operations). In other words, the read signal development portion **410** or latch signal generation portion **420** may or may not be “destructive” from the perspective of a signal development component **250-a**, depending on whether the signal development component **250-a** relies on a latch signal of the sense amplifier **290-a** for rewriting the memory cell **105-b-111**. In some examples (e.g., when a signal development component **250-a** is configured to maintain a charge or other state indicative of an originally-stored logic state of the memory cell **105-b-111**), the rewrite of the memory cell

105-b-111 may occur after some delay period (e.g., of the rewrite signal development portion **430**) depending on a duration that the signal development component **250-a-1** is configured to maintain such a charge or other state, or a type of control logic that implements the write-back (e.g., first-in-first-out (FIFO), least-recently used (LRU), or others).

In some examples of a rewrite operation, the circuit **300** may be configured to couple the memory cell **105-b-111** with a high voltage source (e.g., a high voltage rail, via the signal development component **250-a-1**), which may be a direct coupling by pull-up or pull-down circuitry (e.g., a transistor or other switching component of the signal development component **250-a-1**). In some examples, the signal development component **250-a-1** may be configured with a capacitor or other charge storage component, and the latch signal generation portion **420** or the rewrite signal development portion **430** may include charging or refreshing the capacitor or other charge storage component with a charge that is sufficient to rewrite the memory cell **105-b-111** (e.g., during the rewrite signal development portion **430**). Thus, in various examples, the signal development component **250-a-1** may rewrite the logic state to the memory cell **105-b-111**, which may be performed while the signal development component **250-a-1** is selectively decoupled from the sense amplifier **290-a**, so the sense amplifier **290-a** is free to support operations with other signal development components **250-a**.

The charge sharing of the rewrite signal development portion **430** may be associated with a delay or latency known as a row precharge delay, which may include fully or partially rewriting a logic state originally stored at the memory cell **105-b-111**. For example, to rewrite a logic 0, the digit line **210-a-11** may be biased to a positive voltage (e.g., 1.5 V) and the plate line **215-a-11** may be biased to a ground or negative voltage (e.g., 0 V). To rewrite a logic 1, the digit line **210-a-11** may be biased to a ground or negative voltage (e.g., 0 V) and the plate line **215-a-11** may be biased to a positive voltage (e.g., 1.5 V). In some cases, the biasing of the digit line **210-a-11** and the plate line **215-a-11** may be based at least in part on the generated latch signal (e.g., prior to the sense amplifier **290-a** being selectively isolated from the signal development component **250-a-1**). For example, during the rewrite signal development portion **430**, the signal development component **250-a-1** or the sense amplifier **290-a** may bias the digit line **210-a-11** to either a positive voltage or a ground voltage based at least in part on the latch signal. In some cases, such a bias may be based on a charge or other state maintained at the signal development component **250-a-1**, which may be independent of a generated latch signal (e.g., as generated using the sense amplifier **290-a**).

In a DRAM application, data may be written at a memory cell **105** as electrode charge, and may be relatively fast to respond (e.g., a relatively low latency). In an FeRAM application, data may be written at a memory cell **105** as cell state in form of dipole orientation or polarization. The kinetics of such dipoles may be relatively slow (e.g., a relatively high latency), which may lead to a longer write time for FeRAM applications (e.g., longer than DRAM application). Thus, in some examples (e.g., in an FeRAM application), the rewrite signal development portion **430** may be associated with a relatively high latency or long duration (e.g., in comparison with a latch signal generation portion **420**). At the end of the rewrite signal development portion **430**, all of the digit lines **210-a-11** and all of the plate lines **215-a** of the domain **310-a-1** may be biased with a ground voltage, effectively equalizing a bias across each of

the memory cells **105-b** of the domain **310-a-11**, which may support maintaining logic states stored by the memory cells **105-b** over time.

In some examples, the shunts **330-a** associated with other memory cells **105-b** of the domain **310-a-1**, such as shunts **330-a-12** (not shown, which may be associated with a digit line **210-a-12** or a plate line **215-a-12**) through **330-a-1r**, may be selected or activated during the rewrite signal development portion **430**, which may equalize a bias across memory cells **105-b** that are not being accessed (e.g., equalizing a bias between a digit line **210-a-12** and a plate line **215-a-12**, equalizing a bias between a digit line **210-a-1r** and a plate line **215-a-1r**, and so on). Such an equalization of bias may prevent or reduce a loss of data (e.g., due to charge leakage) of memory cells **105-b** other than the memory cell **105-b-111** that is being rewritten during the rewrite signal development portion **430**.

The read operation **400** may be associated with the reading of a single memory cell **105-b-11** having a total duration of $t_{41}-t_{40}$, which includes the read signal development portion **410**, the latch signal generation portion **420**, and the rewrite signal development portion **430** for reading the single memory cell **105-b-111**. In examples where the read operation **400** does not employ multiplexed signal development techniques (e.g., a sequence of read operations **400** that use the same signal development component **250**), a subsequent read operation that employs the sense amplifier **290-a** may follow the rewrite signal development portion **430**. Thus, performing multiple read operations **400** (e.g., reading multiple memory cells **105-b**) using a same signal development component **250** may involve integer multiples of the duration $t_{41}-t_{40}$ (e.g., at least $2*(t_{41}-t_{40})$ to read two memory cells **105-b**). However, multiplexing signal development components **250-a** (e.g., via the selection component **280-a**) may reduce the amount of time involved for the sense amplifier **290-a** to read multiple memory cells **105-b**.

In some examples, the read operation **400** may be part of a read broadcast operation. A read broadcast may occur from one or more locations of a memory array (e.g., in one or more domains **310-a**) to multiple locations of a set of signal development components **250-a**. In some examples, one or more multiplexers (e.g., of the selection component **280-a**) may act as a distributed routing delivery apparatus to transfer data from locations of the memory array using access lines (e.g., word lines **205-a**, digit lines **210-a**, plate lines **215-a**, etc.) coupled with the locations. For example, to initialize a set of locations in the set of signal development components **250-a** (e.g., in one or more cache blocks) to a desired pattern of data, the set of locations may be concurrently coupled with one or more multiplexers to store the pattern of data from the memory array.

FIG. 4B illustrates an example of a read operation **450** that supports read broadcast operations associated with a memory device in accordance with examples as disclosed herein. The read operation **450** may illustrate portions (e.g., time intervals) of an access operation (e.g., a multi-cell access operation) that are associated with generating cell access signals (e.g., cell read signals, cell write signals) and latch signals when accessing four memory cells **105** (e.g., via four signal development components **250**). For example, the read operation **450** may be divided into read signal development portions **410-a**, latch signal generation portions **420-a**, and rewrite signal development portions **430-a** for each of a set of memory cells **105-b**, which may be examples of corresponding portions described with reference to FIG. 4A. The read operation **450** may employ circuitry that supports read broadcast operations, such as the

circuit 300 described with reference to FIG. 3. The read operation 450 illustrates an example of separating signal development operations from input/output operations, which may improve data throughput in a memory device.

As an illustrative example, the read operation 450 is described with reference to reading a logic state stored by four memory cells 105-b of four different domains 310-a, where each of the different domains is associated with a respective signal development component 250-a that is multiplexed with the sense amplifier 290-a. Read signal development portion 410-a-1, latch signal generation portion 420-a-1, and rewrite signal development portion 430-a-1 may refer to, for example, a read operation of memory cell 105-b-111 (e.g., of a domain 310-a-1, associated with a signal development component 250-a-1). Read signal development portion 410-a-2, latch signal generation portion 420-a-2, and rewrite signal development portion 430-a-2 may refer to, for example, a read operation of a memory cell 105-b-211 (e.g., of a domain 310-a-2, not shown, which may be associated with a signal development component 250-a-2). Read signal development portion 410-a-3, latch signal generation portion 420-a-3, and rewrite signal development portion 430-a-3 may refer to, for example, a read operation of a memory cell 105-b-311 (e.g., of a domain 310-a-3, not shown, which may be associated with a signal development component 250-a-3). Read signal development portion 410-a-4, latch signal generation portion 420-a-4, and rewrite signal development portion 430-a-4 may refer to, for example, a read operation of a memory cell 105-b-411 (e.g., of a domain 310-a-4, not shown, which may be associated with a signal development component 250-a-4). Each of the signal development components 250-a-1, 250-a-2, 250-a-3, and 250-a-4 may be selectively coupled with the same sense amplifier 290-a via a selection component 280-a (e.g., based on a logical selection signal SDCM).

Each of the read signal development portions 410-a may be associated with charge sharing between a respective memory cell 105-b, a respective digit line 210-a and a respective signal development component 250-a, which may occur during overlapping time intervals. The read signal development portions 410-a may be examples of developing a signal (e.g., a cell read signal, a cache signal, a signal state) at a signal development component 250-a of a plurality of signal development components 250-a based at least in part on selectively coupling the signal development component 250-a with a memory cell 105-b of the plurality of memory cells 105-b. The read signal development portion 410-a-1 may be an example of coupling (e.g., via the selection component 280-a, via the selection component 320-a-1), during a first time interval (e.g., and based at least in part on determining to access the memory cell 105-b-111), the memory cell 105-b-111 (e.g., a first memory cell) with the signal development component 250-a-1 (e.g., a first signal development component), and the read signal development portion 410-a-2 may be an example of coupling (e.g., via the selection component 280-a, via a selection component 320-a-2), during a second time interval that overlaps the first time interval (e.g., and based at least in part on determining to access the memory cell 105-b-211), the memory cell 105-b-211 (e.g., a second memory cell) with the signal development component 250-a-2 (e.g., a second signal development component).

Charge may accordingly be shared between the memory cell 105-b-111 and the signal development component 250-a-1, between the memory cell 105-b-211 and the signal development component 250-a-2, between the memory cell 105-b-311 and the signal development component 250-a-3,

and between the memory cell 105-b-411 and the signal development component 250-a-4. In other words, charge may be shared via the signal development components 250-a-1 through 250-a-4 during overlapping time intervals. In some examples, developing the cell read signals at the signal development components 250-a-1 through 250-a-4 is associated with a first latency (e.g., a relatively high latency or long duration).

In some examples of the read signal development portions 410-a, the shunts 330-a associated with other memory cells 105-b of the respective domain 310-a may be selected or activated, which may equalize a bias across memory cells 105-b that are not being accessed. For example, for domain 310-a-1, during the read signal development portion 410-a-1, a bias between a digit line 210-a-12 and a plate line 215-a-12 may be equalized via a shunt 330-a-12, a bias between a digit line 210-a-13 and a plate line 215-a-13 may be equalized via a shunt 330-a-13, and so on. In FeRAM applications, for example, such an equalization of bias may prevent or reduce a loss of data (e.g., due to charge leakage) of memory cells 105-b other than the memory cell 105-b that is being accessed during the respective read signal development portions 410.

The latch signal generation portions 420-a may be associated with a charge sharing between respective ones of the signal development components 250-a-1 and the sense amplifier 290-a, which may occur over non-overlapping time intervals. The latch signal generation portions 420-a may each be an example of generating an output signal of the sense amplifier 290-a based at least in part on the developed signal at the respective signal development component 250-a (e.g., based on the cell read signal, cache signal, or signal state). In some examples, generating the latch signal at the sense amplifier 290-a is associated with a second latency (e.g., a relatively low latency or short duration). The transition from a read signal development portion 410 to the corresponding latch signal generation portion 420-a may include selectively coupling the respective signal development component 250-a with the sense amplifier 290-a.

The latch signal generation portion 420-a-1 may be an example of coupling (e.g., via the selection component 280-a), during a third time interval subsequent to the first time interval, the signal development component 250-a-1 (e.g., the first signal development component) with the sense amplifier 290-a. In some examples, the third time interval may at least partially overlap the second time interval, or the third time interval may be within the second time interval. The latch signal generation portion 420-a-2 may be an example of coupling (e.g., via the selection component 280-a), during a fourth time interval subsequent to the second time interval (e.g., and subsequent to the third time interval), the signal development component 250-a-2 (e.g., the second signal development component) with the sense amplifier 290-a.

The latch signal generation portions 420-a-1 through 420-a-4 may be performed according to a sequence, which may be based at least in part on the sequence of signal development components selected or otherwise indicated by the logical selection signal SDCM. In some examples, each of the latch signal generation portions 420-a may be separated by a gap or delay period (e.g., the period between the latch signal generation portion 420-a-1 and the latch signal generation portion 420-a-2), which may be associated with a gap or delay of the selection component 280-a, a gap or delay associated with changing a value of the logical selection signal SDCM, or a period during which no signal

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development components **250-a** are coupled with the sense amplifier **290-a**. In other words, an access operation may include a gap or delay period between when one signal development component **250-a** is selectively decoupled from the sense amplifier **290-a** and another signal development component **250-a** is selectively coupled with the sense amplifier **290-a**. In other examples, such decoupling and coupling may be configured to occur simultaneously.

In some examples, the latch signal generation portions **420-a** may include “firing” the sense amplifier **290-a**, which may include selectively coupling one or more voltage sources with the sense amplifier **290-a** (e.g., a low voltage source **293**, a high voltage source **294**). Thus, according to the sequence of latch signal generation portions **420-a-1** through **420-a-4**, a sequence of output signals may be generated at the sense amplifier **290-a** that is based at least in part on the respective sequence of cell read signals (e.g., according to the sequence or read signal development portions **410-a-1** through **410-a-4**, based at least in part on the logic states stored by the accessed memory cells **105-b-111** through **105-b-411**).

The output signals may be passed from the sense amplifier **290-a** to another component of a memory device (e.g., an input/output component **160**) via the I/O line **295** to provide an indication of the data stored by the memory cells **105-b**. In some examples, the output signals or some other signals associated with the generated latch signals may also be passed back to, or otherwise shared with the signal development components **250-a-1** through **250-a-4**, which in some examples may support rewrite operations (e.g., following a destructive read operation). For example, based on the generated latch signal or output signal (e.g., based on whether the memory cells **105-b** stored a logic 0 or a logic 1), a rewrite signal may be passed or otherwise shared with the respective one of signal development components **250-a-1** through **250-a-4** as part of the latch signal generation portions **420**.

In some examples of the latch signal generation portions **420-a**, the shunts **330-a** associated with other memory cells **105-b** of the respective domain **310-a** may be selected or activated, which may equalize a bias across memory cells **105-b** that are not being accessed. For example, for domain **310-a-1**, during the latch signal generation portion **420-a-1**, a bias between a digit line **210-a-12** and a plate line **215-a-12** may be equalized via a shunt **330-a-12**, a bias between a digit line **210-a-13** and a plate line **215-a-13** may be equalized via a shunt **330-a-13**, and so on. In FeRAM applications, for example, such an equalization of bias may prevent or reduce a loss of data (e.g., due to charge leakage) of memory cells **105-b** other than the memory cell **105-b** that is being accessed during the respective latch signal generation portions **420**.

The rewrite signal development portions **430-a** may be associated with a charge sharing between the respective one of the memory cells **105-b**, the respective one of the digit lines **210-a**, and the respective one of the signal development components **250-a**. The rewrite signal development portions **430-a** may each be an example of developing a cell access signal (e.g., a cell write signal, a cell rewrite signal) at a signal development component **250-a** based at least in part on a latch signal of the sense amplifier **290-a**, or may be independent of a latch signal of the sense amplifier **290-a**. In some examples, developing the rewrite signals at the signal development components **250-a-1** is associated with a third latency (e.g., a relatively high latency or long duration), which may or may not be equal to the first latency. The transition from a latch signal generation portion **420-a** to a

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corresponding rewrite signal development portion **430-a** may include selectively isolating the respective signal development component **250-a** from the sense amplifier **290-a** (e.g., via the selection component **280-a** or another isolation switching component). Although the rewrite signal development portions **430-a** may support rewriting logic states to memory cell **105** that have been discharged, depolarized, or otherwise destroyed or degraded in a read operation, in examples of non-destructive read operations, the rewrite signal development portions **430-a** (e.g., associated with a charge sharing between a signal development component and a memory cell) may be omitted.

In some examples of the rewrite signal development portions **430-a**, the shunts **330-a** associated with other memory cells **105-b** of the respective domain **310-a** may be selected or activated, which may equalize a bias across memory cells **105-b** that are not being accessed. For example, for domain **310-a-1**, during the rewrite signal development portion **430-a-1**, a bias between a digit line **210-a-12** and a plate line **215-a-12** may be equalized via a shunt **330-a-12**, a bias between a digit line **210-a-13** and a plate line **215-a-13** may be equalized via a shunt **330-a-13**, and so on. Such an equalization of bias may prevent or reduce a loss of data (e.g., due to charge leakage) of memory cells **105-b** other than the memory cell **105-b** that is being accessed during the rewrite signal development portions **430-a**.

Like the read operation **400**, the read operation **450** may also be associated with the reading of a single memory cell **105** (e.g., via the sense amplifier **290-a**) having a total duration of $t_{A1}-t_{A0}$, which may include the read signal development portion **410-a-1**, the latch signal generation portion **420-a-1**, and the rewrite signal development portion **430-a-1** for reading the single memory cell **105-b-111**. However, by employing multiplexed signal development as disclosed herein, performing multiple read operations via the same sense amplifier **290-a** may not take an integer multiple of the duration of $t_{A1}-t_{A0}$ (e.g., where the integer multiple may correspond to the quantity of memory cells **105-b** being accessed in parallel). Rather, by generating cell access signals (e.g., cache signals, signal states) in overlapping time intervals (e.g., the time intervals of read signal development portions **410-a** or rewrite signal development portions **430-a** of the signal development component **250-a-1** that overlap with the time intervals of a read signal development portions **410-a** or rewrite signal development portions **430-a** of the signal development component **250-a-2**, and so on), the multiple memory cells **105-b** may be read in a shorter time than such an integer multiple. In other words, in accordance with the described techniques for multiplexed signal development, the sense amplifier **290-a** may support reading the four memory cells **105-b** in a duration of $t_{A3}-t_{A2}$, a duration which may be shorter than $4*(t_{A1}-t_{A0})$ (e.g., shorter than the corresponding integer multiple of a duration for reading a single memory cell **105-b**).

In one example, the rewrite signal development portions **430-a-1**, **430-a-2**, **430-a-3**, and **430-a-4** of a first set of reads may be followed by read signal development portions **410-a-5**, **410-a-6**, **410-a-7**, and **410-a-8**, respectively, of a second set of reads. The first set of reads may be associated with a first digit line index (e.g., a value of “1” as indicated by logical selection signals DLM_1 , DLM_2 , DLM_3 , and DLM_4), and the second set of reads may be associated with a second digit line index (e.g., a value of “2” as indicated by logical selection signals DLM_1 , DLM_2 , DLM_3 , and DLM_4). Or,

more generally, the first set of reads and the second set of reads may differ based at least in part on selected digit lines **210-a** of the read operations.

In some examples (e.g., where selection components **320-a** across domains **310-a** are independently controllable, where logical selection signals DLM across domains **310-a** are independently controllable), a new digit line **210-a** may be selected for a signal development component **250** (e.g., via a selection component **320-a**) as soon as a rewrite signal development portion **430** is complete for the same signal development component **250**. In other words, as illustrated in the example of operation **450**, a rewrite signal development portion **430-a** of a first set of reads may overlap in time with a read signal development portion **410-a** of a second set of reads for signal development components **250-a** that are multiplexed with the same sense amplifier **290-a** (e.g., the read signal development portion **410-a-5** overlapping the rewrite signal development portion **430-a-4**). Thus, the periodicity for reading four memory cells **105** in the example of operation **450** where domains **310-a-1** through **310-a-4** are independently controllable may be illustrated by the time $t_{A3}-t_{A2}$, which in some examples may be equal or nearly equal to the time $t_{A1}-t_{A0}$, or $t_{A1}-t_{A0}$ plus some delay or gap period (e.g., associated with the selection of a new digit line **210-a** via a selection component **320-a**), or some other duration that is based on the overall duration associated with a read operation (e.g., $t_{A1}-t_{A0}$), the respective latencies of sub-operations (e.g., relative durations of read signal development portions **410**, latch signal generation portions **420**, rewrite signal development portions **430**), and the degree of multiplexing (e.g., a quantity of signal development components **250-a** that are multiplexed with the sense amplifier **290-a**).

In some examples, a subsequent read may be performed on a memory cell **105-b** that is coupled with a different digit line **210-a** than a preceding read operation, but is coupled with a same activated word line **205-a**, which may reduce latency. For example, maintaining a selected word line **205-a** may eliminate a word line deselection operation and a subsequent word line selection operation. Such examples may be accompanied by shunting a digit line **210-a** associated with the earlier read operation (e.g., a digit line **210-a** that was previously un-shunted), and un-shunting a digit line **210-a** associated with the later read operation (e.g., a digit line **210-a** that was shunted during the earlier write operation).

In another example, not shown, a set of reads may be associated with a first common word line (e.g., where logical word lines WL_{11} , WL_{21} , WL_{31} , and WL_{41} are simultaneously activated), and a second set of reads may be associated with a second common word line (e.g., where logical word lines WL_{12} , WL_{22} , WL_{32} , and WL_{42} are simultaneously activated). Or, more generally, the first set of reads and the second set of reads may differ based at least in part on a selected common word line **205-a** of the read operations. In some examples (e.g., where word lines **205-a** across domains **310-a** are not independently controllable), a new word line **205-a** may be selected as soon as a latch signal generation portion **420** is complete or a rewrite signal development portion **430** is complete for all of the multiplexed signal development components **250-a** (e.g., associated with the sense amplifier **290-a**, or other set of domains **310-a** that are not independently controllable). In other words, in some examples, a latch signal generation portion **420** or a rewrite signal development portion **430** of a first set of reads may not overlap in time with a read signal devel-

opment portion **410** of a second set of reads for signal development components multiplexed with the same sense amplifier **290-a**.

For example, when word lines **205-a** are not independently controllable across domains **310-a-1** through **310-a-4**, the read signal development portion **410-a-5** may follow or be otherwise subsequent to the rewrite signal development portion **430-a-4**. Thus, the periodicity for reading four memory cells **105** in the example where the domains **310-a** are not independently controllable may be equal to or nearly equal to the combined time of one read signal development portion **410-a**, each of the latch signal generation portions **420-a-1** through **420-a-4** for the multiplexed signal development components **250-a-1** through **250-a-4**, and one rewrite signal development portion **430-a**, plus any relevant delay or gap periods (e.g., associated with the selection of a new word line **205-a**, or the selection of new signal development components **250-a** via a selection component **280-a**). Accordingly, in some examples, such a periodicity where domains **310-a** are not independently controllable may be longer than the periodicity illustrated by time $t_{A2}-t_{A0}$.

Thus, in accordance with various examples as disclosed herein, the advantages provided by the described signal development multiplexing (e.g., a reduced latency when accessing multiple memory cells **105-b** in parallel) may scale with the relative difference in latency (e.g., durations) of read signal development portions **410**, latch signal generation portions **420**, and rewrite signal development portions **430**. The advantages provided by the described signal development multiplexing may also depend on whether domains **310-a** are configured to be independently controllable, or are controlled via common access lines or common logical signals.

Although the techniques of read operation **450** are described with reference to a single sense amplifier **290-a**, the techniques of read operation **450** may be repeated for each sense amplifier **290** of a sense amplifier array, including various operations being performed concurrently (e.g., in parallel, with simultaneous or offset initiation or triggering), to support further pipelining of read operations in a memory device **100**. For example, the read operation **450**, or another read operation performed concurrently with or offset from the read operation **450**, may include signal development operations including read signal development portions **410-b-1**, **410-b-2**, **410-b-3**, and **410-b-4** (not shown) associated with a different sense amplifier **290** (e.g., of a same sense amplifier array). In some examples, a read signal development portion **410-b-1** may be initiated at the same time as, or otherwise performed concurrently with or offset from, the read signal development portion **410-a-1** (e.g., according to a simultaneous accessing of multiple memory cells of a row, a domain, or a subdomain, according to concurrent signal exchange with a cacheline). Likewise, a read signal development portion **410-b-2** may be initiated at the same time as, or otherwise performed concurrently with or offset from, the read signal development portion **410-a-2**, and so on.

Further, the read operation **450**, or another read operation performed concurrently with the read operation **450**, may include input/output operations including latch signal generation portions **420-b-1**, **420-b-2**, **420-b-3**, and **420-b-4** (not shown) associated with a different sense amplifier **290** (e.g., of a same sense amplifier array). In some examples, a latch signal generation portion **420-b-1** may be initiated at the same time as, or otherwise performed concurrently with or offset from, the latch signal generation portion **420-a-1** (e.g., according to a simultaneous sensing at a sense amplifier array, according to a simultaneous latching at a set of latches

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of a sense component or I/O component, according to concurrent signal exchange with a cacheline). Likewise, a latch signal generation portion **420-b-2** may be initiated at the same time as, or otherwise performed concurrently with or offset from, the latch signal generation portion **420-a-2**, and so on. Although described in the context of two parallel reads associated with two different sense amplifiers **290**, the described techniques may be applied to any quantity of parallel reads. For example, to support a 64-bit information transfer scheme, 64 parallel reads may be performed using 64 sense amplifiers **290** in accordance with examples as disclosed herein.

In some examples, the read operation **450** may be part of a read broadcast operation. A read broadcast may occur from one or more locations of a memory array (e.g., in one or more domains **310-a**) to multiple locations of a set of signal development components **250-a**. In some examples, one or more multiplexers (e.g., of the selection component **280-a**) may act as a distributed routing delivery apparatus to transfer data from locations of the memory array using access lines (e.g., word lines **205-a**, digit lines **210-a**, plate lines **215-a**, etc.) coupled with the locations. For example, to initialize a set of locations in the set of signal development components **250-a** (e.g., in one or more cache blocks) to a desired pattern of data, the set of locations may be concurrently coupled with one or more multiplexers to store the pattern of data from the memory array.

FIG. 5A illustrates an example of a write operation **500** that supports read broadcast operations associated with a memory device in accordance with examples as disclosed herein. The write operation **500** may illustrate portions (e.g., time intervals) of an access operation that are associated with generating latch signals and cell access signals (e.g., cell write signals) when accessing a memory cell **105**. For example, the write operation **500** may be divided into a latch signal generation portion **510** and a write signal development portion **520** (e.g., a cell write portion). The write operation **500** may employ circuitry that supports read broadcast operations, such as the circuit **300** described with reference to FIG. 3. As an illustrative example, the write operation **500** is described with reference to writing a logic state to the memory cell **105-b-111** of the circuit **300**, but the write operation **500** may be illustrative of operations that may be performed on any one or more of the memory cells **105-b** of the circuit **300**.

The latch signal generation portion **510** may be associated with a charge sharing between the signal development component **250-a-1** and the sense amplifier **290-a**. The latch signal generation portion **510** may be an example of generating a latch signal at the sense amplifier **290-a** or the signal development component **250-a-1** (e.g., a cache signal, a signal state) based at least in part on a write command or write signal (e.g., from an input/output component **160** or a memory controller **170**) received via I/O line **295-a**. In some examples, generating the latch signal at the sense amplifier **290-a** or the signal development component **250-a-1** is associated with a fourth latency (e.g., a relatively low latency or short duration), which may be the same as or different than the second latency of the latch signal generation portions **420** described with reference to read operations **400** and **450**.

The latch signal generation portion **510** may include selectively coupling the signal development component **250-a-1** with the sense amplifier **290-a** (e.g., at the beginning of the latch signal generation portion **510**, or at another time after other operations of the latch signal generation portion **510** such as after receiving a write command or write

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signal via I/O line **295-a**). In some examples, selectively coupling the signal development component **250-a-1** with the sense amplifier **290-a** may include a selection via the selection component **280-a**, based on a logical selection signal SDCM. In some examples, selectively coupling the signal development component **250-a-1** with the sense amplifier **290-a** may include a selective coupling via some other switching component (e.g., an isolation switching component) between the signal development component **250-a-1** and the sense amplifier **290-a**.

In some examples, the latch signal generation portion **510** may include “firing” the sense amplifier **290-a**, which may include selectively coupling one or more voltage sources with the sense amplifier **290-a** (e.g., a low voltage source **293**, a high voltage source **294**). Thus, a latch signal may be generated at the sense amplifier **290-a** that is based at least in part on a write command or write signal (e.g., received via the I/O line **295-a**). The generated latch signal or some other signal associated with the generated latch signal may be passed to, or otherwise shared with the signal development component **250-a-1** (e.g., storing a cache signal or signal state at a cache element of the signal development component **250-a-1**) to support the writing of the memory cell **105-b-111**. For example, based on the generated latch signal (e.g., based on whether the memory cell **105-b-111** is to store a logic 0 or a logic 1), a write signal may be passed or otherwise shared or generated with the signal development component **250-a-1** (e.g., via the signal development line **255-a-1**) as part of the latch signal generation portion **510**.

The write signal development portion **520** may be associated with a charge sharing between the memory cell **105-b-111**, the digit line **210-a-11**, and the signal development component **250-a-1**. The write signal development portion **520** may be an example of developing a cell access signal (e.g., a cell write signal) at or using the signal development component **250-a-1** based at least in part on a latch signal of the sense amplifier **290-a**. In some examples, developing the write signal at the signal development component **250-a-1** is associated with a fifth latency (e.g., a relatively high latency or long duration), which may or may not be equal to the third latency of the rewrite signal development portions **430** described with reference to read operations **400** and **450**. The transition from the latch signal generation portion **510** to the write signal development portion **520** may include selectively decoupling or isolating the signal development component **250-a-1** from the sense amplifier **290-a** (e.g., via the selection component **280-a** or an isolation switching component).

In some examples of a write operation, the circuit **300** may be configured to couple the memory cell **105-b-111** with a high voltage source (e.g., a high voltage rail, via the signal development component **250-a-1**), which may be a direct coupling by pull-up or pull-down circuitry (e.g., a transistor or other switching component of the signal development component **250-a-1**). In some examples, the signal development component **250-a-1** may be configured with a capacitor or other charge storage component, and the latch signal generation portion **510** or the write signal development portion **520** may include charging or refreshing the capacitor or other charge storage component with a charge that is sufficient to rewrite the memory cell **105-b-111** (e.g., during the write signal development portion **520**). Thus, in various examples, the signal development component **250-a-1** may write the logic state to the memory cell **105-b-111**, which may be performed while the signal development component **250-a-1** is selectively decoupled from the sense

amplifier 290-a, so the sense amplifier 290-a is free to support operations with other signal development components 250-a.

The charge sharing of the write signal development portion 520 may also be associated with a delay or latency known as a row precharge delay, which may include writing a logic state to the memory cell 105-b-111 based on a write command. For example, to write a logic 0, the digit line 210-a-11 may be biased to a positive voltage (e.g., 1.5 V) and the plate line 215-a-11 may be biased to a ground or negative voltage (e.g., 0 V). To write a logic 1, the digit line 210-a-11 may be biased to a ground or negative voltage (e.g., 0 V) and the plate line 215-a-11 may be biased to a positive voltage (e.g., 1.5 V). The biasing of the digit line 210-a-11 and the plate line 215-a-11 may be based at least in part on the generated latch signal (e.g., prior to the sense amplifier 290-a being selectively isolated from the signal development component 250-a-1). For example, during the write signal development portion 520, the signal development component 250-a-1 may bias the digit line 210-a-11 to either a positive voltage or a ground voltage based at least in part on the latch signal (e.g., based at least in part on a write command). At the end of the write signal development portion 520, all of the digit lines 210-a-11 and all of the plate lines 215-a of the domain 310-a-1 may be biased with a ground voltage, effectively equalizing a bias across each of the memory cells 105-b of the domain 310-a-11, which may support maintaining logic states stored by the memory cells 105-b over time.

In some examples, the shunts 330-a associated with other memory cells 105-b of the domain 310-a-1, such as shunts 330-a-12 through 330-a-1r, may be selected or activated during the write signal development portion 520, which may equalize a bias across memory cells 105-b that are not being accessed (e.g., equalizing a bias between a digit line 210-a-12 and a plate line 215-a-12, equalizing a bias between a digit line 210-a-1r and a plate line 215-a-1r, and so on). Such an equalization of bias may prevent or reduce a loss of data (e.g., due to charge leakage) of memory cells 105-b other than the memory cell 105-b-111 that is being written during the write signal development portion 520.

The write operation 500 may be associated with the writing of a single memory cell 105-b-11 having a total duration of $t_{B1}-t_{B0}$, which includes the latch signal generation portion 510, and the write signal development portion 520 for writing the single memory cell 105-b-111. In examples where the write operation 500 does not employ multiplexed signal development techniques (e.g., a sequence of write operations 500 that use the same signal development component 250), a subsequent write operation that employs the sense amplifier 290-a may follow the write signal development portion 520. Thus, performing multiple write operations 500 (e.g., writing multiple memory cells 105-b) using a same signal development component 250 may involve integer multiples of the duration $t_{B1}-t_{B0}$ (e.g., at least $2^*(t_{B1}-t_{B0})$ to read two memory cells 105-b). However, multiplexing signal development components 250-a (e.g., via the selection component 280-a) may reduce the amount of time involved for the sense amplifier 290-a to write multiple memory cells 105-b.

In some examples, the write operation 500 may follow a read broadcast operation (e.g., as part of a read-write operation, a read-modify-write operation, etc.), where the write operation may be used to write back data to the multiple memory cells 105-b of the circuit 300 following the read broadcast operation. A read broadcast may occur from one or more locations of a memory array (e.g., in one or more

domains 310-a) to multiple locations of a set of signal development components 250-a. In some examples, one or more multiplexers (e.g., of the selection component 280-a) may act as a distributed routing delivery apparatus to transfer data from locations of the memory array using access lines (e.g., word lines 205-a, digit lines 210-a, plate lines 215-a, etc.) coupled with the locations. For example, to initialize a set of locations in the set of signal development components 250-a (e.g., in one or more cache blocks) to a desired pattern of data, the set of locations may be concurrently coupled with one or more multiplexers to store the pattern of data from the memory array.

FIG. 5B illustrates an example of a write operation 550 that supports read broadcast operations associated with a memory device in accordance with examples as disclosed herein. The write operation 550 may illustrate portions (e.g., time intervals) of an access operation (e.g., a multi-cell access operation) that are associated with generating latch signals and cell access signals (e.g., cell write signals) when accessing four memory cells 105 (e.g., via four signal development components 250). For example, the write operation 550 may be divided into latch signal generation portions 510-a and write signal development portions 520-a for each of a set of memory cells 105-b, which may be examples of corresponding portions described with reference to FIG. 5A. The write operation 550 may employ circuitry that supports read broadcast operations, such as the circuit 300 described with reference to FIG. 3. The write operation 550 illustrates an example of separating signal development operations from input/output operations, which may improve data throughput in a memory device.

As an illustrative example, the write operation 550 is described with reference to writing a logic state to four memory cells 105-b of four different domains 310-a, where each of the different domains is associated with a respective signal development component 250-a that is multiplexed with the sense amplifier 290-a. Latch signal generation portion 510-a-1 and write signal development portion 520-a-1 may refer to, for example, a write operation of memory cell 105-b-111 (e.g., of a domain 310-a-1, associated with a signal development component 250-a-1). Latch signal generation portion 510-a-2 and write signal development portion 520-a-2 may refer to, for example, a write operation of a memory cell 105-b-211 (e.g., of a domain 310-a-2, not shown, associated with a signal development component 250-a-2). Latch signal generation portion 510-a-3 and write signal development portion 520-a-3 may refer to, for example, a write operation of a memory cell 105-b-311 (e.g., of a domain 310-a-3, not shown, associated with a signal development component 250-a-3). Latch signal generation portion 510-a-4 and write signal development portion 520-a-4 may refer to, for example, a write operation of a memory cell 105-b-411 (e.g., of a domain 310-a-4, not shown, associated with a signal development component 250-a-4). Each of the signal development components 250-a-1, 250-a-2, 250-a-3, and 250-a-4 may be selectively coupled with a same sense amplifier 290-a via a selection component 280-a (e.g., based on a logical selection signal SDCM).

Each of the latch signal generation portions 510-a may be associated with a charge sharing between respective ones of the signal development components 250-a-1 and the sense amplifier 290-a, which may occur over non-overlapping time intervals. The latch signal generation portions 510-a may each be an example of generating a signal (e.g., a cache signal, a signal state) at a signal development component 250-a based at least in part on selectively coupling the signal development component 250-a with the sense amplifier

290-a (e.g., an amplifier component). In some examples, such a signal may be generated based at least in part on a write command or write signal. In some examples, generating a latch signal, cache signal, or signal state is associated with a fourth latency (e.g., a relatively low latency or short duration).

The latch signal generation portion **510-a-1** may be an example of coupling (e.g., via the selection component **280-a**), during a first time interval and based at least in part on determining to access the memory cell **105-b-111** (e.g., a first memory cell), the signal development component **250-a-1** (e.g., a first signal development component) with the sense amplifier **290-a** (e.g., an amplifier component). The latch signal generation portion **510-a-2** may be an example of coupling (e.g., via the selection component **280-a**), during a second time interval subsequent to the first time interval and based at least in part on determining to access the memory cell **105-b-211** (e.g., a second memory cell), the signal development component **250-a-2** (e.g., a second signal development component) with the sense amplifier **290-a**.

The latch signal generation portions **510-a-1** through **510-a-4** may be performed according to a sequence, which may be based at least in part on a sequence of memory cell write commands or signals (e.g., as received via I/O line **295-a**). Such a sequence may also correspond to the sequence of signal development components **250-a** selected or otherwise indicated by the logical selection signal SDCM. In some examples, each of the latch signal generation portions **510-a** may be separated by a gap or delay period (e.g., the period between the latch signal generation portion **510-a-1** and the latch signal generation portion **510-a-2**), which may be associated with a gap or delay of the selection component **280-a**, a gap or delay associated with changing a value of the logical selection signal SDCM, or a period during which no signal development components **250-a** are coupled with the sense amplifier **290-a**. In other words, an access operation may include a gap or delay period between when one signal development component **250-a** is selectively decoupled from the sense amplifier **290-a** and another signal development component **250-a** is selectively coupled with the sense amplifier **290-a**. In other examples, such decoupling and coupling may be configured to occur simultaneously.

In some examples, the latch signal generation portions **510-a** may include “firing” the sense amplifier **290-a**, which may include selectively coupling one or more voltage sources with the sense amplifier **290-a** (e.g., a low voltage source **293**, a high voltage source **294**). Thus, according to the sequence of latch signal generation portions **510-a-1** through **510-a-4**, a sequence of signals may be generated at the sense amplifier **290-a** or signal development components **250-a** that is based at least in part on the respective sequence of write commands or signals.

One or more signals may be transferred between a sense amplifier **290** and a signal development component **250** as part of or in connection with a write operation. For example, the generated latch signals may also be passed back to, or otherwise shared with the signal development components **250-a-1** through **250-a-4** to support the respective write operations. For example, based on the generated latch signal (e.g., based on whether the memory cells **105-b** are to store a logic 0 or a logic 1), a write signal may be passed or otherwise shared with the respective one of signal development components **250-a-1** through **250-a-4** as part of the latch signal generation portions **510-a**.

The write signal development portions **520-a** may be associated with a charge sharing between a respective one of the memory cells **105-b**, a respective one of the digit lines **210-a**, and a respective one of the signal development components **250-a**. The write signal development portions **520-a** may each be an example of developing a cell access signal (e.g., a cell write signal) at a signal development component **250-a** based at least in part on a latch signal of the sense amplifier **290-a**. The transition from a latch signal generation portion **510** to a corresponding write signal development portion **520-a** may include selectively isolating the respective signal development component **250-a** from the sense amplifier **290-a** (e.g., via the selection component **280-a** or another isolation switching component). The write signal development portion **520-a-1** may be an example of coupling, during a third time interval subsequent to the first time interval, the signal development component **250-a-1** (e.g., the first signal development component) with the memory cell **105-b-111** (e.g., the first memory cell). In some examples, the second time interval is within, or at least partially overlaps the third time interval. The write signal development portion **520-a-2** may be an example of coupling, during a fourth time interval subsequent to the second time interval that overlaps the third time interval, the signal development component **250-a-2** (e.g., the second signal development component) with the memory cell **105-b-211** (e.g., the second memory cell).

In some examples of the write signal development portions **520-a**, the shunts **330-a** associated with other memory cells **105-b** of the respective domain **310-a** may be selected or activated, which may equalize a bias across memory cells **105-b** that are not being accessed. For example, for domain **310-a-1**, during the write signal development portion **520-a-1**, a bias between a digit line **210-a-12** and a plate line **215-a-12** may be equalized via a shunt **330-a-12**, a bias between a digit line **210-a-13** and a plate line **215-a-13** may be equalized via a shunt **330-a-13**, and so on. Such an equalization of bias may prevent or reduce a loss of data (e.g., due to charge leakage) of memory cells **105-b** other than the memory cell **105-b** that is being accessed during the write signal development portions **520-a**.

Like the write operation **500**, the write operation **550** may also be associated with the writing of a single memory cell **105** (e.g., via the sense amplifier **290-a**) having a total duration of $t_{B1}-t_{B0}$, which may include the latch signal generation portion **510-a-1** and the write signal development portion **520-a-1** for writing the single memory cell **105-b-111**. However, by employing multiplexed signal development in accordance with examples as disclosed herein, performing multiple write operations via the same sense amplifier **290-a** may not take an integer multiple of the duration of $t_{B1}-t_{B0}$ (e.g., where the integer multiple may correspond to the quantity of memory cells **105-b** being written in parallel). Rather, by generating cell access signals in overlapping time intervals (e.g., the time intervals of a write signal development portions **520-a** of the signal development component **250-a-1** that overlap with the time intervals of a write signal development portions **520-a** of the signal development component **250-a-2**, and so on), the multiple memory cells **105-b** may be written in a shorter time than such an integer multiple. In other words, in accordance with the described techniques for multiplexed signal development, the sense amplifier **290-a** may support writing the four memory cells **105-b** in a duration of $t_{B2}-t_{B0}$, a duration which may be shorter than $4*(t_{B1}-t_{B0})$ (e.g., shorter than the corresponding integer multiple of duration for writing a single memory cell **105-b**).

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In one example, the write signal development portions **520-a-1**, **520-a-2**, **520-a-3**, and **520-a-4** of a first set of writes may be followed by latch signal generation portions **510-a-5**, **510-a-6**, **510-a-7**, and **510-a-8**, respectively, of a second set of writes. The first set of writes may be associated with a first digit line index (e.g., a value of “1” as indicated by logical selection signals DLM_1 , DLM_2 , DLM_3 , and DLM_4), and the second set of writes may be associated with a second digit line index (e.g., a value of “2” as indicated by logical selection signals DLM_1 , DLM_2 , DLM_3 , and DLM_4). Or, more generally, the first set of writes and the second set of writes may differ based at least in part on selected digit lines **210-a** of the write operations. In some examples (e.g., where selection components **320-a** across domains **310-a** are independently controllable, where logical selection signals DLM across domains **310-a** are independently controllable), a new digit line **210-a** may be selected for a signal development component **250** (e.g., via a selection component **320-a**) as soon as a write signal development portion **520-a** is complete for the same signal development component **250**. In other words, as illustrated in the example of operation **550**, a write signal development portion **520-a** of a first set of writes may overlap in time with a latch signal generation portion **510-a** of a second set of writes for signal development components **250-a** that are multiplexed with the same sense amplifier **290-a** (e.g., the latch signal generation portion **510-a-5** overlapping the write signal development portion **520-a-4**). Thus, the periodicity for writing four memory cells **105** in the example of operation **550** where domains **310-a-1** through **310-a-4** are independently controllable may be illustrated by the time $t_{B2} - t_{B0}$, which may be based on the overall duration associated with a write operation (e.g., $t_{B1} - t_{B0}$), the respective latencies of sub-operations (e.g., relative durations of latch signal generation portions **510-a** and write signal development portions **520-a**), and the degree of multiplexing (e.g., a quantity of signal development components **250-a** that are multiplexed with the sense amplifier **290-a**).

In some examples, a subsequent write may be performed on a memory cell **105-b** that is coupled with a different digit line **210-a** than a preceding write operation, but is coupled with a same activated word line **205-a**, which may reduce latency. For example, maintaining a selected word **205-a** line may eliminate a word line deselection operation and a subsequent word line selection operation. Such examples may be accompanied by shunting a digit line **210-a** associated with the earlier write operation (e.g., a digit line **210-a** that was previously un-shunted), and un-shunting a digit line **210-a** associated with the later write operation (e.g., a digit line **210-a** that was shunted during the earlier write operation).

In another example, not shown, a set of writes may be associated with a first common word line (e.g., where logical word lines WL_{11} , WL_{21} , WL_{31} , and WL_{41} of different domains are simultaneously activated), and a second set of writes may be associated with a second common word line (e.g., where logical word lines WL_{12} , WL_{22} , WL_{32} , and WL_{42} of different domains are simultaneously activated). Or, more generally, the first set of writes and the second set of writes may differ based at least in part on a selected common word line **205-a** of the write operations. In some examples (e.g., where word lines **205-a** across domains **310-a** are not independently controllable), a new word line **205-a** may be selected as soon as a write signal development portion **520** is complete for all of the multiplexed signal development components **250-a** (e.g., associated with the sense amplifier **290-a**, or other set of domains **310-a** that are

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not independently controllable). In other words, in some examples, a write signal development portion **520** of a first set of writes may not overlap in time with a latch signal generation portion **510** of a second set of writes for signal development components **250** that are multiplexed with the same sense amplifier **290-a**.

For example, when word lines **205-a** are not independently controllable across domains **310-a-1** through **310-a-4**, the latch signal generation portion **510-a-5** may follow or be otherwise subsequent to the write signal development portion **520-a-4**. Thus, the periodicity for writing four memory cells **105** in the example where the domains **310-a** are not independently controllable may be equal to or nearly equal to the combined time of each of the latch signal generation portions **510-a-1** through **510-a-4** and one of the write signal development portions **520-a** for the multiplexed signal development components **250-a-1** through **250-a-4**. Accordingly, in some examples, such a periodicity where domains **310-a** are not independently controllable may be longer than the periodicity illustrated by time $t_{B2} - t_{B0}$.

Thus, in accordance with various examples as disclosed herein, the advantages provided by the described signal development multiplexing (e.g., a reduced latency when accessing multiple memory cells **105-b** in parallel) may scale with the relative difference in latency (e.g., durations) of latch signal generation portions **510** and write signal development portions **520**. The advantages of the described signal development multiplexing may also depend on whether domains **310-a** are configured to be independently controllable, or are controlled via common access lines or common logical signals.

Although the techniques of write operation **550** are described with reference to a single sense amplifier **290-a**, the techniques of write operation **550** may be repeated for each sense amplifier **290** of a sense amplifier array, including various operations being performed concurrently (e.g., in parallel, with simultaneous or offset initiation or triggering), to support further pipelining of write operations in a memory device **100**. For example, the write operation **550**, or another write operation performed concurrently with the write operation **550**, may include input/output operations including latch signal generation portions **510-b-1**, **510-b-2**, **510-b-3**, and **510-b-4** (not shown) associated with a different sense amplifier (e.g., of a same sense amplifier array). In some examples, a latch signal generation portion **510-b-1** may be initiated at the same time as, or otherwise performed concurrently with or offset from, the latch signal generation portion **510-a-1** (e.g., according to a simultaneous sensing at a sense amplifier array, according to a simultaneous latching at a set of latches of a sense component or I/O component, according to concurrent signal exchange with a cacheline). Likewise, a latch signal generation portion **510-b-2** may be initiated at the same time as, or otherwise performed concurrently with or offset from, the latch signal generation portion **510-a-2**, and so on.

Further, the write operation **550**, or another write operation performed concurrently with or offset from the write operation **550**, may include signal development operations including write signal development portions **520-b-1**, **520-b-2**, **520-b-3**, and **520-b-4** (not shown) associated with a different sense amplifier (e.g., of a same sense amplifier array). In some examples, a write signal development portion **520-b-1** may be initiated at the same time as, or otherwise performed concurrently with or offset from, the write signal development portion **520-a-1** (e.g., according to a simultaneous accessing of multiple memory cells of a row, a domain, or a subdomain, according to concurrent signal

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exchange with a cacheline). Likewise, a write signal development portion **520-b-2** may be initiated at the same time as, or otherwise performed concurrently with or offset from, the write signal development portion **520-a-2**, and so on. Although described in the context of two parallel writes associated with two different sense amplifiers **290**, the described techniques may be applied to any quantity of parallel writes. For example, to support a 64-bit information transfer scheme, 64 parallel writes may be performed using 64 sense amplifiers **290** in accordance with examples as disclosed herein.

In some examples, the write operation **550** may follow a read broadcast operation (e.g., as part of a read-write operation, a read-modify-write operation, etc.), where the write operation may be used to write back data to the multiple memory cells **105-b** of the circuit **300** following the read broadcast operation. A read broadcast may occur from one or more locations of a memory array (e.g., in one or more domains **310-a**) to multiple locations of a set of signal development components **250-a**. In some examples, one or more multiplexers (e.g., of the selection component **280-a**) may act as a distributed routing delivery apparatus to transfer data from locations of the memory array using access lines (e.g., word lines **205-a**, digit lines **210-a**, plate lines **215-a**, etc.) coupled with the locations. For example, to initialize a set of locations in the set of signal development components **250-a** (e.g., in one or more cache blocks) to a desired pattern of data, the set of locations may be concurrently coupled with one or more multiplexers to store the pattern of data from the memory array.

FIG. 6 illustrates an example of a signal development component **250-b** that supports read broadcast operations associated with a memory device in accordance with examples as disclosed herein. The signal development component **250-b** may be an example of signal development components **250** described with reference to FIGS. 1 through 5. The signal development component **250-b** may be coupled with or between a digit line **210-b** and a signal development line **255-b**. The signal development component **250-b** may include a capacitor **610** (e.g., an integrator capacitor, a storage element, a cache element, a cache storage element) and a transistor **620** that may be configured in an amplifier configuration (e.g., as a charge transfer sensing amplifier, as a cascode).

The capacitor **610** may be an example of a signal storage component or a charge storage component of the signal development component **250-b**. In the example of the signal development component **250-b**, the capacitor **610** may be coupled with or between a line of the signal development component **250-b** (e.g., the signal development line **255-b**) and a voltage source **615** (e.g., a ground voltage source, a voltage source having a reference voltage for the capacitor **610**). Although illustrated as including the capacitor **610**, a signal development component **250** in accordance with examples as disclosed herein may, additionally or alternatively, include or otherwise employ a transistor in a particular state, a diode, or other components that may provide functionality of a signal storage component or charge storage component in the signal development component **250**. In some examples, a set of signal development components **250-b** may include a set of capacitors **610**, which may provide a fast, local, in-memory cache (e.g., a signal development cache) in a device that includes the set of signal development components **250-b**.

In some examples, a memory device that includes the signal development component **250-b** may include memory cells **105** that employ a logic storage element that includes

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a capacitive element (e.g., a linear capacitor in a DRAM application, a ferroelectric capacitor in an FeRAM application). In various examples, the capacitor **610** may include a same capacitive element or technology as a logic storage element (e.g., capacitor **610** may be a linear capacitor in a DRAM application, a capacitor **610** may be a ferroelectric capacitor in an FeRAM application), or a different capacitive element or technology as a logic storage element (e.g., capacitor **610** may be a linear capacitor in an FeRAM application, a PCM application, or a chalcogenide memory application).

The transistor **620** may be an example of an amplifier or voltage regulator of the signal development component **250-b**, and may be configured to transfer charge between the signal development line **255-b** (e.g., a first access line) and the digit line **210-b** (e.g., a second access line) based at least in part on one or both of a voltage of the signal development line **255-b** and a voltage of the digit line **210-b**. For example, a gate node of the transistor **620** may be coupled with a voltage source **625**, and charge may be transferred across the transistor based at least in part on a relationship between a voltage of the voltage source **625** (e.g., V_2) and a voltage of the digit line **210-b**. In various examples, the transistor **620** may be associated with one or more digit lines **210** (e.g., multiplexed digit lines **210**), and may be located outside the illustrative boundaries of the signal development component **250-b** (e.g., in examples of memory devices that include a transistor **620** for each of a set of multiplexed digit lines **210**).

The transistor **620** may provide a conversion of signals between the digit line **210-b** and the signal development line **255-b**. For example, the transistor **620** may permit a flow of charge (e.g., electrical current) from the signal development line **255-b** (e.g., from the capacitor **610**) to the digit line **210-b**, as fed or enabled by the voltage source **625**, upon a reduction in voltage of the digit line **210-b** (e.g., upon selection of a memory cell **105**, upon selection of a digit line **210** via a selection component **320**). A relatively small flow of charge to the digit line **210-b** may be associated with a relatively small change in voltage of the signal development line **255-b**, whereas a relatively large flow of charge to the digit line **210-b** may be associated with a relatively large change in voltage of the signal development line **255-b**. According to the net capacitance of the signal development line **255-b** (e.g., including the capacitor **610**), for example, the signal development line **255-b** may undergo a relatively small change in voltage or a relatively large change in voltage depending on the flow of charge across the transistor **620** after selecting a memory cell **105**. In some examples, the transistor **620** or the signal development component **250-b** may be isolated from the digit line **210-b** by a switching component or a selection component (e.g., a selection component **320**). The transistor **620** may also be referred to as a “voltage regulator” or a “bias component,” relating to how the transistor **620** regulates a flow of charge in response to the voltage of the digit line **210-b**.

In some examples, the signal development component **250-b** may include circuitry configured to support a selective coupling (e.g., of the signal development line **255-b**) with a relatively high voltage (e.g., voltage source **635**). For example, the signal development component **250-b** may include a switching component **630** that is operable based on a logical signal SW_i . In some examples, the voltage source **645** may be coupled with a relatively high voltage rail or supply, which may support charging the capacitor **610** (e.g., for developing a cell access signal).

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In some examples, the signal development component **250-b** may include circuitry configured to support a selective coupling (e.g., of the digit line **210-b**) with a reference voltage (e.g., voltage source **645**). For example, the signal development component **250-b** may include a switching component **640** that is operable based on a logical signal SW_2 . In some examples, the voltage source **645** may be coupled with a ground or virtual ground rail or supply. In some examples, the voltage source **645** may be coupled with a same rail or supply as the voltage source **615** (e.g., V_1 may be equal to V_4).

In some examples, the signal development component **250-b** may include circuitry configured to support a selective coupling (e.g., of the signal development line **255-b**, of the signal development component **250-b**) with another component (e.g., a selection component **280**, a sense amplifier **290**). For example, the signal development component **250-b** may include a switching component **650**, which may be referred to as an isolation switching component, and may be operable based on a logical signal ISO . Additionally or alternatively, an isolation switching component may be included in a sense amplifier **290** in accordance with examples as disclosed herein.

In some examples, the signal development component **250-b** may enable read broadcast operations. A read broadcast may occur from one or more locations of a memory array (e.g., in one or more domains) to multiple locations of a set of signal development components **250-b**. In some examples, one or more multiplexers (e.g., of the selection component **280**) may act as a distributed routing delivery apparatus to transfer data from locations of the memory array using access lines (e.g., word lines **205**, digit lines **210**, plate lines **215**, etc.) coupled with the locations. For example, to initialize a set of locations in the set of signal development components **250-b** (e.g., in one or more cache blocks) to a desired pattern of data, the set of locations may be concurrently coupled with one or more multiplexers to store the pattern of data from the memory array.

FIG. 7 illustrates an example of a sense amplifier **290-b** that supports read broadcast operations associated with a memory device in accordance with examples as disclosed herein. The sense amplifier **290-b** may be an example of sense amplifiers **290** described with reference to FIGS. 1 through 5. The sense amplifier **290-b** may be coupled with or between a signal line **285-b** and a reference line **275-b**. The sense amplifier **290-b** may also be associated with (e.g., coupled with) I/O lines **295-b** and **295-c**. In some examples, the sense amplifier **290-b** may be referred to as an amplifier component of a memory device.

The sense amplifier **290-b** may include a pair of opposed amplifiers **710-a** and **710-b**. Although illustrated as amplifiers **710**, the sense amplifier **290-b** may alternatively or equivalently include pairs of cross-coupled transistors (e.g., a pair of cross-coupled p-type transistors and a pair of cross-coupled n-type transistors).

In some examples, the sense amplifier **290-b** may include circuitry configured to support a selective coupling (e.g., of the amplifiers **710-a** and **710-b**) with sense amplifier low and high voltage sources (e.g., voltage sources **293-b** and **294-b**). For example, the sense amplifier **290-b** may include switching components **730-a** and **730-b** that are operable based on logical signals SW_3 and SW_4 , respectively. In some examples, activating or selecting logical signals SW_3 and SW_4 may be referred to as activating or latching the sense amplifier **290-b**.

In some examples, the sense amplifier **290-b** may include circuitry configured to support a selective coupling with or

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decoupling from another component (e.g., a signal development component **250**, a selection component **280**, a reference component **270**). For example, the sense amplifier **290-b** may include switching components **720-a** and **720-b**, which may be referred to as an isolation switching component, and may be operable based on a logical signals ISO_1 and ISO_2 . Additionally or alternatively, an isolation switching component may be included in a signal development component **250** or a selection component **280** in accordance with examples as disclosed herein.

In some examples (e.g., in support of a read operation), the sense amplifier **290-b** may generate an output signal based at least in part on a cell read signal. For example, a signal development component **250** (e.g., a selected one of a set of signal development components **250**) may pass a cell access signal, or otherwise share a charge with the sense amplifier **290-b** that is based at least in part on a cell access signal, via the signal line **285-b**. A reference component **270** may pass a reference signal, or otherwise share a charge with the sense amplifier **290-b** that is based at least in part on a reference signal, via the reference line **275-b**. When the signal line **285-b** has a higher voltage than the reference line **275-b**, the output signal may be generated with the I/O line **295-b** having a relatively higher voltage (e.g., V_H) and the I/O line **295-c** having a relatively lower voltage (e.g., V_L). When the reference line **275-b** has a higher voltage than the signal line **285-b**, the output signal may be generated with the I/O line **295-c** having a relatively higher voltage (e.g., V_H) and the I/O line **295-b** having a relatively lower voltage (e.g., V_L). In some examples, the switching components **720-a** and **720-b** may be closed to receive cell read signals or cell reference signals, and subsequently opened when activating the sense amplifier **290-b** (e.g., “latching”).

In some examples, a generated sense or latch signal, or otherwise generated output signal, may be shared or otherwise associated with a write signal or rewrite signal passed to the selected signal development component **250** via the signal line **285-b** (e.g., after closing the switching component **720-a**). In some examples, a write command or write signal may be received at the sense amplifier **290-b** (e.g., from an input/output component **160** via I/O lines **295-b** and **295-c**), and the received write command or write signal may be latched, shared (e.g., via the signal line **285-b**), or otherwise associated with a cell write signal generated by the selected signal development component **250**. In some examples, a write command or write signal associated with the sense amplifier **290-b** may bypass signal development components **250** (e.g., via a bypass line **260**).

In some examples, a read command or read signal associated with a sense amplifier **290-b**, for example, may initiate a read broadcast operation. A read broadcast may occur from one or more locations of a memory array (e.g., in one or more domains **310**) to multiple locations of a set of signal development components **250**. In some examples, one or more multiplexers (e.g., of the selection component **280**) may act as a distributed routing delivery apparatus to transfer data from locations of the memory array using access lines (e.g., word lines **205**, digit lines **210**, plate lines **215**, etc.) coupled with the locations. For example, to initialize a set of locations in the set of signal development components **250** (e.g., in one or more cache blocks) to a desired pattern of data, the set of locations may be concurrently coupled with one or more multiplexers to store the pattern of data from the memory array.

FIG. 8 shows a block diagram of a system **800** that supports read broadcast operations associated with a memory device in accordance with examples as disclosed

herein. The system **800** may include a memory array **805**, a selection component **815**, a signal development component array **825**, a selection component **835**, and a sense amplifier array **845**. In some examples, these and other components may be included in a data path **860** of the system **800**.

The memory array **805** may include a set of memory cells **105**, which may be associated with access lines such as those described with reference to FIGS. **1** through **3** (e.g., word lines **205**, digit lines **210**, plate lines **215**). In some examples, the memory array may be associated with A rows (e.g., A independently accessible word lines **205**) and B columns (e.g., B independently accessible digit lines **210**). In one example, the memory array **805** may be associated with 1,048,576 memory cells **105**, arranged according to 1,024 word lines **205** and 1,024 digit lines **210**. Each of the memory cells **105** may be configured to store a respective logic state, which may alternatively be referred to as a memory state.

In some examples, the memory array **805** may be arranged in a set of domains, which may be similar to domains **310** described with reference to FIG. **3**. In one example, the memory array **805** may be split among 4 domains, and each of the four domains may have four independent zones with plate control (e.g., each domain of the memory array **805** may have four zones, which may be an example of subdomains, having commonly or individually biased plate lines **215**). In such examples, the memory array **805** may be arranged according to 16 control zones, which may be associated with selecting 64-bit data.

The signal development component array **825** may include a set of signal development components **250**, which may include aspects of signal development components **250** described with reference to FIGS. **2** through **7**. The signal development component array **825**, or components thereof (e.g., cache elements of the signal development component array **825**) may be an example of a signal development cache in accordance with examples as disclosed herein. In some examples, signal development components **250**, or cache elements thereof, of the signal development component array **825** may be arranged in a grid having C rows and D columns. In some examples, each of the D columns may be associated with a cache block, and each of the C rows may be associated with a position in a respective cache block. In one example, the signal development component array **825** may be associated with 8 cache blocks, each having 64 positions. Each of the positions of each of the cache blocks may correspond to a single signal development component **250**, or cache element of a signal development component **250**.

The selection component **815** may include various components that support mapping memory cells **105** of the memory array **805** with signal development components **250** of the signal development component array **825**. For example, the selection component **815** may provide for selective coupling and decoupling of individual digit lines **210** of the memory array **805** with individual signal development components **250** of the signal development component array **825** to support various examples of read broadcast operations described herein.

The selection component **815** may be coupled with the memory array **805** via a bus **810** having N signal paths, and the selection component **815** may be coupled with the signal development component array **825** via a bus **820** having M signal paths. In some examples, the selection component **815** may be coupled with each of the digit lines **210** of the memory array **805** (e.g., where N=B). In some examples, the bus **820** may have fewer signal paths than the bus **810**, where

M is associated with the quantity of cache blocks of the signal development component array. For example, the bus **810** may have N=1,024 signal paths, and the bus **820** may have M=8 signal paths, or M=4 signal paths, or some other quantity of signal paths.

In various examples, each digit line **210** of the memory array **805** may be configured for selective coupling with a particular one of the signal development components **250** of the signal development component array **825**, a particular set of the signal development components **250** of the signal development component array **825**, or may be configured for selective coupling with any one of the signal development components **250** of the signal development component array. Additionally or alternatively, a signal development component **250** of the signal development component array **825** may be configured for selective coupling with a particular one of the digit lines **210** of the memory array **805**, a particular set of the digit lines **210** of the memory array, or may be configured for selective coupling with any one of the digit lines **210** of the memory array **805**. In other words, the mapping between digit lines **210** and signal development components **250** in accordance with the described techniques may include a one-to-many mapping, a many-to-one mapping, or a many-to-many mapping.

The sense amplifier array **845** may include a set of sense amplifiers **290**, which may include aspects of sense amplifiers **290** described with reference to FIGS. **2** through **7**. In some examples, sense amplifiers of the sense amplifier array **845** may be arranged in a strip or other grouped arrangement. The selection component **835** may be coupled between the signal development component array **825** (e.g., via a bus **830**) and the sense amplifier array **845** (e.g., via a bus **840**) to support various mappings between signal development components **250** and sense amplifiers **290**. In various examples, the sense amplifiers **290** (e.g., of the sense amplifier array **845**) may be integrated between cache blocks (e.g., of the signal development component array **825**) or may be external to the signal development component cache region (e.g., external to the signal development component array **825**). In some examples, the sense amplifier array **845** may be coupled with a bus **850**, which may support communication of information with an I/O component (not shown), which may be considered to be within out outside the illustrative boundary of the data path **860**.

In some examples, the signal development component array **825** may be coupled with a strip or other group of sense amplifiers **290** (e.g., of the sense amplifier array **845**), each of which may also be independently accessible. For example, each of a strip of sense amplifiers **290** may be configured for selective coupling with a particular one of the signal development components **250** of the signal development component array **825**, a particular set of the signal development components **250** of the signal development component array **825**, or may be configured for selective coupling with any one of the signal development components **250** of the signal development component array. Additionally or alternatively, a signal development component **250** of the signal development component array **825** may be configured for selective coupling with a particular one of the sense amplifiers **290** of the strip of sense amplifiers, a particular set of the sense amplifiers of the strip of sense amplifiers, or may be configured for selective coupling with any one of the sense amplifiers **290** of the strip of sense amplifiers. In other words, the mapping (e.g., via the selection component **835**) between signal development components **250** of the signal development component array **825** and sense amplifiers **290** of the sense amplifier array **845** in

accordance with the described techniques may include a one-to-many mapping, a many-to-one mapping, or a many-to-many mapping.

In an illustrative example where the memory array **805** is associated with 1,024 digit lines **210**, each of the 1,024 digit lines **210** may be coupled with a multiplexer (e.g., of the selection component **815**), where they may be reduced to $64 \times 4 = 256$ digit lines. This may support signal transfer of 4 sets of 64 digit lines overlapping in time (e.g., participating in simultaneous transfer between a memory cell **105** and a signal development component **250**). In some examples, each of these 4 sets can be routed to any of 8 cache blocks (e.g., of the signal development component array **825**), where each cache block may include 8 lines by 64 bits. In other words, the total cache size associated with such a signal development component array **825** may be 64×64 bits. According to this example of array routing, any 64 bit sub-row from memory array may be routed to any of 64 bit signal development component cache lines.

In another illustrative example, the system **800** may include several domains (e.g., of the memory array **805**) each with 1,048,576 memory cells **105** arranged in 1,024 uniquely addressed rows and 1,024 columns. Each of the domains of the system **800** may be mapped (e.g., via the selection component **815**) with 64 signal development components (e.g., of the signal development component array **825**). In other words, 64 signal development components may be mapped to 1,024 digit lines **210** within each domain. In some examples, a particular signal development component **250** may be mapped to 16 digit lines **210** within each domain (e.g., 1,024 digit lines **210** divided by 64 signal development components **250**). In some examples, such a mapping may be fixed (e.g., where groups of 16 digit lines **210** are mapped to a respective signal development component **250** within each domain) which, in some examples, may reduce multiplexing or selection circuit complexity. In various other examples, a signal development component **250** may be mapped to more than one domain, more than one set of digit lines **210** (e.g., of a domain), or other configurations. Additionally or alternatively, a domain or a set of digit lines **210** may be mapped to more than one signal development component **250**. In other words, a memory device may include various configurations of signal development components **250** to support examples of the read broadcast operations described herein.

In this illustrative example, a row of 1024 memory cells **105** (e.g., spanning one domain **310**) may be selected by a single word line **205** in each domain. With 64 signal development components **250** per domain, 64 of the set of 1,024 memory cells **105** may be accessed at a time in each domain (e.g., by selectively coupling a respective digit line **210** with each of the 64 signal development components **250-a** via the selection component **815**). During such accessing, other digit lines **210** may be selectively isolated from the signal development components **250** interfacing the same domain. Further, the other digit lines **210** may be shunted or masked as described herein.

Although the system **800** is illustrated with a selection component **815** operable to selectively couple the memory array **805** with the signal development component array **825**, and a selection component **835** operable to selectively couple the signal development component array **825** with the sense amplifier array **845**, other configurations are possible for supporting the described techniques for memory accessing. For example, in some cases, the memory array **805** may be selectively coupled with the sense amplifier array **845** in a manner that bypasses the signal development

component array **825**, or components thereof. In some examples, a coupling between the memory array **805** and the sense amplifier array **845** may be supported by way of one or more bypass lines, such as the bypass line **260** described with reference to FIG. 2. In some examples, the system **800** may include another selection component operable for selectively coupling the memory array **805** with the sense amplifier array **845** (e.g., bypassing the signal development component array **825**, or components thereof), which may include or otherwise support the functionality of the switching component **265** described with reference to FIG. 2, among other features or functions.

More generally, the system **800** may include various selection components or other circuitry operable to selectively couple the components of the system **800** to support various access techniques in accordance with examples as disclosed herein. For example, the system **800** may include various selection components or other circuitry operable to selectively couple the memory array **805**, or components thereof (e.g., a plurality of access lines of the memory array **805**), with the signal development component array **825** or components thereof (e.g., cache elements of a signal development cache). Additionally or alternatively, the system **800** may include various selection components or other circuitry operable to selectively couple the signal development component array **825** with the sense amplifier array **845**, or components thereof (e.g., a plurality of sense amplifiers of the sense amplifier array). Additionally or alternatively, the system **800** may include various selection components or other circuitry operable to selectively couple the plurality of access lines of the memory array with the plurality of sense amplifiers of the sense amplifier array, or any combination thereof.

In some examples, operations of one or more components of the system **800** may be controlled by a memory controller, such as memory controller **870**. The memory controller **870** may be an example of, or otherwise be associated with performing operations of a memory controller **170** as described with reference to FIG. 1. The memory controller **870** may be illustrative of a controller or other circuitry that is configured to control various components or operations of the system **800**. For example, the system **800** may include various components or circuitry of a data path **860**, which may include the memory array **805**, the selection component **815**, the signal development component array **825**, the selection component **835**, and the sense amplifier array **845**, among other components along a path of information transfer in the system **800** (e.g., a row component **125**, a column component **135**, a plate component **145**, an I/O component **160**, and others). In various examples, the memory controller **870** may be in communication with any one or more of the components of the data path **860** for controlling the associated components or operations.

The memory controller **870** may be configured (e.g., by one or more commands received from a host device) for performing one or more write operations, read operations, eviction operations, or bypass operations, among other examples of memory operations of the system **800**. In various examples of such operations, the memory controller **870** may be configured for transferring data between one or more portions of the memory array **805**, one or more portions of the signal development component array **825** (e.g., a cache block of the signal development component array **825**), or one or more portions of the sense amplifier array **845** in accordance with the one or more memory operations.

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In some examples, the memory controller **870** may be configured for performing a read operation, which may include transferring data from the signal development component array **825** to the sense amplifier array **845** (e.g., when requested data is stored in the signal development component array **825**). In some examples, the memory controller **870** may be configured for transferring the data from the memory array **805** to the signal development component array **825** (e.g., when requested data is not found in the signal development component array **825**). Additionally or alternatively, the memory controller **870** may be configured for performing an eviction operation. The eviction operation may include transferring data stored in the signal development component array **825** to the memory array **805** prior to transferring other data (e.g., data associated with a read operation) from the memory array **805** to the signal development component array **825**. In some examples, the memory controller **870** may be configured for performing a cache bypass operation, which may include transferring data directly from the memory array **805** to the sense amplifier array **845**, which may facilitate, as an example, streaming read operations (e.g., performing multiple read operations in parallel).

In some examples, the memory controller may be configured for performing a write-back operation, which may include transferring data from the sense amplifier array **845** to the signal development component array **825** (e.g., after performing a read operation). Additionally or alternatively, the memory controller **870** may be configured for performing a write-through operation. The write through operation may include transferring data directly from the sense amplifier array **845** to the memory array **805** based on determining that the data is stored at the signal development component array **825** in accordance with a write command. In some examples, the memory controller **870** may be configured for performing a bypass operation. For example, the bypass operation may include transferring data directly from the sense amplifier array **845** to the memory array **805** based on determining that the data is not stored in the signal development cache in accordance with a write command. Such examples of bypass operations may facilitate streaming write operations (e.g., performing multiple write operations in parallel). In some cases, one or more of the write operations described herein may include an eviction operation. For example, the memory controller **870** may transfer data stored in the signal development component array **825** to the memory array **805** based on determining that data corresponding to a write command (e.g., a write-back command) is not currently stored in the signal development component array **825**.

The system **800** may support various examples of read broadcast operations. A read broadcast may occur from one or more locations of the memory array **805** (e.g., from one or more domains **310**) to multiple locations of the signal development component array **825**. In some examples, one or more multiplexers (e.g., of the selection component **315**) may act as a distributed routing delivery apparatus to transfer data from locations of the memory array **805** using access lines (e.g., word lines **205**, digit lines **210**, plate lines **215**, etc.) coupled with the locations. For example, the one or more multiplexer may capture the data from the memory array **805** as a pattern of logic states, for example in a sensing or latching operation.

To initialize a set of locations in the signal development component array **825** (e.g., in one or more cache blocks) to a desired pattern of data (e.g., based on the pattern of logic states), the set of locations may be concurrently coupled

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with the one or more multiplexers to store the pattern of data from the memory array **805**. In some examples, the one or more multiplexers may perform some processing on the pattern of data, in addition to coupling with the set of locations in the signal development component array **825** to store the pattern of data. In some examples, the processing may include a shuffling operation, a mathematical operation, a logical operation (e.g., a logic operation such as AND, OR, XOR, NOT, etc.). Based on the processing, a diverse patterning may be achieved along the data path **860** in a read broadcast operation. For example, a read broadcast operation may initialize data for use with multiple threads of a central processing unit (CPU), such as for single instruction, multiple data (SIMD) operations.

In some examples, activating multiple access lines (e.g., word lines **205**, digit lines **210**, etc.) to store data in locations in the signal development component array **825** may lead to data being overwritten (e.g., signal development components **250** may charge share with one another). Accordingly, in some examples, data overwrite may be reduced or avoided using masking techniques, which may include a masking technique referred to as a "silk screening" technique. In some example masking techniques, storing the pattern of data in the signal development component array **825** may occur in multiple passes, where each pass may include one or more modifications to a portion of a pattern written to the signal development component array **825** during a previous pass. In some examples, patterns associated the multiple passes may be stored in the memory array **805**, a second memory array (not shown), or a processor (not shown), or any combination thereof.

In some examples, the pattern of data may be stored column-wise across digit lines (e.g., digit lines **210**, signal lines, etc.) associated with the locations of the signal development component array **825**. Along each digit line, a respective bit pattern of data may be known. The pattern of data may be written to the locations using write operations based on the pattern of data.

As part of a first operation, a first set of word lines may be activated, and a first signal (e.g., a first voltage or a low voltage) may be applied to a set of digit lines (e.g., a set of signal lines) to store a signal state representing a first logic state (e.g., a memory state such as a logic 0) in a first set of signal development components **250**.

As part of a second operation (e.g., following the first operation), a second set of word lines may be activated, and a second signal (e.g., a second voltage or a high voltage) may be applied to the set of digit lines to store a signal state representing a second logic state (e.g., a memory state such as a logic 1) in a second set signal development components **250**.

In some examples, the first set of word lines may be deactivated before the second operation is initiated. In some examples, storing the pattern of data may include additional write operations, which may include activating additional sets of word lines and additional sets of digit lines. In some examples, the second set of word lines may include one or more word lines of the first set of word lines.

In some examples, activating access lines (e.g., word lines **205**, digit lines **210**, plate lines **215**, etc.) to read out data from the locations of the memory array **805** may also lead to data loss (e.g., due to charge sharing). Accordingly, in some examples, the pattern of logic states may be written back to the memory array **805** after capturing the data at the one or more multiplexers. In some examples, the pattern of logic states may be written using a write operation as described herein.

In some examples of a read broadcast operation, the data to be read out may be based on a command, for example, received from a host device or from the memory controller **870**. In some examples, the command may include a read command, a write command, or a modify command, another command, or any combination thereof. The command may identify a pattern of logic states to be stored in the memory array **805**. In some examples, the pattern of logic states may be configured to be stored as a pattern of signal states at a set of locations of the signal development component array **825** before subsequently being read out (e.g., to the sense amplifier array **845** via the selection component **835**). Additionally or alternatively, the pattern of logic states may include one or more logic states, and the one or more logic states may be configured to be stored as one or more signal states at each location of the set of locations of the signal development component array **825**. In other words, copies of the data may be configured to be stored at multiple locations of the signal development component array **825**.

In some examples, the system **800** may be configured to modify the pattern of logic states in the memory array **805**, for example based on a modify command. The access lines may be coupled (e.g., via the selection component **815** and the bus **820**) with the signal development component array **825**. Based on the coupling, the pattern of logic states may be read out from at least one location in the memory array **805**. In some examples, the set of signal states may be stored at the signal development component array **825** based on the pattern of logic states read out from the memory array **805**.

In some examples, the one or more multiplexers (or the selection component **815**, or the selection component **835**) may include one or more signal processors (e.g., a digital signal processor (DSP)). The one or more signal processors may enable signal processing operations for processing the signal states stored at the signal development component array **825**, for example in response to a command.

FIG. 9 illustrates examples of signal development component arrays **825-a** that support read broadcast operations associated with a memory device in accordance with examples as disclosed herein. The signal development component arrays **825-a** may be examples of the signal development component array **825** described with reference to FIG. 8. The signal development component arrays **825-a** may include a set of signal development components **250**, which may include aspects of signal development components **250** described with reference to FIGS. 2 through 7. The signal development component arrays **825-a**, or components thereof (e.g., cache elements of the signal development component arrays **825-a**) may be examples of a signal development cache in accordance with examples as disclosed herein.

In some examples, the signal development component arrays **825-a** may enable read broadcast operations. A read broadcast may occur from one or more locations of a memory array (e.g., in one or more domains) to multiple locations of a signal development component array **825-a**. In some examples, one or more multiplexers (e.g., of a selection component) may act as a distributed routing delivery apparatus to transfer data from locations of the memory array using access lines (e.g., word lines, digit lines, plate lines, etc.) coupled with the locations. For example, to initialize a set of locations in a signal development component array **825-a** (e.g., in one or more cache blocks) to a desired pattern of data, the set of locations may be concurrently coupled with one or more multiplexers to store the pattern of data from the memory array.

In some examples, a pattern of logic states from the memory array may include one or more logic states, and the one or more logic states may be configured to be stored as one or more signal states in patterns **905** at multiple locations of a signal development component array **825-a-1** before subsequently being read out. In other words, copies of the data may be configured to be stored at multiple locations of the signal development component array **825-a-1**, for example as patterns **905-a**, **905-b**, and **905-c**.

The patterns **905** may be initialized as part of a read broadcast operation for use with multiple threads of a CPU. For example, in CPU multiprocessing, each process or thread may be configured to independently process one of the patterns **905**, for example based on a read command or a write command. In some examples, the CPU may process the patterns **905** through a sense amplifier coupled with the signal development component array **825-a-1**.

In some examples, a pattern of logic states from the memory array may be configured to be stored as a pattern **910** of signal states at a set of locations of a signal development component array **825-a-2** before subsequently being read out. In some examples, the pattern **910** may be stored at the signal development component array **825-a-2** using a masking technique such as a silk screening technique. For example, the pattern **910** may be stored at the signal development component array **825-a-2** in multiple passes, where each pass may include one or more modifications to a portion of a pattern written to the signal development component array **825-a-2** during a previous pass. After the final pass, the pattern **910** may be available at the signal development component array **825-a-2** for read out, for example based on a read command from a requesting device. In some examples, the pattern **910** may be stored column-wise across digit lines (e.g., digit lines **210**, signal lines, etc.) associated with locations of the signal development component array **825-a-2**, as described with reference to FIG. 8.

FIG. 10 shows a block diagram **1000** of a memory device **1005** that supports read broadcast operations associated with a memory device in accordance with examples as disclosed herein. The memory device **1005** may be an example of aspects of a memory device as described with reference to FIGS. 1 through 9. The memory device **1005** may include a read data manager **1010**, an access line manager **1015**, a selection component manager **1020**, a storage location manager **1025**, a signal state manager **1030**, a write operation manager **1035**, and a command manager **1040**. Each of these modules may communicate, directly or indirectly, with one another (e.g., via one or more buses). In some examples, the memory device **1005** may enable read broadcast operations. A read broadcast may occur from a memory array to multiple locations of a signal development cache, for example via one or more multiplexers.

The read data manager **1010** may identify a pattern of logic states stored in a second component of a memory device. In some examples, the read data manager **1010** may transfer, via the first component, data associated with the pattern of logic states from one or more locations of the set of locations to a sense component array based on the command. In some cases, the second component includes a set of domains or subdomains. In some cases, each domain or subdomain of the set of domains or subdomains includes one or more memory cells storing at least a portion of the pattern of logic states. In some cases, the second component of the memory device includes a memory array different from the first component of the memory device.

The access line manager **1015** may couple a set of access lines with a selection component, where the set of access

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lines are coupled with the second component. In some examples, the access line manager **1015** may activate a first set of word lines associated with the first component. In some examples, the access line manager **1015** may apply a first signal to a set of signal lines coupled with the first component after activating the first set of word lines, where the first signal corresponds to a first logic state. In some examples, the access line manager **1015** may activate a second set of word lines associated with the first component based on applying the first signal. In some examples, the access line manager **1015** may apply a second signal to the set of signal lines after activating the second set of word lines, where the second signal corresponds to a second logic state.

In some cases, each domain or subdomain of the set of domains or subdomains includes one or more access lines of the set of access lines coupled with the second component, the one or more access lines coupled with the one or more memory cells. In some cases, the pattern of logic states is captured at the selection component based on coupling the one or more access lines with the selection component.

The selection component manager **1020** may capture the pattern of logic states at the selection component based on coupling the set of access lines with the selection component. In some examples, the selection component manager **1020** may process, at the selection component, the pattern of logic states based on capturing the pattern of logic states at the selection component, where processing the pattern of logic states includes a shuffling operation, a mathematical operation, a logic operation, or any combination thereof. In some examples, the selection component manager **1020** may process, at the selection component, the pattern of signal states based on capturing the pattern of logic states at the selection component. In some cases, the selection component includes a multiplexer, a digital signal processor, or any combination thereof.

The storage location manager **1025** may couple the selection component with a set of locations of a first component of the memory device based on capturing the pattern of logic states. In some cases, the first component of the memory device includes a capacitor array, a signal development cache component, or any combination thereof.

The signal state manager **1030** may store, at the set of locations of the first component and based on coupling the selection component with the set of locations of the first component, a pattern of signal states associated with the pattern of logic states. In some examples, the signal state manager **1030** may store the pattern of signal states at the set of locations of the first component based on applying the first signal and the second signal to the set of signal lines. In some examples, the signal state manager **1030** may store one or more signal states at each location of the set of locations of the first component based on the pattern of signal states.

The write operation manager **1035** may write the pattern of logic states to the second component as part of a write operation based on capturing the pattern of logic states at the selection component.

The command manager **1040** may receive, from a requesting device, a command associated with the pattern of logic states, where determining the pattern of logic states is based on receiving the command. In some cases, the command includes a read command, a write command, a modify command, or any combination thereof.

FIG. **11** shows a flowchart illustrating a method or methods **1100** that supports read broadcast operations associated with a memory device in accordance with examples as disclosed herein. The operations of method **1100** may be

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implemented by a memory device or its components as described herein. For example, the operations of method **1100** may be performed by a memory device as described with reference to FIG. **10**. In some examples, a memory device may execute a set of instructions to control the functional elements of the memory device to perform the described functions. Additionally or alternatively, a memory device may perform aspects of the described functions using special-purpose hardware. In some examples, the memory device described herein may enable read broadcast operations. A read broadcast may occur from a memory array to multiple locations of a signal development cache, for example via one or more multiplexers.

At **1105**, the memory device may identify a pattern of logic states stored in a second component of a memory device. The operations of **1105** may be performed according to the methods described herein. In some examples, aspects of the operations of **1105** may be performed by a read data manager as described with reference to FIG. **10**.

At **1110**, the memory device may couple a set of access lines with a selection component, where the set of access lines are coupled with the second component. The operations of **1110** may be performed according to the methods described herein. In some examples, aspects of the operations of **1110** may be performed by an access line manager as described with reference to FIG. **10**.

At **1115**, the memory device may capture the pattern of logic states at the selection component based on coupling the set of access lines with the selection component. The operations of **1115** may be performed according to the methods described herein. In some examples, aspects of the operations of **1115** may be performed by a selection component manager as described with reference to FIG. **10**.

At **1120**, the memory device may couple the selection component with a set of locations of a first component of the memory device based on capturing the pattern of logic states. The operations of **1120** may be performed according to the methods described herein. In some examples, aspects of the operations of **1120** may be performed by a storage location manager as described with reference to FIG. **10**.

At **1125**, the memory device may store, at the set of locations of the first component and based on coupling the selection component with the set of locations of the first component, a pattern of signal states associated with the pattern of logic states. The operations of **1125** may be performed according to the methods described herein. In some examples, aspects of the operations of **1125** may be performed by a signal state manager as described with reference to FIG. **10**.

In some examples, an apparatus as described herein may perform a method or methods, such as the method **1100**. The apparatus may include features, means, or instructions (e.g., a non-transitory computer-readable medium storing instructions executable by a processor) for identifying a pattern of logic states stored in a second component of a memory device, coupling a set of access lines with a selection component, where the set of access lines are coupled with the second component, capturing the pattern of logic states at the selection component based on coupling the set of access lines with the selection component, coupling the selection component with a set of locations of a first component of the memory device based on capturing the pattern of logic states, and storing, at the set of locations of the first component and based on coupling the selection component with the set of locations of the first component, a pattern of signal states associated with the pattern of logic states.

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Some examples of the method **1100** and the apparatus described herein may further include operations, features, means, or instructions for writing the pattern of logic states to the memory array as part of a write operation based on capturing the pattern of logic states at the selection component.

Some examples of the method **1100** and the apparatus described herein may further include operations, features, means, or instructions for activating a first set of word lines associated with the first component, applying a first signal to a set of signal lines coupled with the first component after activating the first set of word lines, where the first signal corresponds to a first logic state, activating a second set of word lines associated with the first component based on applying the first signal, applying a second signal to the set of signal lines after activating the second set of word lines, where the second signal corresponds to a second logic state, and storing the pattern of signal states at the set of locations of the first component based on applying the first signal and the second signal to the set of signal lines.

Some examples of the method **1100** and the apparatus described herein may further include operations, features, means, or instructions for processing, at the selection component, the pattern of logic states based on capturing the pattern of logic states at the selection component, where processing the pattern of logic states includes a shuffling operation, a mathematical operation, a logic operation, or any combination thereof.

Some examples of the method **1100** and the apparatus described herein may further include operations, features, means, or instructions for processing, at the selection component, the pattern of signal states based on capturing the pattern of logic states at the selection component.

In some examples of the method **1100** and the apparatus described herein, the pattern of signal states may include one or more signal states, and storing the pattern of signal states at the set of locations of the first component may include operations, features, means, or instructions for storing the one or more signal states at each location of the set of locations of the first component based on the pattern of signal states.

Some examples of the method **1100** and the apparatus described herein may further include operations, features, means, or instructions for receiving, from a requesting device, a command associated with the pattern of logic states, where determining the pattern of logic states may be based on receiving the command.

Some examples of the method **1100** and the apparatus described herein may further include operations, features, means, or instructions for transferring, via the first component, data associated with the pattern of logic states from one or more locations of the set of locations to a sense component array based on the command.

In some examples of the method **1100** and the apparatus described herein, the command may include a read command, a write command, a modify command, or any combination thereof.

In some examples of the method **1100** and the apparatus described herein, the second component may include a set of domains or subdomains, each domain or subdomain of the set of domains or subdomains includes one or more memory cells storing at least a portion of the pattern of logic states and one or more access lines of the set of access lines coupled with the one or more memory cells, and the pattern of logic states may be captured at the selection component based on coupling the one or more access lines with the selection component.

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In some examples of the method **1100** and the apparatus described herein, the first component of the memory device may include a capacitor array, a signal development cache component, or any combination thereof. In some examples of the method **1100** and the apparatus described herein, the second component of the memory device may include a memory array different from the first component of the memory device. In some examples of the method **1100** and the apparatus described herein, the selection component may include a multiplexer, a digital signal processor, or any combination thereof.

FIG. **12** shows a flowchart illustrating a method or methods **1200** that supports read broadcast operations associated with a memory device in accordance with aspects of the present disclosure. The operations of method **1200** may be implemented by a memory device or its components as described herein. For example, the operations of method **1200** may be performed by a memory device as described with reference to FIG. **10**. In some examples, a memory device may execute a set of instructions to control the functional elements of the memory device to perform the described functions. Additionally or alternatively, a memory device may perform aspects of the described functions using special-purpose hardware. In some examples, the memory device described herein may enable read broadcast operations. A read broadcast may occur from a memory array to multiple locations of a signal development cache, for example via one or more multiplexers.

At **1205**, the memory device may identify a pattern of logic states stored in a second component of a memory device. The operations of **1205** may be performed according to the methods described herein. In some examples, aspects of the operations of **1205** may be performed by a read data manager as described with reference to FIG. **10**.

At **1210**, the memory device may couple a set of access lines with a selection component, where the set of access lines are coupled with the second component. The operations of **1210** may be performed according to the methods described herein. In some examples, aspects of the operations of **1210** may be performed by an access line manager as described with reference to FIG. **10**.

At **1215**, the memory device may capture the pattern of logic states at the selection component based on coupling the set of access lines with the selection component. The operations of **1215** may be performed according to the methods described herein. In some examples, aspects of the operations of **1215** may be performed by a selection component manager as described with reference to FIG. **10**.

At **1220**, the memory device may process, at the selection component, the pattern of logic states based on capturing the pattern of logic states at the selection component, where processing the pattern of logic states includes a shuffling operation, a mathematical operation, a logic operation, or any combination thereof. The operations of **1220** may be performed according to the methods described herein. In some examples, aspects of the operations of **1220** may be performed by a selection component manager as described with reference to FIG. **10**.

At **1225**, the memory device may couple the selection component with a set of locations of a first component of the memory device based on capturing the pattern of logic states. The operations of **1225** may be performed according to the methods described herein. In some examples, aspects of the operations of **1225** may be performed by a storage location manager as described with reference to FIG. **10**.

At **1230**, the memory device may store, at the set of locations of the first component and based on coupling the

selection component with the set of locations of the first component, a pattern of signal states associated with the pattern of logic states. The operations of 1230 may be performed according to the methods described herein. In some examples, aspects of the operations of 1230 may be performed by a signal state manager as described with reference to FIG. 10.

It should be noted that the methods described herein are possible implementations, and that the operations and the steps may be rearranged or otherwise modified and that other implementations are possible. Furthermore, portions from two or more of the methods may be combined.

An apparatus is described. The apparatus may include a second component operable to store a pattern of logic states, a set of access lines coupled with the second component, a selection component coupleable with the set of access lines and operable to capture the pattern of logic states based on coupling the set of access lines with the selection component, and a first component including a set of locations coupleable with the selection component, the first component operable to store a pattern of signal states associated with the pattern of logic states based on coupling the selection component with the set of locations.

Some examples of the apparatus may include a first set of word lines associated with the first component and a second set of word lines associated with the first component, the first set of word lines and the second set of word lines operable to be activated as part of storing the pattern of signal states at the set of locations of the first component, and a set of signal lines coupled with the set of locations.

Some examples of the apparatus may include a signal source operable to apply a first signal corresponding to a first logic state to the set of signal lines after activating the first set of word lines, and apply a second signal corresponding to a second logic state to the set of signal lines after activating the second set of word lines.

In some examples, the selection component may be further operable to process the pattern of logic states based on capturing the pattern of logic states at the selection component, where processing the pattern of logic states includes a shuffling operation, a mathematical operation, a logic operation, or any combination thereof.

In some examples, the selection component may be further operable to process the pattern of signal states based on capturing the pattern of logic states at the selection component.

In some examples, the pattern of signal states may include one or more signal states, and the first component may be operable to store the one or more signal states at each location of the set of locations based on the pattern of signal states.

Some examples of the apparatus may include a sense component array coupleable with the first component, where the first component may be further operable to transfer data associated with the pattern of logic states from one or more locations of the set of locations to the sense component array based on a read command.

In some examples, the second component may further include a set of domains or subdomains, where each domain or subdomain of the set of domains or subdomains includes one or more memory cells storing at least a portion of the pattern of logic states.

Another apparatus is described. The apparatus may include a first component, a second component, and a controller operable to cause the apparatus to identify a pattern of logic states stored in the second component, couple a set of access lines with a selection component,

where the set of access lines are coupled with the second component, capture the pattern of logic states at the selection component based on coupling the set of access lines with the selection component, couple the selection component with a set of locations of a first component of the apparatus based on capturing the pattern of logic states, and store, at the set of locations of the first component and based on coupling the selection component with the set of locations of the first component, a pattern of signal states associated with the pattern of logic states.

In some examples of the apparatus, the controller may be further operable to cause the apparatus to activate a first set of word lines associated with the first component, apply a first signal to a set of signal lines coupled with the first component after activating the first set of word lines, where the first signal corresponds to a first logic state, activate a second set of word lines associated with the first component based on applying the first signal, apply a second signal to the set of signal lines after activating the second set of word lines, where the second signal corresponds to a second logic state, and store the pattern of signal states at the set of locations of the first component based on applying the first signal and the second signal to the set of signal lines.

In some examples of the apparatus, the controller may be further operable to cause the apparatus to process, at the selection component, the pattern of logic states based on capturing the pattern of logic states at the selection component, where processing the pattern of logic states includes a shuffling operation, a mathematical operation, a logic operation, or any combination thereof. In some examples of the apparatus, the controller may be further operable to cause the apparatus to process, at the selection component, the pattern of signal states based on capturing the pattern of logic states at the selection component.

In some examples, the pattern of signal states may include one or more signal states, and the controller may be further operable to cause the apparatus to store the one or more signal states at each location of the set of locations of the first component based on the pattern of signal states.

In some examples of the apparatus, the controller may be further operable to cause the apparatus to receive, from a requesting device, a command associated with the pattern of logic states, where determining the pattern of logic states is based on receiving the command. In some examples of the apparatus, the controller may be further operable to cause the apparatus to transfer, via the first component, data associated with the pattern of logic states from one or more locations of the set of locations to a sense component array based on the command.

Information and signals described herein may be represented using any of a variety of different technologies and techniques. For example, data, instructions, commands, information, signals, bits, symbols, and chips that may be referenced throughout the above description may be represented by voltages, currents, electromagnetic waves, magnetic fields or particles, optical fields or particles, or any combination thereof. Some drawings may illustrate signals as a single signal; however, it will be understood by a person of ordinary skill in the art that the signal may represent a bus of signals, where the bus may have a variety of bit widths.

The terms "electronic communication," "conductive contact," "connected," and "coupled" may refer to a relationship between components that supports the flow of signals between the components. Components are considered in electronic communication with (or in conductive contact with or connected with or coupled with) one another if there is any conductive path between the components that can, at

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any time, support the flow of signals between the components. At any given time, the conductive path between components that are in electronic communication with each other (or in conductive contact with or connected with or coupled with) may be an open circuit or a closed circuit based on the operation of the device that includes the connected components. The conductive path between connected components may be a direct conductive path between the components or the conductive path between connected components may be an indirect conductive path that may include intermediate components, such as switches, transistors, or other components. In some examples, the flow of signals between the connected components may be interrupted for a time, for example, using one or more intermediate components such as switches or transistors.

The term “coupling” refers to condition of moving from an open-circuit relationship between components in which signals are not presently capable of being communicated between the components over a conductive path to a closed-circuit relationship between components in which signals can be communicated between components over the conductive path. When a component, such as a controller, couples other components together, the component initiates a change that allows signals to flow between the other components over a conductive path that previously did not permit signals to flow.

The term “isolated” refers to a relationship between components in which signals are not presently capable of flowing between the components. Components are isolated from each other if there is an open circuit between them. For example, two components separated by a switch that is positioned between the components are isolated from each other when the switch is open. When a controller isolates two components from one another, the controller affects a change that prevents signals from flowing between the components using a conductive path that previously permitted signals to flow.

The term “layer” or “level” used herein refers to a stratum or sheet of a geometrical structure (e.g., relative to a substrate). Each layer or level may have three dimensions (e.g., height, width, and depth) and may cover at least a portion of a surface. For example, a layer or level may be a three dimensional structure where two dimensions are greater than a third, e.g., a thin-film. Layers or levels may include different elements, components, and/or materials. In some examples, one layer or level may be composed of two or more sublayers or sublevels.

As used herein, the term “substantially” means that the modified characteristic (e.g., a verb or adjective modified by the term substantially) need not be absolute but is close enough to achieve the advantages of the characteristic.

As used herein, the term “electrode” may refer to an electrical conductor, and in some examples, may be employed as an electrical contact to a memory cell or other component of a memory array. An electrode may include a trace, wire, conductive line, conductive layer, or the like that provides a conductive path between elements or components of the memory array.

The devices discussed herein, including a memory array, may be formed on a semiconductor substrate, such as silicon, germanium, silicon-germanium alloy, gallium arsenide, gallium nitride, etc. In some examples, the substrate is a semiconductor wafer. In other cases, the substrate may be a silicon-on-insulator (SOI) substrate, such as silicon-on-glass (SOG) or silicon-on-sapphire (SOS), or epitaxial layers of semiconductor materials on another substrate. The conductivity of the substrate, or sub-regions of

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the substrate, may be controlled through doping using various chemical species including, but not limited to, phosphorous, boron, or arsenic. Doping may be performed during the initial formation or growth of the substrate, by ion-implantation, or by any other doping means.

A switching component or a transistor discussed herein may represent a field-effect transistor (FET) and comprise a three terminal device including a source, drain, and gate. The terminals may be connected to other electronic elements through conductive materials, e.g., metals. The source and drain may be conductive and may comprise a heavily-doped, e.g., degenerate, semiconductor region. The source and drain may be separated by a lightly-doped semiconductor region or channel. If the channel is n-type (i.e., majority carriers are electrons), then the FET may be referred to as a n-type FET. If the channel is p-type (i.e., majority carriers are holes), then the FET may be referred to as a p-type FET. The channel may be capped by an insulating gate oxide. The channel conductivity may be controlled by applying a voltage to the gate. For example, applying a positive voltage or negative voltage to an n-type FET or a p-type FET, respectively, may result in the channel becoming conductive. A transistor may be “on” or “activated” when a voltage greater than or equal to the transistor’s threshold voltage is applied to the transistor gate. The transistor may be “off” or “deactivated” when a voltage less than the transistor’s threshold voltage is applied to the transistor gate.

The description set forth herein, in connection with the appended drawings, describes example configurations and does not represent all the examples that may be implemented or that are within the scope of the claims. The term “exemplary” used herein means “serving as an example, instance, or illustration,” and not “preferred” or “advantageous over other examples.” The detailed description includes specific details to providing an understanding of the described techniques. These techniques, however, may be practiced without these specific details. In some instances, well-known structures and devices are shown in block diagram form to avoid obscuring the concepts of the described examples.

In the appended figures, similar components or features may have the same reference label. Further, various components of the same type may be distinguished by following the reference label by a dash and a second label that distinguishes among the similar components. If just the first reference label is used in the specification, the description is applicable to any one of the similar components having the same first reference label irrespective of the second reference label.

Information and signals described herein may be represented using any of a variety of different technologies and techniques. For example, data, instructions, commands, information, signals, bits, symbols, and chips that may be referenced throughout the above description may be represented by voltages, currents, electromagnetic waves, magnetic fields or particles, optical fields or particles, or any combination thereof.

The various illustrative blocks and modules described in connection with the disclosure herein may be implemented or performed with a general-purpose processor, a DSP, an ASIC, an FPGA or other programmable logic device, discrete gate or transistor logic, discrete hardware components, or any combination thereof designed to perform the functions described herein. A general-purpose processor may be a microprocessor, but in the alternative, the processor may be any processor, controller, microcontroller, or state machine. A processor may also be implemented as a combination of computing devices (e.g., a combination of a DSP

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and a microprocessor, multiple microprocessors, one or more microprocessors in conjunction with a DSP core, or any other such configuration).

The functions described herein may be implemented in hardware, software executed by a processor, firmware, or any combination thereof. If implemented in software executed by a processor, the functions may be stored on or transmitted over as one or more instructions or code on a computer-readable medium. Other examples and implementations are within the scope of the disclosure and appended claims. For example, due to the nature of software, functions described above can be implemented using software executed by a processor, hardware, firmware, hardwiring, or combinations of any of these. Features implementing functions may also be physically located at various positions, including being distributed such that portions of functions are implemented at different physical locations. Also, as used herein, including in the claims, “or” as used in a list of items (for example, a list of items prefaced by a phrase such as “at least one of” or “one or more of”) indicates an inclusive list such that, for example, a list of at least one of A, B, or C means A or B or C or AB or AC or BC or ABC (i.e., A and B and C). Also, as used herein, the phrase “based on” shall not be construed as a reference to a closed set of conditions. For example, an exemplary step that is described as “based on condition A” may be based on both a condition A and a condition B without departing from the scope of the present disclosure. In other words, as used herein, the phrase “based on” shall be construed in the same manner as the phrase “based at least in part on.”

The description herein is provided to enable a person skilled in the art to make or use the disclosure. Various modifications to the disclosure will be apparent to those skilled in the art, and the generic principles defined herein may be applied to other variations without departing from the scope of the disclosure. Thus, the disclosure is not limited to the examples and designs described herein but is to be accorded the broadest scope consistent with the principles and novel features disclosed herein.

What is claimed is:

1. A memory device, comprising:
 - a first component;
 - a second component;
 - a selection component; and
 - a controller configured to cause the memory device to:
 - couple the first component with the selection component based at least in part on identifying a pattern of logic states stored at the first component; and
 - store, at a plurality of locations of the second component, a pattern of signal states associated with the pattern of logic states based at least in part on coupling the first component with the selection component and coupling the selection component with the plurality of locations of the second component.
2. The memory device of claim 1, wherein the controller is further configured to cause the memory device to:
 - activate a first set of word lines associated with the second component;
 - apply a first signal corresponding to a first logic state to a set of signal lines coupled with the second component after activating the first set of word lines;
 - activate a second set of word lines associated with the second component after applying the first signal;
 - apply a second signal corresponding to a second logic state to the set of signal lines after activating the second set of word lines; and

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store the pattern of signal states at the plurality of locations of the second component based at least in part on applying the first signal and the second signal to the set of signal lines.

3. The memory device of claim 1, wherein the controller is further configured to cause the memory device to:

- write the pattern of logic states to the first component as part of a write operation based at least in part on capturing the pattern of logic states at the selection component.

4. The memory device of claim 1, wherein the controller is further configured to cause the memory device to:

- process the pattern of logic states at the selection component based at least in part on coupling the first component with the selection component, the processing comprising a shuffling operation, a mathematical operation, a logic operation, or any combination thereof.

5. The memory device of claim 1, wherein the controller is further configured to cause the memory device to:

- process the pattern of signal states at the selection component based at least in part on coupling the first component with the selection component, wherein storing the pattern of signal states is based at least in part on processing the pattern of signal states.

6. The memory device of claim 1, wherein the pattern of signal states comprises one or more signal states, and to store the pattern of signal states at the plurality of locations of the second component, the controller is configured to cause the memory device to:

- store the one or more signal states at each location of the plurality of locations of the second component based at least in part on the pattern of signal states.

7. The memory device of claim 1, wherein the controller is further configured to cause the memory device to:

- identify the pattern of logic states based at least in part on receiving a command; and

- transfer, via the second component, data associated with the pattern of logic states from one or more locations of the plurality of locations of the second component to a sense amplifier array based at least in part on receiving the command.

8. The memory device of claim 7, wherein the command comprises a read command, a write command, a modify command, or any combination thereof.

9. The memory device of claim 1, wherein:

- the first component comprises a plurality of domains or subdomains, each of the plurality of domains or subdomains including one or more memory cells storing at least a portion of the pattern of logic states and one or more access lines coupled with the one or more memory cells; and

- the pattern of logic states is captured at the selection component based at least in part on coupling the one or more access lines of the plurality of domains or subdomains with the selection component.

10. The memory device of claim 1, wherein:

- the second component comprises a capacitor array, a signal development cache component, or a combination thereof;

- the first component comprises a memory array having a different type of storage element than the second component; and

- the selection component comprises a multiplexer, a digital signal processor, or a combination thereof.

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11. A method at a memory device, comprising:
coupling a first component of the memory device with a
selection component of the memory device based at
least in part on identifying a pattern of logic states
stored at the first component; and
storing, at a plurality of locations of a second component
of the memory device, a pattern of signal states asso-
ciated with the pattern of logic states based at least in
part on coupling the first component with the selection
component and coupling the selection component with
the plurality of locations of the second component.
12. The method of claim 11, further comprising:
processing the pattern of logic states at the selection
component based at least in part on coupling the first
component with the selection component, the process-
ing comprising a shuffling operation, a mathematical
operation, a logic operation, or any combination
thereof.
13. The method of claim 11, further comprising:
processing the pattern of signal states at the selection
component based at least in part on coupling the first
component with the selection component, wherein stor-
ing the pattern of signal states is based at least in part
on processing the pattern of signal states.
14. The method of claim 11, wherein the pattern of signal
states comprises one or more signal states, and storing the
pattern of signal states at the plurality of locations of the
second component comprises:
storing the one or more signal states at each location of the
plurality of locations of the second component based at
least in part on the pattern of signal states.
15. The method of claim 11, wherein:
the second component comprises a capacitor array, a
signal development cache component, or a combination
thereof;
the first component comprises a memory array having a
different type of storage element than the second com-
ponent; and
the selection component comprises a multiplexer, a digital
signal processor, or a combination thereof.

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16. The method of claim 11, further comprising:
activating a first set of word lines associated with the
second component;
applying a first signal corresponding to a first logic state
to a set of signal lines coupled with the second com-
ponent after activating the first set of word lines;
activating a second set of word lines associated with the
second component after applying the first signal;
applying a second signal corresponding to a second logic
state to the set of signal lines after activating the second
set of word lines; and
storing the pattern of signal states at the plurality of
locations of the second component based at least in part
on applying the first signal and the second signal to the
set of signal lines.
17. The method of claim 11, further comprising:
write the pattern of logic states to the first component as
part of a write operation based at least in part on
capturing the pattern of logic states at the selection
component.
18. The method of claim 11, further comprising:
identifying the pattern of logic states based at least in part
on receiving a command; and
transferring, via the second component, data associated
with the pattern of logic states from one or more
locations of the plurality of locations of the second
component to a sense amplifier array based at least in
part on receiving the command.
19. The method of claim 18, wherein the command
comprises a read command, a write command, a modify
command, or any combination thereof.
20. The method of claim 12, wherein:
the first component comprises a plurality of domains or
subdomains, each of the plurality of domains or sub-
domains including one or more memory cells storing at
least a portion of the pattern of logic states and one or
more access lines coupled with the one or more
memory cells; and
the pattern of logic states is captured at the selection
component based at least in part on coupling the one or
more access lines of the plurality of domains or sub-
domains with the selection component.

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